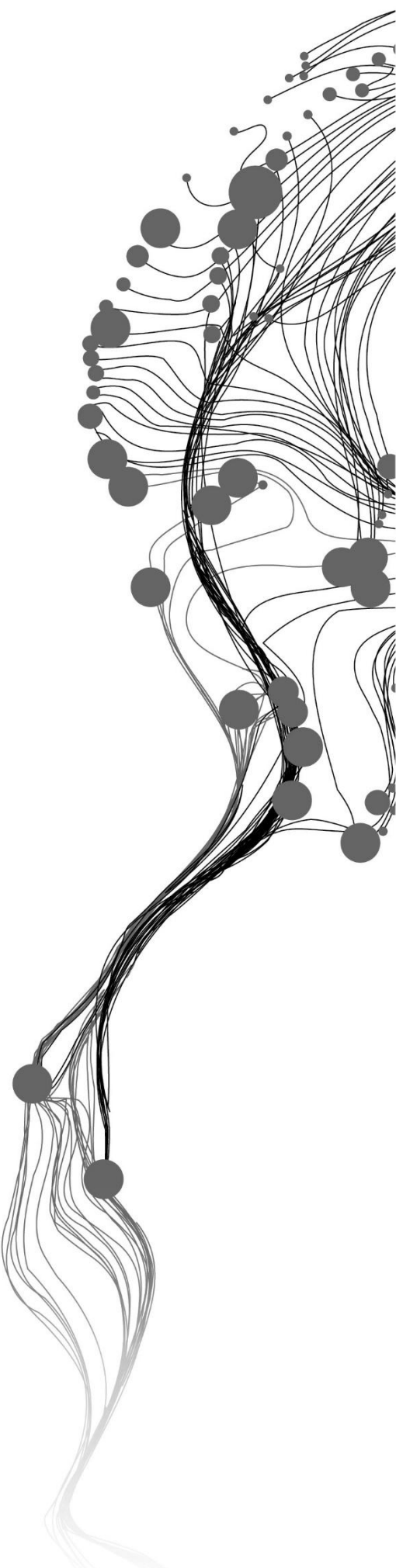


UNCERTAINTY-BASED DETECTION OF LABEL ERRORS IN DEFORESTATION MAPPING

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August 2026

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ABSTRACT

Accurate reference labels are essential for reliable land-use and land-cover classification, yet discrepancies between satellite imagery and mapped reference data can arise from temporal mismatch, cloud contamination, rapid land-cover change, geolocation error, and annotation uncertainty. These discrepancies make it difficult to determine whether an apparent prediction error originates from the model or from the reference labels. This study evaluated the use of uncertainty estimation for distinguishing label errors, model errors, and correct samples in semantic segmentation of deforestation using Landsat imagery and PRODES reference data from the Brazilian Amazon.

A baseline U-Net model was trained for pixel-wise classification, followed by Monte Carlo Dropout inference to derive predictive, aleatoric, and epistemic uncertainty together with additional uncertainty, confidence, calibration, and loss-based metrics. Samples were manually interpreted as label errors, model errors, or correct samples. These classes were then analysed using descriptive statistics, a decision tree classifier, random forest feature importance, and the Hilbert–Schmidt Independence Criterion. Decision tree-generated label-error masks were subsequently used to filter the original training labels before retraining the U-Net.

Using the combined metrics, the decision tree distinguished the three sample classes with a macro-average F1-score of 0.77 and an overall accuracy of 0.84. The decision tree identified label errors with a precision of 0.83 and a recall of 0.60, indicating that uncertainty-guided classification can support the prioritisation of candidate label errors. Across the feature analyses, the Brier score was consistently the most informative metric, while cross-entropy loss and CE – entropy were also highly informative. Reference-dependent metrics incorporating prediction–reference disagreement were generally more informative than metrics based solely on predictive uncertainty.

Filtering identified label errors did not meaningfully improve retraining performance, with F1-score, intersection over union, and overall accuracy remaining comparable to those of the baseline model. Overall, the study demonstrates that uncertainty estimation can support the systematic distinction of label errors, model errors, and correct samples and provides a practical workflow for uncertainty-guided label-quality assessment in remote-sensing semantic segmentation.

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1. INTRODUCTION

1.1. Background

Land use and land cover (LULC) classification plays a significant role in addressing several real-world challenges. For instance, it can be used to identify areas impacted by deforestation and biodiversity loss (Baidoo et al., 2023; Mekasha et al., 2020) and to estimate an area's carbon stock (Fernandes et al., 2021). In this way, it can help assess the overall health of an ecosystem (Zhao & Deng, 2024). The data produced through LULC classification support evidence-based policy formulation, providing valuable insights for the various stakeholders involved.

As one of the most biodiverse countries, Brazil is a key focus for LULC classification. Its vast territory encompasses six terrestrial biomes (Amazon, Caatinga, Cerrado, Pantanal, Atlantic Forest, and Pampa), as well as a large marine biome formed by its extensive coastline (Ellwanger et al., 2022). Consequently, it is home to about 15% of the world's living species (Valli et al., 2018). However, these biomes are being increasingly threatened by human activities such as agricultural expansion, mining, urbanisation, and the development of infrastructure including roads and dams (Azevedo-Santos et al., 2021).

To inform public policy and environmental management, Brazil maintains several monitoring programmes that generate large-scale spatial datasets. PRODES produces annual deforestation maps for Brazil, TerraClass produces land-use and land-cover maps for the Amazon and Cerrado, and MapBiomas provides annual LULC maps at the national scale (Santos et al., 2025). Their broad coverage and high reliability make them valuable reference products for both environmental monitoring and machine-learning applications. Their spatially explicit land-cover information can also be used as reference labels for training and evaluating machine-learning models, reducing the need to manually annotate large volumes of satellite imagery.

However, challenges emerge when these products are used as training and validation data for machine-learning models. Label noise can arise when reference labels do not accurately represent the conditions observed in the corresponding satellite imagery. Such discrepancies may occur because of temporal mismatches between image acquisition and deforestation map production, persistent cloud cover, rapid land-use transitions, seasonal variability, geolocation errors, or imperfections in the annotation process (Hell & Brandmeier, 2024; Santos et al., 2025). As a result, an apparent prediction error may reflect either a genuine model mistake or an error in the reference data itself. These ambiguities complicate both model training and performance assessment and may reduce the reliability of machine-learning systems for LULC mapping.

The presence of label noise highlights the importance of data quality in machine-learning workflows. While traditional model-centric approaches primarily focus on improving algorithms, architectures, and hyperparameter configurations, increasing attention has been directed towards data-centric approaches

that seek to improve the quality, consistency, and reliability of training data (Kumar et al., 2024; Malerba & Pasquadibisceglie, 2024; Roscher et al., 2024; Zha et al., 2025). In remote-sensing applications, improving data quality is particularly important because large-scale reference datasets are often generated from multiple sources collected at different times and under varying environmental conditions (Ferchichi et al., 2017).

A key challenge in this context is distinguishing between uncertainty caused by the model and uncertainty caused by the data. In uncertainty-based sampling, samples associated with high model uncertainty are often considered informative when the uncertainty reflects observations that are difficult for the model or insufficiently represented in the training data. For example, a recently deforested area with a spectral response intermediate between intact forest and cleared land may be difficult for the model to classify and therefore exhibit high uncertainty despite being correctly labelled. However, high uncertainty may also arise from problems in the reference data rather than from genuinely difficult observations. Annotation errors occur when an incorrect label is assigned despite the underlying class being sufficiently clear, such as an intact forest pixel being labelled as deforestation. In contrast, ambiguous labels occur when the appropriate class is itself uncertain, such as a pixel containing a mixture of forest and recently cleared vegetation for which class membership is not clearly defined. Other sources include temporal mismatch between satellite imagery and reference labels, cloud contamination, and spatial misalignment between imagery and mapped deforestation boundaries. Uncertainty quantification provides a framework for analysing prediction reliability and investigating the potential causes of uncertainty. In particular, the distinction between aleatoric uncertainty, which originates from noise and ambiguity in the observations, and epistemic uncertainty, which originates from incomplete model knowledge, offers a promising approach for understanding the causes of prediction errors and distinguishing potentially noisy samples from genuinely informative but difficult observations (Deshpande et al., 2025; Kendall & Gal, 2017; Kondylatos et al., 2025; Ren et al., 2022; Valdenegro-Toro & Mori, 2022).

This study investigates the use of uncertainty-based methods for distinguishing potentially noisy samples from samples for which the reference label is correct but the model prediction is incorrect. These cases are operationalised as label errors and model errors, respectively, together with correctly labelled samples. Particular emphasis is placed on uncertainty decomposition and uncertainty metrics to determine whether these three sample categories can be distinguished. By separating potential label errors from model errors, the study aims to identify unreliable training samples without unnecessarily removing correctly labelled samples that may contain useful information for model learning. In this way, the study aims to support more robust and data-centric approaches to LULC classification and contribute to the development of more reliable machine-learning systems for environmental monitoring.

1.2. Research Gap

The literature demonstrates that substantial progress has been made in both uncertainty quantification and label-noise mitigation. Bayesian neural networks, Monte Carlo dropout, deep ensembles, evidential learning, and conformal prediction provide mechanisms for estimating predictive uncertainty (He et al., 2026). Similarly, numerous approaches have been proposed for learning under noisy labels, including robust architectures, robust regularisation techniques, robust loss functions, loss-adjustment methods, and sample-selection approaches (Song et al., 2023).

Uncertainty estimates are not only used to assess prediction reliability but are also widely employed in uncertainty-sampling and active-learning strategies. In uncertainty sampling, samples for which the model

produces the most uncertain predictions are prioritised for annotation or inclusion in the training process. This approach is founded on the assumption that highly uncertain samples are informative because they contain information that is not yet well captured by the current model (Ren et al., 2021).

However, uncertainty does not always originate from informative sample complexity. For example, a pixel located at the boundary between intact forest and a recently cleared area may be difficult for a model to classify because it contains spectrally mixed information and therefore represents a genuinely difficult observation. In contrast, high uncertainty may also occur when the reference label indicates forest but the corresponding Landsat imagery shows that the area has already been cleared because of a temporal mismatch between the imagery and reference data. In many real-world datasets, uncertainty may also arise from annotation errors, ambiguous labels, temporal inconsistencies, cloud contamination, geolocation inaccuracies, or other forms of data corruption (Ferchichi et al., 2017). As a result, highly uncertain samples may represent either genuinely difficult observations or problematic samples affected by label noise.

Although uncertainty quantification and label-noise mitigation have both been studied extensively, they are often treated as separate research problems. Existing uncertainty-quantification research focuses primarily on estimating and characterising predictive uncertainty (He et al., 2026), whereas label-noise research has developed methods for robust learning, loss adjustment, and sample selection under noisy labels (Song et al., 2023). This leaves comparatively limited attention to the use of uncertainty information for distinguishing whether an erroneous prediction arises from an incorrect reference label or from the model itself, particularly in remote-sensing applications.

In deforestation mapping, prediction errors may originate either from inaccuracies in the model or from inaccuracies in the reference labels caused by temporal mismatch, cloud contamination, annotation inconsistencies, or land-cover transitions (Ferchichi et al., 2017). Existing uncertainty metrics, such as predictive entropy, Monte Carlo disagreement, and mutual information, can identify predictions associated with high uncertainty, but they do not directly explain whether the underlying source of uncertainty originates from the model, the data, or genuine sample complexity.

Furthermore, despite growing interest in data-centric AI, limited research has investigated whether uncertainty measures can be used to distinguish potentially noisy samples from correctly labelled samples on which the model produces erroneous predictions in large-scale deforestation datasets.

1.3. Research Objectives and Research Questions

Objective 1. To analyse whether uncertainty measures distinguish label errors, model errors, and correct samples, and to determine which uncertainty metrics are most informative for this task.

- **RQ1.** Can uncertainty measures distinguish label errors, model errors, and correct samples?
- **RQ2.** Which uncertainty metrics are most useful for distinguishing these sample classes?

Objective 2. To evaluate the effectiveness of uncertainty metrics for identifying label errors in semantic segmentation.

- **RQ3.** How effectively can uncertainty metrics be used to identify label errors in semantic segmentation?

Objective 3. To assess whether uncertainty-guided sample filtering improves model performance through retraining.

- **RQ4.** Does filtering identified label errors improve model retraining?

1.4. Research Contribution

This study makes the following empirical contributions within the context of remote-sensing semantic segmentation:

1. It evaluates whether model-derived uncertainty and error-related information can distinguish label errors, model errors, and correct samples as three separate classes in remote-sensing semantic segmentation.
2. It compares multiple uncertainty and reference-dependent metrics to determine which metrics are most informative for distinguishing label errors, model errors, and correct samples.
3. It evaluates how effectively uncertainty-guided classification can identify candidate label errors at the pixel level when model errors are explicitly retained as a competing source of prediction–reference disagreement.
4. It evaluates whether filtering uncertainty-guided candidate label errors improves downstream semantic-segmentation performance through model retraining.

2. LITERATURE REVIEW

2.1. Uncertainty in Machine Learning

Understanding uncertainty is essential for interpreting machine-learning predictions and assessing their reliability. Uncertainty in machine learning is commonly divided into aleatoric uncertainty and epistemic uncertainty. Aleatoric uncertainty arises from inherent randomness or noise in the data-generating process, such as measurement errors, sensor noise, ambiguous observations, or overlapping class distributions. Because this uncertainty is intrinsic to the data, it cannot be eliminated simply by collecting additional training samples. In contrast, epistemic uncertainty originates from incomplete knowledge of the model or the underlying data-generating process. It reflects uncertainty about the correct model or its parameters and can generally be reduced by acquiring more representative data or improving the model (Kendall & Gal, 2017; Valdenegro-Toro & Mori, 2022; Deshpande et al., 2025; Kondylatos et al., 2025).

Aleatoric uncertainty may be categorised as homoscedastic, in which the observation noise is assumed to have a constant variance for all inputs, or heteroscedastic, in which the observation noise has an input-dependent variance. The observed values remain random under both assumptions; only the variance of the underlying noise distribution differs. Heteroscedastic uncertainty is particularly relevant in computer vision and remote sensing because some observations are inherently more ambiguous than others, leading to larger predictive variances for those inputs (Kendall & Gal, 2017; Kondylatos et al., 2025).

Mathematically, aleatoric uncertainty is represented as uncertainty in the model output conditioned on an input. It is therefore associated with the conditional predictive distribution

$$p(y | x)$$

where x denotes the input sample and y the corresponding output or class label. This conditional distribution describes the variability that remains even if the true data-generating process were known.

In regression problems, aleatoric uncertainty is commonly modelled as

$$Y = f(x) + \epsilon$$

where Y is the target variable, $f(x)$ is the model prediction, and ϵ represents the observation noise. The noise is often assumed to follow a Gaussian distribution,

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

where σ^2 denotes the noise variance, or more generally,

$$\epsilon \sim p(\epsilon)$$

For heteroscedastic regression, the observation noise depends on the input,

$$Y = f(x) + \epsilon(x)$$

with

$$\epsilon(x) \sim \mathcal{N}(0, \sigma^2(x))$$

where $\sigma^2(x)$ is an input-dependent variance.

For classification tasks, aleatoric uncertainty is represented through predictive class probabilities, often obtained using the softmax function or by modelling Gaussian uncertainty over the logits before applying the softmax layer (Kendall & Gal, 2017; Valdenegro-Toro & Mori, 2022; Kondylatos et al., 2025).

Epistemic uncertainty, in contrast, represents uncertainty about the model itself rather than its outputs. In Bayesian learning, this is modelled by placing a probability distribution over the model parameters,

$$p(W \mid X, Y)$$

or more generally,

$$p(\theta \mid D)$$

where W denotes the neural-network weights, θ represents the model parameters, and $D = (X, Y)$ is the training dataset containing inputs X and corresponding outputs Y .

Predictions are then obtained by marginalising over the posterior distribution,

$$p(y \mid x, D) = \int p(y \mid x, W) p(W \mid D) dW$$

where the integral averages predictions over all plausible model parameters.

In practice, this integral is typically approximated using Monte Carlo sampling, Bayesian neural networks, MC Dropout, or Deep Ensembles.

Consequently, aleatoric uncertainty is represented through the predictive distribution

$$p(y \mid x, W)$$

whereas epistemic uncertainty is represented through the posterior distribution

$$p(W \mid D)$$

(Kendall & Gal, 2017; Valdenegro-Toro & Mori, 2022; Deshpande et al., 2025)

Most deep-learning approaches represent both aleatoric and epistemic uncertainty probabilistically. Aleatoric uncertainty is typically described using conditional probability distributions, Gaussian likelihoods, Bernoulli or categorical distributions, and predictive distributions over the model outputs. Similarly, Bayesian methods represent epistemic uncertainty through probability distributions over model parameters or hypotheses (Kendall & Gal, 2017; Valdenegro-Toro & Mori, 2022; Deshpande et al., 2025; Kondylatos et al., 2025)

However, Hüllermeier & Waegeman (2021) argue that epistemic uncertainty is not inherently probabilistic. Besides Bayesian posterior distributions, it can also be represented using non-probabilistic frameworks, such as version spaces, credal sets, possibility distributions, fuzzy sets, or evidence theory. They further argue that probability distributions are well suited for representing randomness but are less appropriate for representing ignorance, since ignorance does not necessarily imply equal probabilities. Consequently, while aleatoric uncertainty is naturally probabilistic, epistemic uncertainty may be represented either probabilistically or non-probabilistically depending on the uncertainty framework employed.

Several Bayesian deep-learning methods express predictive uncertainty as the sum of aleatoric and epistemic components. For regression, this decomposition is commonly derived from the law of total variance,

$$\sigma_*^2(x) = \mathbb{E}[\sigma_i^2(x)] + \text{Var}[\mu_i(x)]$$

where $\sigma_*^2(x)$ is the total predictive variance for input x , $\mathbb{E}[\cdot]$ denotes the expectation operator, $\text{Var}[\cdot]$ denotes the variance operator, $\sigma_i^2(x)$ is the predicted variance from the i -th model sample or ensemble member, and $\mu_i(x)$ is the corresponding predicted mean. The expected predictive variance represents aleatoric uncertainty, whereas the variance of the predictive means represents epistemic uncertainty (Valdenegro-Toro & Mori, 2022).

A similar decomposition was presented by Kendall and Gal (2017) in the context of Bayesian neural networks and Monte Carlo Dropout:

$$\text{Var}(y) \approx \sigma^2 + \frac{1}{T} \sum_{t=1}^T f_{W_t}(x)^T f_{W_t}(x) - \mathbb{E}(y)^T \mathbb{E}(y)$$

where σ^2 is the observation noise variance, T is the number of Monte Carlo samples, $f_{W_t}(x)$ is the prediction obtained using sampled weights W_t , and $\mathbb{E}(y)$ is the predictive mean. The first term corresponds to aleatoric uncertainty arising from noise in the data, while the second term captures epistemic uncertainty arising from variability across model predictions (Kendall & Gal, 2017).

For classification tasks, uncertainty can also be decomposed into aleatoric and epistemic components. Deshpande et al. (2025) expressed total predictive uncertainty as

$$U = \frac{1}{T} \sum_{t=1}^T \hat{y}_t (1 - \hat{y}_t) + \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \bar{y})^2$$

where $\hat{y}_t$ is the prediction from the t -th Monte Carlo sample, $\bar{y}$ is the mean prediction, and T is the number of Monte Carlo samples. The first term represents aleatoric uncertainty modelled through the Bernoulli predictive distribution, whereas the second term represents epistemic uncertainty arising from variability among Monte Carlo predictions (Deshpande et al., 2025).

Nevertheless, this decomposition is not universal. Hüllermeier and Waegeman (2021) emphasise that additive decompositions depend on the chosen uncertainty measure and modelling framework. They argue that predictive uncertainty may arise from multiple sources, including aleatoric uncertainty, model uncertainty, and approximation uncertainty, with the latter two jointly constituting epistemic uncertainty. Moreover, the distinction between aleatoric and epistemic uncertainty is context-dependent because changes in the hypothesis space, available information, or modelling assumptions may alter whether uncertainty is considered reducible or irreducible (Hüllermeier & Waegeman, 2021).

2.2. Sources of Uncertainty

After distinguishing between aleatoric and epistemic uncertainty, the next step is to examine the specific sources from which these uncertainties arise in remote-sensing applications. Unless otherwise specified, the sources discussed below are assumed to be potentially present in the data and modelling framework used in this study, although their relative contributions are not quantified individually. Spectral parameters are primarily associated with aleatoric uncertainty because they arise from inherent variability in the observed data and measurement process. Sources of uncertainty include spectral reflectance variability (e.g., differences in reflectance between dense forest, degraded forest, and pasture), sensor noise (e.g., random fluctuations in Landsat pixel values), mixed pixels (e.g., a 30 m pixel containing both forest and pasture), channel shifts (e.g., slight displacement of spectral band positions between sensors), image registration errors (e.g., misalignment between multi-temporal Landsat scenes), atmospheric conditions (e.g., varying water vapour over the Amazon), haze (e.g., smoke haze from biomass burning), clouds (e.g.,

cloud-covered forest areas), topography (e.g., terrain-induced reflectance differences in hilly regions), and viewing geometry (e.g., different sensor viewing angles between acquisitions). These factors can alter the recorded spectral response even when the underlying land cover remains unchanged, thereby introducing uncertainty into deforestation mapping and land-cover classification (Ferchichi et al., 2017). In this study, sensor noise, topography, and viewing geometry are present but likely secondary sources of uncertainty, while channel shifts are scientifically valid but likely negligible for explaining the observed label-noise patterns.

Texture parameters are generally considered a source of epistemic uncertainty because they depend on how spatial patterns are represented and interpreted. Important sources include object-size versus sensor-resolution interactions (e.g., narrow logging roads smaller than a Landsat pixel), border effects (e.g., pixels along the boundary between forest and deforested land), and object-background ambiguity (e.g., sparse forest appearing similar to shrubland). Such uncertainties arise because the available image resolution may not adequately capture the true spatial characteristics of landscape features, resulting in ambiguity during classification (Ferchichi et al., 2017).

Shape parameters also contribute to epistemic uncertainty because they influence how objects are perceived and modelled in remotely sensed imagery. Relevant factors include solar position (e.g., different sun angles between dry- and wet-season acquisitions), illumination conditions (e.g., shaded versus sunlit forest patches), and scene illumination variability (e.g., changing illumination caused by scattered cloud cover). These factors can alter the apparent shape and appearance of land-cover features, potentially leading to misinterpretation of forest and non-forest areas (Ferchichi et al., 2017). In this study, solar position, illumination conditions, and scene illumination variability are present but likely secondary sources of uncertainty.

Vegetation parameters are predominantly associated with aleatoric uncertainty because they reflect natural variability in vegetation properties. Sources include soil background variability (e.g., exposed soils of different colours after deforestation), red and NIR band variability (e.g., seasonal changes in vegetation reflectance), and sub-pixel structural variability (e.g., varying canopy cover within a single Landsat pixel). Such variations influence the spectral response of vegetation and can make it difficult to distinguish between intact forest, degraded forest, and recently cleared areas (Ferchichi et al., 2017).

Climate parameters introduce additional aleatoric uncertainty through environmental and atmospheric processes that affect image acquisition. These include atmospheric correction uncertainty (e.g., imperfect correction of Amazon atmospheric effects), land-surface emissivity uncertainty (e.g., uncertainty in emissivity estimates over mixed forest–soil surfaces), aerosols and gaseous absorbers (e.g., smoke particles from forest fires), angular effects (e.g., bidirectional reflectance effects caused by changing sun-sensor geometry), wavelength uncertainty (e.g., uncertainty in the exact spectral response of a sensor band), and sensor band-pass effects (e.g., different Landsat sensors integrating slightly different wavelength ranges). These factors can alter the radiometric properties of imagery and therefore affect the reliability of land-cover classifications (Ferchichi et al., 2017). In this study, angular effects are present but likely secondary, while land-surface emissivity uncertainty, wavelength uncertainty, and sensor band-pass effects are scientifically valid but likely negligible for explaining the observed label-noise patterns.

Model structure represents a major source of epistemic uncertainty because it originates from imperfect knowledge about the most appropriate model for a given problem. Sources include model specification errors (e.g., omitting spatial-context features important for detecting forest degradation), physical model assumptions (e.g., assuming spectral signatures remain stable across years), choice of hypothesis space (e.g., selecting a decision tree instead of a random forest or neural network), and model-form uncertainty (e.g., uncertainty arising from choosing U-Net rather than another segmentation architecture). These

uncertainties arise from modelling decisions and could potentially be reduced through improved model design, additional knowledge, or alternative methodological choices (Kiureghian & Ditlevsen, 2009; Hüllermeier & Waegeman, 2021). In this study, physical model assumptions are present but likely a secondary source of uncertainty.

Parameter estimation uncertainty is another form of epistemic uncertainty that arises during model training. It reflects statistical uncertainty in estimated model parameters due to limited observations (e.g., uncertainty in learned model weights due to a small number of verified label-error polygons). When training data are limited or noisy, multiple parameter configurations may explain the observed data equally well, leading to uncertainty in the resulting predictions (Kiureghian & Ditlevsen, 2009).

Training data uncertainty arises when the available samples do not adequately represent the true data-generating process. This includes approximation uncertainty caused by finite (e.g., only a limited number of annotated deforestation samples), incomplete (e.g., missing examples of forest degradation), insufficient (e.g., too few samples to capture regional variability across Pará), or unrepresentative training samples (e.g., training data concentrated in clear-cut areas but not selective logging areas). Such uncertainty is epistemic because it stems from incomplete knowledge and could potentially be reduced by collecting more representative training data (Hüllermeier & Waegeman, 2021).

Measurements and calibration can contribute both aleatoric and epistemic uncertainty. Sources include measurement errors (e.g., inaccurate reflectance measurements caused by sensor limitations), proxy measurements (e.g., using MapBiomas or PRODES labels as proxies for ground truth), sensor calibration errors (e.g., incorrect radiometric calibration coefficients), and uncertainty in calibration data (e.g., calibration references affected by measurement noise). In deforestation mapping, these uncertainties are particularly important because reference labels and sensor measurements often serve as the basis for model training and validation (Kiureghian & Ditlevsen, 2009). In this study, measurement errors and uncertainty in calibration data are present but likely secondary sources of uncertainty, while sensor calibration errors are scientifically valid but likely negligible for explaining the observed label-noise patterns.

Computation introduces epistemic uncertainty through the numerical methods used to implement models. Relevant sources include numerical approximations (e.g., approximating a complex optimisation problem during model training), convergence tolerances (e.g., stopping optimisation before reaching the exact optimum), truncation errors (e.g., errors introduced by finite-precision numerical calculations), and computational simplifications (e.g., reducing model complexity to decrease training time). These uncertainties arise from practical limitations in computation and algorithm design rather than from the observed data themselves (Kiureghian & Ditlevsen, 2009). In this study, numerical approximations, convergence tolerances, and computational simplifications are present but likely secondary sources of uncertainty, while truncation errors are scientifically valid but likely negligible for explaining the observed label-noise patterns.

Finally, intrinsic process variability represents aleatoric uncertainty arising from the inherent randomness of the underlying phenomenon. Sources include non-deterministic relationships between inputs and outputs (e.g., forest areas with similar spectral signatures experiencing different future land-use outcomes) and stochastic behaviour of the underlying phenomenon (e.g., unpredictable human decisions leading to deforestation in one area but not another). Unlike epistemic uncertainty, this form of uncertainty cannot be eliminated through additional data collection or improved modelling because it reflects genuine variability in the real world (Hüllermeier & Waegeman, 2021).

2.3. Methods of Uncertainty Quantification

Once uncertainty sources have been identified, they must be quantified using appropriate uncertainty-estimation methods. He et al. (2026) classify uncertainty quantification methods in machine learning into three broad categories: model uncertainty, data uncertainty, and combined model–data uncertainty.

2.3.1. Model uncertainty

Methods for quantifying model uncertainty focus on estimating uncertainty arising from limited knowledge about model parameters, structure, or training data. He et al. (2026) classify model uncertainty methods into three broad categories: Bayesian neural networks (BNNs), ensemble models, and sample distribution-related methods.

2.3.1.1. Bayesian Neural Networks

Bayesian neural networks quantify model uncertainty by treating the parameters of a neural network as probability distributions rather than as fixed values. In a conventional frequentist neural network, training aims to identify one optimal parameter set, expressed as

$$\theta^* = \operatorname{argmin}_{\theta} \mathcal{L}(Y, f(X, \theta)),$$

where θ^* is the estimated optimal parameter set, $\mathcal{L}$ is the loss function, X is the training input, Y is the training target, and $f(X, \theta)$ is the model output. This point-estimation approach can lead to overfitting and overconfident predictions. In contrast, Bayesian neural networks assign a prior distribution to the parameters, $p(\theta)$, and infer a posterior distribution after observing the training data:

$$p(\theta|X, Y) = \frac{p(Y|X, \theta)p(\theta)}{p(Y|X)}.$$

Here, $p(\theta|X, Y)$ represents the posterior distribution of the model parameters given the training data. This posterior expresses how uncertain the model is about its parameters. If the parameters are assumed to follow a Gaussian distribution, a larger posterior variance indicates higher model uncertainty, often due to limited training data. For a new sample, prediction is obtained by marginalising over the parameter posterior:

$$p(y^*|x^*, X, Y) = \int p(y^*|x^*, \theta)p(\theta|X, Y)d\theta.$$

Here, x^* is a test input and y^* is its predicted output. The uncertainty of the prediction is reflected in the variance of the predictive distribution. Although Bayesian neural networks can theoretically represent both data uncertainty through $p(y^*|x^*, \theta)$ and model uncertainty through $p(\theta|X, Y)$, many practical implementations simplify the prediction term and mainly estimate parameter uncertainty (He et al., 2026).

The main contribution of Bayesian neural network methods is that they provide a probabilistic framework for estimating uncertainty in model parameters. Since the exact posterior distribution is usually intractable, several approximation methods are used. Variational inference and Laplace approximation approximate the posterior using a simpler distribution family, such as factorised Gaussian distributions. Markov Chain Monte Carlo estimates the posterior through sampling rather than analytic approximation. Monte Carlo dropout offers a more practical approximation by keeping dropout active during inference and generating multiple stochastic predictions using different dropout masks. The variance among these predictions is then used as an estimate of uncertainty. In this way, these methods make Bayesian-style uncertainty estimation more feasible for deep neural networks (He et al., 2026).

The strength of Bayesian neural networks is that they are theoretically well grounded and directly connect model uncertainty to uncertainty in the learned parameters. Approximation methods such as variational inference, Laplace approximation, and MC dropout reduce the computational burden compared with exact Bayesian inference. MC dropout is especially useful because it requires only minor architectural changes and can be applied to existing neural networks. However, Bayesian neural networks also have important limitations. The true posterior is difficult to compute, so approximations are required, and these approximations may produce less accurate uncertainty estimates. MCMC is flexible but computationally expensive, while MC dropout is simple and scalable but can be poorly calibrated compared with other uncertainty quantification methods (He et al., 2026).

Among the Bayesian approximation methods reviewed above, Monte Carlo dropout was selected for this study because it provides a computationally practical means of estimating model uncertainty through repeated stochastic forward passes without requiring the training of multiple independent models.

2.3.1.2. Ensemble Models

Ensemble models quantify model uncertainty by combining predictions from multiple neural networks. Instead of relying on one trained model, ensemble approaches construct several models and use the variation among their outputs as an uncertainty estimate. In regression tasks, this uncertainty can be measured using the variance of the ensemble predictions. In classification tasks, it can be measured using predictive entropy or the average uncertainty across class-probability outputs. Conceptually, if different models produce similar predictions, the model uncertainty is low; if their predictions diverge, the model uncertainty is high (He et al., 2026).

The contribution of ensemble methods is that they capture model uncertainty from several possible sources. Bootstrap ensembles train different networks on resampled versions of the training dataset, allowing uncertainty from limited or variable training data to be reflected in the prediction distribution. Architecture-based ensembles vary the structure of the neural networks, such as the number of layers, hidden units, or activation functions, thereby capturing uncertainty from model misspecification. Other ensemble strategies use different initialisations or data-shuffling procedures, allowing each model to use the full training dataset while still producing diverse models. Hyperensembles extend this idea by varying hyperparameters such as learning rate, optimisation strategy, or training procedure, which allows uncertainty from training choices to be incorporated into the final uncertainty estimate (He et al., 2026).

The main strength of ensemble models is that they are conceptually simple and can capture uncertainty from multiple modelling decisions, including architecture, parameters, data sampling, and hyperparameters. They are often easier to implement than fully Bayesian neural networks and can provide strong empirical uncertainty estimates. However, their main weakness is computational cost. Multiple independent networks must be trained, stored, and evaluated, which increases both training time and memory requirements. In addition, ensemble uncertainty depends on diversity among the models; if the ensemble members are too similar, the resulting uncertainty estimates may be unreliable or underestimated (He et al., 2026).

2.3.1.3. Sample Distribution-Related Methods

Sample distribution-related methods quantify model uncertainty by considering how a test sample relates to the training data distribution. This category focuses on cases where uncertainty arises because a test sample is far from the training samples or lies in a sparse region of the feature space. The basic idea is that predictions should be less certain when the model receives samples that are poorly represented in the

training data. He et al. (2026) distinguish this from broader out-of-distribution detection and focus specifically on sample-density or distance-aware uncertainty within the feature space.

The methods in this category contribute by making uncertainty sensitive to the location of a sample relative to the training distribution. Gaussian process hybrid neural networks use covariance between test and training samples to estimate uncertainty. Since Gaussian processes condition predictions on similarity to observed data, they naturally produce lower uncertainty in dense regions and higher uncertainty in sparse or boundary regions. However, standard Gaussian processes scale poorly because covariance-matrix inversion has computational complexity

$$\mathcal{O}(n^3),$$

where n is the number of training samples. Sparse approximations reduce this to

$$\mathcal{O}(m^2n),$$

where m is the number of inducing variables and $m \ll n$. Deep kernel learning further improves this category by using a neural network to learn useful latent representations before applying a Gaussian process kernel. This changes the kernel from comparing raw inputs, $K_\theta(x_i, x_j)$, to comparing learned representations, $K_\theta(g(x_i; w), g(x_j; w))$, where $g(\cdot; w)$ is the neural network representation function. Deep or compositional Gaussian processes extend this further by stacking Gaussian process layers:

$$y = f_L(f_{L-1}(\dots f_1(x))).$$

Each $f_i(\cdot)$ represents a Gaussian process layer, allowing uncertainty to propagate through multiple levels of representation (He et al., 2026).

Distance-aware neural networks contribute to this category by encouraging the learned feature space to preserve meaningful distances between samples. Their aim is to avoid feature collapse, where in-distribution and distant or out-of-distribution samples are mapped to similar latent representations. This problem can cause overconfident predictions for unfamiliar samples. To reduce this risk, distance-aware methods introduce sensitivity and smoothness constraints, often expressed through bi-Lipschitz bounds:

$$L_1|x - x'| * \mathcal{X} < |h * \theta(x) - h_\theta(x')| * \mathcal{H} < L_2|x - x'| * \mathcal{X}.$$

Here, $h_\theta(x)$ is the learned representation of input x , $|\cdot| * \mathcal{X}$ denotes distance in the input space, $|\cdot| * \mathcal{H}$ denotes distance in the hidden feature space, and L_1 and L_2 are constants. These constraints aim to ensure that changes in the input space are reflected appropriately in the learned feature space. Spectral normalisation enforces this efficiently by normalising layer weights, while gradient penalty adds a soft regularisation term based on input gradients, although it is more computationally expensive (He et al., 2026).

The strength of sample distribution-related methods is that they explicitly address uncertainty caused by sparse training coverage and distance from the training distribution. Gaussian process-based methods offer a strong probabilistic foundation, while distance-aware neural networks provide a simpler and more scalable way to incorporate sample-distance information into deep learning models. However, this category also has limitations. Gaussian processes are difficult to scale to large datasets and depend heavily on suitable kernel functions. Deep kernel learning and compositional Gaussian processes reduce some of these issues but introduce additional modelling and optimisation complexity. Distance-aware neural networks are more efficient, but their reliability depends on whether distances in the learned embedding space truly represent meaningful similarity between samples (He et al., 2026).

2.3.2. Data Uncertainty

In contrast to model uncertainty, data uncertainty methods focus on quantifying uncertainty that originates from the observations themselves. He et al. (2026) organise data uncertainty methods into two main categories: deep discriminative models and deep generative models.

2.3.2.1. Deep Discriminative Models

Deep discriminative models quantify data uncertainty by directly producing a predictive distribution from a neural network. Instead of returning only a single deterministic prediction, the model outputs either the parameters of a probability distribution or a prediction interval that represents the range of plausible target values. He et al. (2026) divide discriminative approaches into parametric and non-parametric methods. Parametric methods assume that the output follows a predefined probability distribution, whereas non-parametric methods avoid specifying an explicit output distribution.

In classification, a parametric discriminative model commonly represents the prediction as a categorical distribution over k classes:

$$p(y|x, \theta) = \text{Categorical}(y; \pi), \quad \pi = f(x; \theta), \quad \sum_{c=1}^k \pi_c = 1, \quad \pi_c > 0.$$

Here, $p(y|x, \theta)$ is the predictive distribution, x is the input sample, y is the class label, θ denotes the model parameters, and $\pi = (\pi_1, \dots, \pi_k)$ is the vector of predicted class probabilities. A common implementation uses the softmax function to obtain the class probabilities:

$$\pi_i = \frac{\exp(h_i(x; \theta))}{\sum_{c=1}^k \exp(h_c(x; \theta))}.$$

Here, $h_i(x; \theta)$ is the logit for class i . However, softmax outputs can be overconfident because they often push probabilities towards values close to zero or one. Temperature scaling addresses this by introducing a calibration parameter T :

$$p_i = \frac{\exp(h_i(x)/T)}{\sum_{c=1}^k \exp(h_c(x)/T)}.$$

Here, T controls the sharpness of the predicted probability distribution. Although temperature scaling is simple, it still requires calibration and may not fully remove overconfidence (He et al., 2026).

In regression, parametric discriminative models usually represent data uncertainty as sample-dependent noise. The target is commonly written as

$$y_i = f_{\theta}(x_i) + \epsilon(x_i),$$

where $f_{\theta}(x_i)$ is the predicted mean and $\epsilon(x_i)$ is heterogeneous Gaussian noise. This gives the predictive distribution

$$p(y|x, \theta) = \mathcal{N}(f_{\theta}(x), \sigma_{\theta}(x)),$$

where $f_{\theta}(x)$ is the predicted mean and $\sigma_{\theta}(x)$ is the predicted uncertainty. The model can be trained by optimising the negative log-likelihood:

$$\mathcal{L}_{NN}(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2\sigma_{\theta}^2(x_i)} \|y_i - f_{\theta}(x_i)\|^2 + \frac{1}{2} \log \sigma_{\theta}^2(x_i).$$

Here, n is the number of training samples. This formulation is known as heteroscedastic regression because the noise variance changes across samples. It is useful when some observations are inherently more uncertain than others. Other parametric choices are also possible, such as Gaussian mixture distributions implemented through mixture density networks. These can represent multiple plausible prediction modes but require more computation and depend on choosing an appropriate output distribution (He et al., 2026).

Non-parametric discriminative methods estimate data uncertainty without assuming a specific predictive distribution. A common approach is the prediction interval, which outputs a lower and upper bound $[y_l, y_u]$ such that the true value is expected to fall inside the interval with a specified confidence level:

$$p(y \in [y_l, y_u]) > 1 - \alpha.$$

Here, $1 - \alpha$ is the required confidence level. The aim is to produce intervals that are wide enough to cover the true value but narrow enough to remain informative. Lower upper bound estimation directly optimises interval quality by considering both interval width and coverage probability, but its objective can be non-differentiable. A differentiable alternative uses a coverage-width-based loss:

$$\text{Loss} = \text{MPIW} + \lambda \max(0, (1 - \alpha) - \text{PICP})^2.$$

Here, MPIW is the mean prediction interval width, PICP is the prediction interval coverage probability, and λ controls the penalty for insufficient coverage. More recent work formulates prediction interval learning as constrained optimisation. From the primal perspective, the objective is to minimise interval width while enforcing the required coverage:

$$\min_{L, U \in \mathcal{H}, L < U} \mathbb{E}_{x \sim \pi(x)} (U(x) - L(x)) \quad \text{s.t.} \quad p_{\pi}(y \in [L(x), U(x)]) > 1 - \alpha.$$

Here, $L(x)$ and $U(x)$ are the lower and upper interval functions, $\mathcal{H}$ is the function class, and p_{π} denotes probability under the input-output distribution. From the dual perspective, the objective is to minimise expected prediction error while imposing a budget on the total interval width:

$$\min_{f \in \mathcal{F}} \mathbb{E}_{(x,y) \sim \pi(x,y)} L(y, U(x), L(x)) \quad \text{s.t.} \quad \sum_i (U(x_i) - L(x_i)) < B.$$

Here, B is the allowed global interval-width budget. This formulation can construct intervals for a batch of test samples simultaneously, reducing computational overhead (He et al., 2026).

The main contribution of deep discriminative methods is that they provide direct and relatively simple mechanisms for estimating data uncertainty within standard neural network prediction pipelines. Parametric models contribute by allowing the network to learn distributional parameters, such as class probabilities, means, variances, or mixture components. Non-parametric prediction interval methods contribute by estimating uncertainty ranges without forcing the output to follow a fixed probability distribution. In this category, the central aim is not to represent uncertainty in model parameters, but to represent uncertainty in the observed target or prediction outcome itself (He et al., 2026).

The main strength of deep discriminative models is their simplicity. They can often be incorporated into existing neural networks with limited architectural change, especially when the output layer is modified to

predict distribution parameters or interval bounds. Parametric methods are straightforward to train, while non-parametric methods are more flexible because they do not impose a rigid output distribution. However, this category also has limitations. Parametric methods depend heavily on choosing an appropriate probability distribution, which can be difficult without prior knowledge. Softmax-based uncertainty can be overconfident, even after calibration. Non-parametric interval methods avoid distributional assumptions, but they often require specialised loss functions or constrained optimisation. In addition, discriminative approaches are less suitable for structured outputs, such as images or segmentation maps, because they do not naturally capture dependencies among output elements (He et al., 2026).

2.3.2.2. Deep Generative Models

Deep generative models quantify data uncertainty by learning a probability distribution over complex, high-dimensional data or outputs. Rather than predicting only a point estimate or a simple distributional parameter, they learn how possible outputs are generated. A deep generative model maps a latent variable $z \in \mathbb{R}^d$, usually sampled from a simple distribution such as $\mathcal{N}(0, I)$, into the original data space $\mathbb{R}^n$ through a neural network mapping

$$g_\theta(\cdot): \mathbb{R}^d \rightarrow \mathbb{R}^n.$$

Here, d is the dimension of the latent space, n is the dimension of the original data space, and g_θ is the generative model. For uncertainty quantification, conditional deep generative models are used to learn the predictive distribution $p(y|x)$ from supervised input-output pairs. In this setting, the model learns a conditional generator $g_\theta(z, x)$ such that samples generated from $z \sim p(z)$ approximate the unknown true conditional distribution $p_{true}(y|x)$. During inference, multiple outputs can be generated for the same input, and the variability among these generated outputs is used to estimate prediction uncertainty (He et al., 2026).

Variational autoencoder-based models contribute to this category by providing a probabilistic encoder-decoder framework. A VAE contains an encoder $q_\phi(z|x)$ that maps high-dimensional data to a latent representation and a decoder $p_\theta(x|z)$ that reconstructs the data from that latent variable. For data uncertainty, VAE-based methods can model either uncertainty in the input or uncertainty in the target output. For input uncertainty, each sample can be embedded as a Gaussian distribution rather than as a fixed vector, with the mean representing the feature embedding and the variance representing uncertainty in that embedding. For target-output uncertainty, conditional VAEs model the predictive distribution by integrating over latent variables:

$$p(y|x) = \int p(y|x, z)p(z|x)dz \approx \frac{1}{n} \sum_{j=1}^n p(y|x, z_j), \quad z_j \sim p(z).$$

Here, z_j denotes a sampled latent variable. This sampling process allows multiple plausible predictions to be generated for the same input. Probabilistic U-Net extends this idea by combining a conditional VAE with the U-Net architecture, making it suitable for uncertainty estimation in image segmentation, where several plausible segmentation maps may exist (He et al., 2026).

Generative adversarial network-based models contribute by learning output distributions through adversarial training. A GAN consists of a generator and a discriminator. In a conditional GAN, the generator receives both the input x and random noise z and produces a target output:

$$\mathcal{G}: (x, z) \rightarrow y.$$

The discriminator learns to distinguish generated outputs from real target samples. For uncertainty quantification, repeated sampling of z can produce multiple plausible outputs for the same input, and their variability represents predictive uncertainty. GAN-based uncertainty methods are expressive because they can represent complex output distributions without requiring an explicit probability density function. Extensions such as Wasserstein GANs with gradient penalty can improve convergence, and GANs can also be combined with physics-informed neural networks for uncertainty estimation in physical systems (He et al., 2026).

Diffusion-based generative models contribute by estimating uncertainty through an iterative noise-removal process. In the forward process, noise is gradually added to the data according to a variance schedule β_t :

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I).$$

The reverse process learns to denoise the data step by step:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)).$$

Here, μ_θ and Σ_θ are learned neural network functions that reconstruct less noisy data from noisier inputs. Conditional diffusion models can generate multiple plausible outputs given the same input, allowing uncertainty to be measured from the diversity of generated samples. Latent diffusion models improve efficiency by applying the diffusion process in a lower-dimensional latent space rather than directly in the full-resolution data space (He et al., 2026).

The main contribution of deep generative models is their ability to represent uncertainty in structured, high-dimensional outputs. Unlike simpler discriminative methods, generative models can capture dependencies among output elements, such as spatial relationships between pixels in an image or segmentation map. VAE-based models contribute stable probabilistic training and structured latent representations. GAN-based models contribute expressive sample generation without requiring an explicit output distribution. Diffusion models contribute highly expressive probabilistic generation and can produce high-quality samples from complex output distributions (He et al., 2026).

The main strength of deep generative models is that they are well suited to structured uncertainty quantification, especially when multiple plausible outputs may exist for the same input. This makes them useful for image-based tasks, medical image analysis, segmentation, video, time series, and scientific simulations. VAE-based models are relatively stable and provide probabilistic outputs, GANs are highly expressive, and diffusion models combine strong expressiveness with a probabilistic generation framework. However, these advantages come with important limitations. VAE-based methods may be less expressive than GANs or diffusion models. GANs can be difficult to train and may suffer from unstable convergence because their optimisation depends on adversarial learning. Diffusion models are computationally expensive because training and inference usually require many iterative denoising steps. More generally, deep generative models require multiple samples during inference to estimate uncertainty, which increases computational cost (He et al., 2026).

2.3.3. Model and Data Uncertainty

Because many real-world applications involve both model-related and data-related uncertainty, several methods have been developed to estimate these components jointly.

2.3.3.1. Approaches Combining Data and Model Uncertainty

Approaches that combine data and model uncertainty aim to estimate both uncertainty caused by noisy or ambiguous observations and uncertainty caused by limited knowledge of the model parameters, architecture, or training process. Conceptually, these approaches join one method designed for data uncertainty with another method designed for model uncertainty within the same predictive framework. Data uncertainty is usually represented through a predictive distribution or prediction interval, while model uncertainty is estimated through Bayesian neural networks, Monte Carlo dropout, or ensemble models. The purpose is to obtain a total uncertainty estimate that reflects both the variability inherent in the data and the variability arising from the model itself (He et al., 2026).

One common approach combines a Bayesian neural network approximation with a predictive distribution. In this framework, model uncertainty is estimated using Monte Carlo dropout, where each stochastic forward pass corresponds to a different sampled set of network weights, $W_t \sim \text{Bernoulli}(p)$, with p denoting the dropout rate. For each pass, the model outputs a predictive distribution rather than a single point prediction. In regression, this can be written as $[y_t, \sigma_t^2] = p(y|x; W_t)$, where y_t is the predicted mean and σ_t^2 is the predictive variance for the t -th forward pass. After T stochastic forward passes, total predictive uncertainty can be approximated using the law of total variance:

$$\text{Var}(y) \approx \frac{1}{T} \sum_{t=1}^T y_t^2 - \left(\frac{1}{T} \sum_{t=1}^T y_t \right)^2 + \frac{1}{T} \sum_{t=1}^T \sigma_t^2.$$

Here, the first part captures model uncertainty through disagreement among the stochastic model predictions, while the final term captures average data uncertainty through the predicted variances. This method contributes to combined uncertainty estimation by making it possible to separate uncertainty caused by model disagreement from uncertainty caused by noisy observations within a single regression framework (He et al., 2026). This combined uncertainty framework was adopted in this study, with repeated Monte Carlo dropout predictions used to derive predictive uncertainty together with aleatoric and epistemic uncertainty components for each pixel.

Another approach combines ensemble models with predictive distributions. In this case, an ensemble $\mathcal{M} = \{\mathcal{M}_i\}_{i=1}^K$ is constructed using different model parameters, architectures, training subsets, or other modelling choices. Each ensemble member outputs a predictive distribution $p(y|x, \mathcal{M}_i)$. For classification, total uncertainty can be measured using the entropy of the average predictive distribution, while data uncertainty is measured using the average entropy of the individual model predictions. Model uncertainty is then expressed as the mutual information between the prediction and the ensemble model:

$$MI(y, \mathcal{M}) = \mathcal{H}(\mathbb{E}_{p(\mathcal{M}_i)} p(y|x, \mathcal{M}_i)) - \mathbb{E}_{p(\mathcal{M}_i)} \mathcal{H}(p(y|x, \mathcal{M}_i)).$$

Here, $\mathcal{H}(\cdot)$ denotes entropy, the first term represents total predictive uncertainty, and the second term represents expected data uncertainty. The difference between them captures model uncertainty. The contribution of this approach is that it uses ensemble disagreement to quantify model uncertainty while preserving the predictive distribution of each model to represent data uncertainty (He et al., 2026).

A further approach combines ensemble models with prediction intervals. Prediction intervals alone mainly represent data noise, because they estimate a lower and upper range within which the true value is expected to fall. To incorporate model uncertainty, several models are trained with different specifications or different subsets of the training data. For sample i and model j , the lower and upper interval bounds

are denoted as $[y_{l(ij)}, y_{u(ij)}]$. The model uncertainty is estimated from the variance of the lower and upper bounds across ensemble members:

$$\sigma_l^2(i) = \frac{1}{m-1} \sum_{j=1}^m (y_{l(ij)} - \hat{y}_l(i))^2, \quad \text{where} \quad \hat{y}_l(i) = \frac{1}{m} \sum_{j=1}^m y_{l(ij)},$$

$$\sigma_u^2(i) = \frac{1}{m-1} \sum_{j=1}^m (y_{u(ij)} - \hat{y}_u(i))^2, \quad \text{where} \quad \hat{y}_u(i) = \frac{1}{m} \sum_{j=1}^m y_{u(ij)}.$$

For a 95% confidence level, the final interval can then be widened using the ensemble-based uncertainty estimates:

$$\tilde{y}_l = y_l - 1.96\sigma_l^2(i), \quad \tilde{y}_u = y_u + 1.96\sigma_u^2(i).$$

This method contributes to combined uncertainty estimation by allowing prediction intervals to reflect both data uncertainty and model uncertainty. More recent aggregation approaches improve this idea by fitting split normal distributions to prediction intervals and deriving final intervals from the resulting mixture distribution, rather than treating lower and upper bounds independently (He et al., 2026).

The main strength of combination approaches is that they are flexible and can use already established methods for data uncertainty and model uncertainty. They can represent noisy observations through predictive distributions or intervals while also accounting for parameter, architecture, or training-related uncertainty through Bayesian approximations or ensembles. However, these approaches also have important weaknesses. Bayesian and ensemble-based combinations require multiple forward passes, which increases computational cost and storage requirements. In addition, simple combinations of data and model uncertainty may lack strong theoretical justification, so post hoc calibration is often needed to improve the reliability of the resulting uncertainty estimates (He et al., 2026).

2.3.3.2. Evidential Deep Learning

Evidential deep learning is a category of methods designed to estimate both data and model uncertainty using a single deterministic neural network. Instead of producing class probabilities directly, the model predicts evidence for each class. This evidence is then used to parameterise a Dirichlet distribution over categorical probability vectors. Conceptually, the categorical probabilities describe the prediction itself, while the Dirichlet distribution describes uncertainty about those categorical probabilities. This allows the method to distinguish uncertainty caused by ambiguous class probabilities from uncertainty caused by insufficient evidence (He et al., 2026).

For classification, a standard neural network usually outputs a categorical probability vector $\pi = (\pi_1, \pi_2, \dots, \pi_K)$, where $\sum_i \pi_i = 1$. This gives only a point estimate of the class-probability distribution. In evidential deep learning, the network instead predicts class-wise evidence, represented as Dirichlet concentration parameters $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_K\}$, with $\alpha_i > 0$. The Dirichlet distribution defines a probability density over possible categorical probability vectors $\mu = [\mu_1, \dots, \mu_K]$:

$$\text{Dir}(\mu|\alpha) = \frac{\Gamma(\alpha_0)}{\prod_{c=1}^K \Gamma(\alpha_c)} \prod_{c=1}^K \mu_c^{\alpha_c-1}, \quad \alpha_c > 0, \quad \alpha_0 = \sum_{c=1}^K \alpha_c.$$

Here, μ represents possible categorical distribution parameters, K is the number of classes, α_c is the evidence for class c , and α_0 is the total evidence or precision of the Dirichlet distribution. The expected class probability is obtained as

$$p(w_c|x, \theta) = \int p(w_c|\mu)p(\mu|x, \theta)d\mu = \frac{\alpha_c}{\alpha_0}.$$

Data uncertainty is associated with the entropy of the expected class-probability distribution, whereas model uncertainty is associated with the amount of total evidence. When the total evidence $\sum_i \alpha_i$ is high, the Dirichlet distribution becomes sharper and model uncertainty decreases. When the class evidence is balanced across classes, the predicted class distribution has high entropy, indicating higher data uncertainty. Therefore, evidential deep learning separates uncertainty due to ambiguous class membership from uncertainty due to lack of evidence (He et al., 2026).

The main contribution of evidential deep learning is that it avoids repeated stochastic sampling during inference. Unlike Monte Carlo dropout or ensembles, it estimates both data and model uncertainty in a single forward pass. Dirichlet prior networks learn concentration parameters directly from the input, while posterior networks and other second-order extensions further develop this idea for more reliable uncertainty estimation and for regression settings. The unique contribution of this category is therefore computational efficiency combined with an explicit probabilistic distinction between data and model uncertainty (He et al., 2026).

The main strength of evidential deep learning is efficiency. Because it uses one deterministic model, it avoids the repeated forward passes required by Bayesian sampling or ensemble methods. It also provides a principled way to separate data uncertainty from model uncertainty through the Dirichlet distribution. However, the category also has weaknesses. Training is more complex than standard neural network training, and predictive accuracy is not necessarily guaranteed to match a conventional model. It also depends on assumptions about the prior distribution and often requires out-of-distribution samples during training to learn effective uncertainty behaviour. In addition, it does not yet fully benefit from some recent advances in modern deep learning, such as large-scale pretraining and sampling-based architectures (He et al., 2026).

2.3.3.3. Conformal Prediction

Conformal prediction is a post-processing approach that constructs prediction sets or intervals with formal statistical coverage guarantees. Unlike many uncertainty quantification methods that estimate uncertainty without guaranteed correctness, conformal prediction provides a distribution-free guarantee that the true label will be included in the prediction set with a specified probability. It is model-agnostic, meaning that it can be applied on top of different machine learning or deep learning models without requiring major changes to the model architecture (He et al., 2026).

Mathematically, conformal prediction aims to construct a prediction set $\mathcal{C}(x)$ for a sample x such that

$$\mathbb{P}(y \in \mathcal{C}(x)) \geq 1 - \alpha.$$

Here, y is the true label, $\mathcal{C}(x)$ is the prediction set for input x , and $1 - \alpha$ is the desired coverage level. This means that, on average, the prediction set contains the correct label with probability at least $1 - \alpha$. Unlike traditional confidence intervals, conformal prediction does not require a specific parametric assumption about the data distribution or residual distribution, which makes it useful for complex deep learning models where such assumptions are often unrealistic (He et al., 2026).

The contribution of conformal prediction is that it adds formal coverage guarantees to uncertainty quantification. It can be applied directly to pre-trained deep learning models, used with credal Bayesian deep learning to produce prediction sets for total uncertainty, or combined with density-based methods that account for sample distances and out-of-distribution behaviour. Recent extensions also adapt conformal prediction beyond the standard independent and identically distributed setting, including graph data through permutation-invariance assumptions and time series through adaptive conformal methods (He et al., 2026).

The main strength of conformal prediction is that it provides a statistically grounded uncertainty output without imposing a rigid distributional form. It is also flexible because it can be used as a post-processing layer over many types of models. However, its main weakness is that standard conformal prediction relies on exchangeability, which is closely related to the assumption that calibration and test samples follow the same distribution. This assumption may not hold in settings with distribution shift, temporal dependence, spatial dependence, or structured data. Although newer variants relax this limitation for graphs and time series, these extensions usually require additional assumptions or methodological adjustments (He et al., 2026).

2.4. Uncertainty and Calibration Metrics

The uncertainty-quantification methods reviewed above can be operationalised through various uncertainty and calibration metrics. This section reviews the metrics used in this study.

2.4.1. Monte Carlo Dropout

Monte Carlo Dropout is an uncertainty estimation procedure in which dropout remains active at test time, rather than being disabled after training. Each stochastic forward pass applies a different dropout mask, so the same input is evaluated through a slightly different thinned version of the neural network. The resulting set of predictions can be interpreted as samples from an approximate Bayesian posterior predictive distribution. Conceptually, this means that MC Dropout estimates predictive uncertainty by repeatedly sampling plausible model configurations and averaging their outputs. This extends the original dropout idea, where randomly removing units during training acts as regularisation and can be viewed as sampling from many thinned networks. In the Bayesian interpretation, these stochastic passes approximate integration over model weights, allowing uncertainty to be estimated without changing the trained network architecture. (Srivastava et al., 2014; Gal & Ghahramani, 2016)

Mathematically, for a test input x^* , the Bayesian predictive distribution can be written as:

$$q(y^* | x^*) = \int p(y^* | x^*, \omega) q(\omega) d\omega,$$

where ω denotes the model parameters and $q(\omega)$ is the approximate posterior induced by dropout. Since this integral is generally intractable for deep neural networks, MC Dropout approximates it using T stochastic forward passes:

$$\mathbb{E}[y^*] \approx \frac{1}{T} \sum_{t=1}^T \hat{y}^*(x^*, W_1^t, \dots, W_L^t).$$

For regression, the predictive variance can be estimated as:

$$\text{Var}(y^*) \approx \tau^{-1} I_D + \frac{1}{T} \sum_{t=1}^T \hat{y}_t \hat{y}_t^T - \bar{y} \bar{y}^T,$$

where $\tau^{-1}I_D$ represents the observation noise term and the remaining terms estimate the variance across stochastic model predictions. In classification, the same idea is usually applied by averaging the softmax probability vectors from T stochastic forward passes (Gal & Ghahramani, 2016).

MC Dropout is mainly associated with epistemic uncertainty because the stochasticity comes from sampling different effective model parameter configurations. If the sampled dropout models produce similar predictions, the model is relatively certain; if they disagree, the model is uncertain about which parameter setting best explains the data. This form of uncertainty can decrease when more representative training data are available. However, MC Dropout predictions may also contain aleatoric uncertainty indirectly, because the final predictive distribution can still be uncertain for inherently ambiguous or noisy observations. Therefore, MC Dropout is best understood as a practical Bayesian approximation for estimating model-related uncertainty, while the resulting predictive distribution may still reflect both model and data uncertainty (Gal & Ghahramani, 2016; Kendall & Gal, 2017).

2.4.2. Predictive Entropy

Predictive entropy is a scalar measure of uncertainty computed from the predictive probability distribution of a classifier. Conceptually, it measures how spread out the predicted class probabilities are. If most probability mass is assigned to one class, entropy is low and the model prediction is confident. If probability mass is distributed across several classes, entropy is high and the prediction is uncertain. Under MC Dropout, predictive entropy is computed after averaging the probability vectors from multiple stochastic forward passes, so it summarises the uncertainty in the final posterior predictive distribution (Kendall & Gal, 2017; Smith & Gal, 2018).

Mathematically, for a classification problem with C classes and predictive probability vector $p = (p_1, \dots, p_C)$, predictive entropy is defined as:

$$H(p) = - \sum_{c=1}^C p_c \log p_c.$$

When MC Dropout is used, the predictive probability for class c can be approximated by:

$$p(y = c \mid x, X, Y) \approx \frac{1}{T} \sum_{t=1}^T \text{Softmax}(f_{W_t}(x))_c,$$

where W_t is the sampled masked weight configuration at stochastic pass t . Predictive entropy is then applied to this averaged probability vector (Kendall & Gal, 2017).

Predictive entropy is often interpreted as a measure of total predictive uncertainty because it can increase for different reasons. It can be high because the model has not seen enough relevant training data, which corresponds to epistemic uncertainty. It can also be high because the input is genuinely ambiguous or noisy, which corresponds to aleatoric uncertainty. For this reason, predictive entropy is useful as a general uncertainty score, but it cannot by itself separate epistemic from aleatoric uncertainty. In other words, a high entropy prediction indicates uncertainty, but not necessarily why the uncertainty occurs (Kendall & Gal, 2017; Smith & Gal, 2018).

2.4.3. Mutual Information

Mutual information is an uncertainty metric designed to isolate the part of predictive uncertainty that comes from uncertainty over the model parameters. Conceptually, it measures how much information

would be gained about the model parameters if the label of a given input were observed. If different sampled models make confident but different predictions, mutual information is high, indicating strong epistemic uncertainty. If all sampled models produce similarly uncertain predictions, mutual information can remain low even when predictive entropy is high, suggesting that the uncertainty is more likely due to inherent ambiguity in the data (Smith & Gal, 2018).

Mathematically, mutual information between the prediction y and model parameters ω , conditioned on input x and training data D , is defined as:

$$I(\omega, y \mid D, x) = H[p(y \mid x, D)] - \mathbb{E}_{p(\omega \mid D)}[H[p(y \mid x, \omega)]].$$

With MC Dropout, the predictive distribution is approximated using T stochastic samples:

$$p(y \mid D, x) \approx \frac{1}{T} \sum_{t=1}^T p(y \mid x, \omega_t),$$

so mutual information can be estimated as:

$$I(\omega, y \mid D, x) \approx H\left[\frac{1}{T} \sum_{t=1}^T p(y \mid x, \omega_t)\right] - \frac{1}{T} \sum_{t=1}^T H[p(y \mid x, \omega_t)]$$

The first term is predictive entropy, and the second term is the expected entropy of predictions under individual sampled model parameter settings (Smith & Gal, 2018).

Mutual information is closely related to epistemic uncertainty because it becomes large when different plausible model parameter samples imply different class predictions. This means the model would learn substantially from seeing the true label of that input. In contrast, mutual information is less sensitive to purely aleatoric uncertainty. If an input is inherently ambiguous, each sampled model may produce a similarly high-entropy prediction, so the predictive entropy and expected entropy are both high and their difference remains small. Therefore, mutual information is commonly interpreted as an epistemic uncertainty metric rather than a total uncertainty metric (Kendall & Gal, 2017; Smith & Gal, 2018).

2.4.4. Monte Carlo Disagreement

Monte Carlo disagreement is calculated as the proportion of stochastic predictions that differ from the modal class prediction (i.e., the variation ratio). It is closely related to the classical variation ratio. Conceptually, it quantifies lack of consensus across multiple stochastic forward passes. If nearly all MC Dropout samples predict the same class, disagreement is low. If many samples predict different classes, disagreement is high. This makes the metric especially intuitive for classification because it treats each stochastic pass as a vote and measures the fraction of votes that do not belong to the majority class (Gal et al., 2017).

Mathematically, the variation ratio is defined as:

$$VR(x) = 1 - \max_y p(y \mid x, D_{train}).$$

In an MC Dropout implementation, this can be estimated from class votes across T stochastic forward passes. If f_m is the number of MC samples that predict the modal class, then the sample-based disagreement fraction is:

$$\text{Disagreement}(x) = 1 - \frac{f_m}{T}.$$

This is the Monte Carlo estimate of the variation ratio, because f_m/T estimates the predictive probability of the most frequently predicted class. Therefore, disagreement fraction and variation ratio describe the same underlying idea: uncertainty as lack of agreement around the most probable class (Gal et al., 2017).

Disagreement fraction is mainly related to epistemic uncertainty because disagreement across MC Dropout samples arises from variation across sampled model parameter configurations. When different posterior weight samples lead to different class decisions, the model is uncertain about the appropriate decision boundary. This is characteristic of model uncertainty and can often be reduced with more informative training data. However, disagreement fraction is not a pure epistemic measure in all cases. Ambiguous or noisy observations can also cause unstable class votes, so aleatoric uncertainty may contribute to disagreement when the class boundary is intrinsically unclear. Still, compared with predictive entropy, disagreement fraction is more directly tied to model-sample disagreement and is therefore often used as a practical epistemic-style uncertainty indicator (Gal et al., 2017; Kendall & Gal, 2017; Smith & Gal, 2018).

2.4.5. Expected Calibration Error

Expected Calibration Error (ECE) is a summary measure of how well a classifier’s stated confidence agrees with its empirical correctness. A model is well calibrated when predictions made with a given confidence level are correct at approximately the same frequency. For example, among samples predicted with confidence close to **0.80**, around 80% should be classified correctly. ECE operationalises this idea by grouping predictions into confidence intervals and comparing the average confidence with the observed accuracy inside each interval. Therefore, ECE does not measure classification accuracy directly; instead, it measures the reliability of the model’s probability estimates. A low ECE indicates that the predicted probabilities are close to observed frequencies, while a high ECE indicates miscalibration, such as overconfidence or underconfidence (Bohdal et al., 2021; Wang, 2023).

Mathematically, suppose a dataset contains n samples and the predictions are divided into M confidence bins $B_1, \dots, B_M$. For each bin, $\text{acc}(B_m)$ is the fraction of correctly classified samples, and $\text{conf}(B_m)$ is the mean confidence of the model in that bin. ECE is then defined as

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|.$$

The bin accuracy and confidence are usually computed as

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(\hat{y}_i = y_i),$$

and

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i,$$

where $\hat{y}_i$ is the predicted class, y_i is the true class, and $\hat{p}_i$ is the confidence assigned to the predicted class. In a perfectly calibrated model, the difference between $\text{acc}(B_m)$ and $\text{conf}(B_m)$ would be zero for all bins, giving $\text{ECE} = 0$ (Bohdal et al., 2021; Wang, 2023).

ECE is related to uncertainty because calibrated probabilities make uncertainty estimates more trustworthy. However, ECE itself should not be interpreted as a direct measure of either aleatoric or epistemic uncertainty. It evaluates whether the model's reported probabilities match observed outcomes. Aleatoric uncertainty, such as label ambiguity or data noise, can contribute to calibration problems because inherently ambiguous samples may produce less reliable confidence estimates. Epistemic uncertainty can also affect calibration when the model is uncertain because it has limited knowledge, insufficient training data, or encounters inputs unlike those seen during training. Therefore, ECE is best understood as a diagnostic of probability reliability, not as a decomposition of uncertainty into aleatoric and epistemic components (Bohdal et al., 2021; Wang, 2023).

2.4.6. Brier Score

The Brier Score is a proper scoring rule that evaluates the accuracy of probabilistic predictions. Unlike ordinary accuracy, which only checks whether the predicted class is correct, the Brier Score considers the full probability assigned to the outcome. A prediction is penalised more when it assigns high probability to the wrong outcome, and penalised less when its probability estimate is close to the observed label. In this sense, the Brier Score combines information about correctness, confidence, and calibration. It is therefore useful for assessing whether a model's probability estimates are both accurate and reliable (Mehrtash et al., 2020; Yang et al., 2022).

For binary classification, let $Y_i \in \{0,1\}$ be the observed outcome for sample i , and let r_i be the model's predicted probability that $Y_i = 1$. The Brier Score is defined as

$$BS = \frac{1}{N} \sum_{i=1}^N (r_i - Y_i)^2.$$

For multi-class classification with K classes, it can be written as

$$Br = \frac{1}{N} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K (p(\hat{y}_i = y_k | x_i, \theta) - \mathbf{1}(y_i = y_k))^2,$$

where $p(\hat{y}_i = y_k | x_i, \theta)$ is the predicted probability of class k , and $\mathbf{1}(y_i = y_k)$ indicates whether class k is the true label. Lower Brier Scores indicate better probabilistic predictions (Mehrtash et al., 2020; Yang et al., 2022).

The Brier Score has an important relationship with aleatoric uncertainty because part of the score is affected by the intrinsic variability of the outcome. Yang et al. (2022) decompose the expected squared prediction error into a model-performance component and a data-variability component:

$$E[(r - Y)^2] = (r - p)^2 + E[(p - Y)^2],$$

where p is the true outcome probability. The first term, $(r - p)^2$, reflects how close the predicted probability is to the true probability, while the second term, $E[(p - Y)^2]$, is the variance of the outcome determined by the data-generating mechanism. This second component is closely related to aleatoric uncertainty because it represents variability inherent in the data rather than an error that can be removed by improving the model. The Brier Score is only indirectly related to epistemic uncertainty: if a model is poorly specified or trained with insufficient data, the model-performance term may increase, but the Brier Score does not explicitly separate parameter uncertainty from data uncertainty (Mehrtash et al., 2020; Yang et al., 2022).

2.4.7. Softmax Confidence

Softmax confidence refers to the confidence score derived from the output probabilities of a neural network classifier after applying the softmax function. In standard classification, the neural network produces logits, and the softmax function converts these logits into a probability distribution over classes. The highest softmax probability is often interpreted as the model's confidence in its predicted class. For example, if the largest softmax probability is 0.95, the model is expressing high confidence in that class. However, softmax confidence should be interpreted carefully because neural networks can assign high confidence to incorrect or out-of-distribution inputs (Subramanya et al., 2017; Pearce et al., 2021).

Given a logit vector $\mathbf{z}$ with K classes, the softmax probability for class i is

$$\sigma(\mathbf{z})_i = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)}.$$

The maximum softmax confidence is then

$$SC(\mathbf{x}) = \max_i \sigma(\mathbf{z})_i.$$

Some studies define uncertainty from softmax confidence by taking the negative maximum probability or by using entropy over the softmax distribution:

$$U_{max}(\mathbf{z}) = -\max_i \sigma(\mathbf{z})_i,$$

and

$$U_{entropy}(\mathbf{z}) = -\sum_{i=1}^K \sigma(\mathbf{z})_i \log \sigma(\mathbf{z})_i.$$

Thus, softmax confidence can be used either as a confidence metric through the maximum class probability or as a basis for uncertainty metrics such as predictive entropy (Subramanya et al., 2017; Pearce et al., 2021).

Softmax confidence can reflect aleatoric uncertainty when low confidence occurs because the input is genuinely ambiguous or lies in an overlapping region between classes. In that case, the model may assign probability mass to more than one class because the data itself does not clearly support a single label. However, softmax confidence is weaker as a measure of epistemic uncertainty. Pearce et al. (2021) emphasise that a softmax layer can represent uncertainty caused by overlapping classes, but it may fail to reduce confidence for inputs far from the training distribution. Therefore, softmax confidence is more naturally connected to aleatoric ambiguity within the learned class distribution, while its use as a proxy for epistemic uncertainty is limited and can be unreliable, especially for out-of-distribution samples (Subramanya et al., 2017; Pearce et al., 2021).

2.4.8. Cross-Entropy Loss

Cross-entropy loss is one of the most widely used training objectives for classification models. Conceptually, it measures how well the predicted probability distribution matches the true class label. When the model assigns high probability to the correct class, the loss is low. When the model assigns low probability to the correct class, the loss becomes high. This makes cross-entropy especially sensitive to confident wrong predictions. During optimisation, samples whose predictions disagree strongly with their

labels receive larger gradient contributions, meaning that cross-entropy places more emphasis on difficult or misclassified samples (Zhang & Sabuncu, 2018; Gordon-Rodriguez et al., 2020).

For a multi-class problem with C classes and one-hot labels y_{ij} , cross-entropy risk can be written as

$$R_L(f) = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^C y_{ij} \log f_j(x_i; \theta),$$

where $f_j(x_i; \theta)$ is the predicted probability for class j for sample x_i . Equivalently, for a categorical distribution with probability vector π , the cross-entropy loss for a one-hot label y is

$$\ell(\pi; y) = -\sum_{k=1}^K y_k \log \pi_k.$$

This is equivalent to the negative log-likelihood of the observed class under a categorical probability model. Therefore, minimising cross-entropy can be interpreted as maximum likelihood estimation for classification (Zhang & Sabuncu, 2018; Gordon-Rodriguez et al., 2020).

Cross-entropy loss is related to uncertainty because it increases when the model assigns low probability to the provided label. In practical error analysis, high cross-entropy can indicate difficult, ambiguous, mislabelled, or incorrectly predicted samples. Its relationship to aleatoric uncertainty is therefore indirect: if a sample is inherently ambiguous, noisy, or affected by label error, the model may struggle to assign high probability to the provided label, producing a larger loss. Its relationship to epistemic uncertainty is also indirect: if the model has not learned a region of the feature space well, or if the sample lies outside the model’s knowledge, the predicted probability for the correct class may be low. However, cross-entropy does not itself distinguish whether the high loss comes from irreducible data ambiguity, label noise, or model uncertainty. It is a training and evaluation loss, not a formal aleatoric–epistemic decomposition (Zhang & Sabuncu, 2018; Gordon-Rodriguez et al., 2020).

2.5. Uncertainty-Based Sampling

The uncertainty-quantification methods and metrics discussed in the previous sections provide a means of estimating how confident a model is in its predictions. Beyond their use for model evaluation and reliability assessment, these uncertainty estimates can also be used to guide the selection of training samples. This provides an important link between uncertainty quantification and active learning, in which samples are selectively acquired or annotated to improve the learning process (Ren et al., 2021).

Uncertainty-based sampling methods are founded on the principle that the most useful training examples are those that provide the greatest amount of information to the learning algorithm. Rather than selecting samples at random, these approaches prioritise observations for which the model exhibits uncertainty, as such samples are expected to contribute more effectively to improving model performance. The underlying assumption is that highly uncertain examples contain information that is not yet well captured by the current model, whereas confidently classified examples are likely to provide less additional learning value. Consequently, annotating uncertain samples can help reduce the space of plausible hypotheses considered by the model and accelerate improvements in predictive performance. For this reason, uncertainty-based strategies have become one of the most widely used approaches in active learning and sample-selection research (Ren et al., 2021).

Although uncertainty sampling has been widely adopted because uncertain samples often contain valuable information for learning, uncertainty does not always originate from informative sample complexity. High uncertainty may also result from annotation errors, ambiguous labels, or other forms of data corruption. Consequently, uncertainty can indicate either genuinely informative observations or problematic samples affected by label noise (Ferchichi et al., 2017; Ren et al., 2021). Distinguishing between these two possibilities represents an important challenge for data-centric machine learning and motivates the discussion of data quality and label noise in the following sections.

2.6. Model-Centric and Data-Centric AI

Artificial intelligence (AI) systems rely on two fundamental components: algorithms and data. Historically, machine-learning research has been dominated by a model-centric paradigm, in which performance improvements are primarily pursued through the development of new algorithms, model architectures, loss functions, and hyperparameter configurations while treating the training data as largely fixed. Within this paradigm, the dataset often serves as a static benchmark against which models are evaluated, and research efforts concentrate on optimising predictive accuracy and computational efficiency. Consequently, advances in AI have traditionally been associated with increasingly sophisticated models rather than with systematic improvements to the underlying data (Roscher et al., 2024; Malerba & Pasquadibisceglie, 2024; Zha et al., 2025; Kumar et al., 2024).

Model-centric AI has achieved considerable success across many application domains; however, several limitations have become increasingly apparent. The assumption that data are clean, static, and representative of the target problem often does not hold in real-world settings. As a result, models developed under a model-centric paradigm are frequently specialised to particular datasets and tasks, making them difficult to transfer across domains or adapt to changing environments. Furthermore, reliance on fixed datasets may overlook important aspects of the problem space, including data quality, representativeness, coverage, and temporal evolution. These limitations have led researchers to recognise that improvements in model architecture alone may not always translate into improved real-world performance (Malerba & Pasquadibisceglie, 2024; Zha et al., 2025).

In response to these challenges, the data-centric AI paradigm has emerged. Rather than focusing primarily on model optimisation, data-centric AI emphasises the systematic engineering, development, maintenance, and evaluation of data throughout the machine-learning lifecycle. The central premise is that improvements in data quality can often yield greater benefits than further modifications to model architecture. Data-centric approaches therefore focus on activities such as data acquisition, data curation, data creation, annotation, validation, maintenance, and evaluation, with the objective of producing high-quality datasets that better support learning and generalisation. Under this perspective, data are no longer viewed as static inputs but as dynamic assets that require continuous improvement and monitoring (Roscher et al., 2024; Malerba & Pasquadibisceglie, 2024; Zha et al., 2025; Kumar et al., 2024).

A key distinction between the two paradigms lies in their primary optimisation target. Whereas model-centric approaches seek to improve performance by refining algorithms and model parameters, data-centric approaches seek to improve performance by enhancing the quality, reliability, diversity, and representativeness of the training and evaluation data. Data-centric AI assumes that better data lead to better AI systems and therefore prioritises the systematic identification and correction of data-related issues, including annotation errors, inconsistencies, missing information, class imbalance, and distributional shifts. In addition, data-centric AI extends beyond training data development to include inference-data design and data maintenance, thereby ensuring that data remain reliable throughout deployment and operation (Malerba & Pasquadibisceglie, 2024; Zha et al., 2025).

An important motivation for data-centric AI is the growing recognition that data quality is as important as data quantity. Several studies have demonstrated that not all samples contribute equally to model learning and generalisation. Simply increasing dataset size does not necessarily improve performance and may even introduce diminishing returns when additional samples are redundant, noisy, or incorrectly labelled. Consequently, effective AI systems require attention not only to the amount of available data but also to their quality, relevance, and suitability for the target task. Data-centric AI therefore advocates systematic processes for identifying high-value data, improving data quality, and maintaining reliable datasets over time (Roscher et al., 2024; Malerba & Pasquadibisceglie, 2024).

Although model-centric and data-centric AI are often presented as contrasting paradigms, several authors argue that they should be viewed as complementary rather than mutually exclusive. AI performance ultimately depends on both the quality of the learning algorithm and the quality of the data used to train it. Consequently, the most effective AI systems are likely to emerge from approaches that combine advances in model development with systematic improvements in data quality. Such hybrid perspectives recognise that robust and generalisable AI systems require continuous refinement of both models and data throughout the machine-learning lifecycle (Kumar et al., 2024; Roscher et al., 2024).

From a mathematical perspective, the distinction between model-centric and data-centric AI is less pronounced because both paradigms generally rely on the same underlying learning objective. Given a dataset $D = \{(x_i, y_i)\}_{i=1}^N$, the goal is to learn a function $f(x; \theta)$ that maps input features x to outputs y by optimising model parameters θ . This is typically formulated as a loss-minimisation problem,

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(y_i, f(x_i; \theta)).$$

where $\mathcal{L}$ is a loss function measuring the discrepancy between predictions and ground-truth labels. The parameters are commonly updated using gradient-based optimisation methods. Data-centric AI does not fundamentally alter this mathematical objective; instead, it seeks to improve the quality of the dataset D on which optimisation is performed. Consequently, the distinction between the two paradigms lies primarily in where effort is invested: model-centric AI concentrates on improving the function f and its parameters θ , whereas data-centric AI concentrates on improving the data used to estimate them (Kumar et al., 2024).

Overall, the emergence of data-centric AI reflects a broader shift in machine-learning research from an exclusive focus on model development towards a more holistic consideration of the entire AI lifecycle. As datasets continue to grow in scale and complexity, and as AI systems are increasingly deployed in dynamic real-world environments, systematic data engineering is becoming a critical component of building accurate, robust, and generalisable machine-learning systems (Roscher et al., 2024; Malerba & Pasquadibisceglie, 2024; Zha et al., 2025; Kumar et al., 2024).

2.7. Noise in Machine-Learning Data

A major source of uncertainty in machine learning is noise in the data used for model training and evaluation, which can affect both the observed features and the associated labels. Real-world datasets frequently contain imperfections introduced during measurement, collection, or processing. In the machine learning literature, these imperfections are commonly referred to as noise and are understood as factors that obscure the relationship between the attributes of an instance and its true class (Ishii & Ljunggren, 2021; Fréney & Verleysen, 2014). Noise in machine learning datasets can affect either the input features or the target labels. Feature noise describes errors or uncertainty in the observed attributes of a sample, such as those introduced during data collection (e.g., cloud contamination or haze affecting

Landsat observations over the Amazon), atmospheric correction (e.g., residual atmospheric effects altering surface reflectance values), image registration (e.g., slight spatial misalignment between multi-temporal satellite images), feature computation (e.g., inaccuracies in vegetation indices such as NDVI caused by noisy input bands), or data encoding (e.g., incorrect pixel values introduced during raster conversion or resampling). Label noise, on the other hand, refers to inaccuracies in the class labels assigned to samples, resulting in a discrepancy between the observed label and the true class. This may appear as incorrect labels (e.g., an intact forest area labelled as deforested or a deforested area labelled as forest), missing labels (e.g., a recent clearing omitted from the reference dataset), or a combination of both (e.g., a deforestation polygon incorrectly mapped while nearby deforested areas remain unlabelled), on the classification task (Frénay & Verleysen, 2014; Pelletier et al., 2017; Kondylatos et al., 2025).

Several studies have reported that label noise typically has a greater negative effect on machine learning performance than feature noise. This difference is often attributed to the fact that each sample contains many input features but only a single target label. While learning algorithms can reduce the influence of noisy features by assigning them lower importance, the label directly guides the learning process and therefore has a much stronger impact on model training. As a result, errors in class labels are generally more damaging than errors in feature values, highlighting the importance of identifying and addressing label noise in training datasets (Frénay & Verleysen, 2014; Pelletier et al., 2017; Karimi et al., 2020).

2.8. Types of Label Noise

Having defined label noise, it is useful to distinguish between the principal forms that label corruption can take. The literature generally distinguishes three main types of label noise: uniform, class-dependent, and feature-dependent noise. Uniform label noise, also referred to as class-independent noise or noisy completely at random (NCAR), occurs when labels are corrupted without regard to the true class or the characteristics of the sample. Under this assumption, every instance has the same probability of being assigned an incorrect label, and any incorrect class is selected randomly from the remaining classes. For example, in a deforestation-mapping dataset, uniform noise could occur if a fixed proportion of labels were randomly corrupted during dataset preparation, causing forest and deforested pixels to be flipped irrespective of their location, spectral characteristics, or true class. Because the corruption process is independent of both the class label and the input features, uniform noise represents the simplest form of label noise and is frequently used in experimental studies through random label flipping (Frénay & Verleysen, 2014; Algan & Ulusoy, 2020; Karimi et al., 2020; Pelletier et al., 2017).

Class-dependent label noise, also known as noisy at random (NAR), arises when the likelihood of mislabelling depends on the true class of an instance. In this case, some classes are more susceptible to annotation errors than others, resulting in asymmetric label corruption patterns. For example, in deforestation mapping, the deforested class may be more likely to be mislabelled than the intact-forest class because recently cleared areas often share characteristics with bare soil or pasture. Such noise is commonly represented through a label-transition matrix that specifies the probability of one class being mistaken for another. Misclassifications often occur between classes that share similar characteristics, although systematic corruption may also consistently map instances from one class to a particular alternative class (Frénay & Verleysen, 2014; Algan & Ulusoy, 2020; Karimi et al., 2020; Pelletier et al., 2017).

Feature-dependent label noise, often referred to as noisy not at random (NNAR), represents the most complex and realistic form of label corruption. In this setting, the probability of mislabelling depends not only on the class but also on the characteristics of individual samples. For example, in Amazon deforestation mapping, annotation errors may occur more frequently along forest–non-forest boundaries, in areas affected by cloud shadows, or in regions containing small clearings embedded within intact forest,

where the visual distinction between classes is less clear. As a result, errors are more likely to occur for observations located near class boundaries, in regions with low sample density, or when instances closely resemble samples from other classes. This form of noise reflects situations in which certain observations are inherently more difficult to classify and therefore more prone to annotation mistakes (Frénay & Verleysen, 2014; Algan & Ulusoy, 2020; Karimi et al., 2020).

2.9. Sources of Label Noise

Label noise can originate from a wide range of human, technical, environmental, and data-management factors. Unless otherwise specified, the sources discussed below are assumed to be potentially present in the data used in this study, although their relative contributions are not quantified individually. The most commonly reported sources are discussed below. Human and expert annotation errors arise when labels are assigned incorrectly because of mistakes made during the annotation process. Such errors may result from human error (e.g., an interpreter accidentally labels an intact forest patch as deforested), lack of expertise (e.g., a non-specialist confuses secondary forest with primary forest), fatigue (e.g., errors introduced after reviewing large numbers of image tiles), subjective judgement (e.g., differing opinions on whether sparse tree cover constitutes forest), disagreement between annotators (e.g., one annotator labels a clearing as pasture while another labels it as deforestation), field-survey limitations (e.g., inaccessible Amazon locations preventing field verification), non-expert annotation (e.g., labels produced by students with limited land-cover knowledge), crowdsourcing (e.g., volunteers assigning inconsistent land-cover labels), and other unreliable labelling procedures (e.g., rapid visual interpretation without quality control). These factors can introduce inconsistencies into reference datasets and reduce the reliability of the labels used for model training and evaluation (Karimi et al., 2020; Frénay & Verleysen, 2014; Kondylatos et al., 2025; Pelletier et al., 2017). In this study, lack of expertise, fatigue, field-survey limitations, and other unreliable labelling procedures are present but likely secondary or minor sources of uncertainty, whereas non-expert annotation and crowdsourcing are scientifically valid sources of label noise but likely negligible for explaining the observed label-noise and uncertainty patterns.

Automated label-generation errors occur when labels are produced by algorithmic rather than manual procedures. These errors may arise from automated classification systems (e.g., a rule-based algorithm incorrectly mapping flooded forest as deforestation), computer-generated labels (e.g., labels derived automatically from a previous land-cover product), semi-supervised methods (e.g., pseudo-labels propagated from a small labelled dataset), or unsupervised methods (e.g., clustering algorithms assigning incorrect land-cover classes). Because the generated labels are subsequently used as reference data, any errors produced by these methods may propagate through the training process and affect downstream models (Karimi et al., 2020; Bernhardt et al., 2022; Frénay & Verleysen, 2014; Kondylatos et al., 2025). In this study, computer-generated labels, automated classification systems, semi-supervised methods, and unsupervised methods are scientifically valid sources of label noise but likely negligible for explaining the observed label-noise and uncertainty patterns.

Label noise may also arise when the information available to the annotator is insufficient, ambiguous, or of poor quality. This includes insufficient information (e.g., limited spatial context around a forest clearing), poor or variable data quality (e.g., low-contrast or partially obscured Landsat imagery), ambiguous input or output spaces (e.g., uncertainty regarding whether a pixel belongs to forest or shrubland), land-cover complexity (e.g., heterogeneous forest–pasture mosaics), low inter-class variability (e.g., young secondary forest appearing spectrally similar to mature forest), high intra-class variability (e.g., large spectral differences among deforested areas due to different land uses), unclear class definitions (e.g., inconsistent definitions of forest degradation versus deforestation), and difficulties in disentangling target signals from confounding patterns (e.g., distinguishing recent clear-cutting from seasonal vegetation

changes). Such conditions make it difficult to assign labels consistently and increase the likelihood of annotation errors (Bernhardt et al., 2022; Frénay & Verleysen, 2014; Kondylatos et al., 2025; Pelletier et al., 2017). In this study, unclear class definitions are present but likely a secondary or minor source of uncertainty.

Atmospheric and acquisition-related confounding occurs when image characteristics unrelated to the target phenomenon influence the interpretation of observations. Examples include atmospheric effects (e.g., haze over Amazon imagery), coherence variations (e.g., SAR coherence changes caused by rainfall rather than land-cover change), sensor artefacts (e.g., striping or sensor noise in satellite imagery), and acquisition conditions that mimic the target phenomenon (e.g., cloud shadows resembling deforested areas). These factors may obscure or alter the appearance of land-cover classes, thereby increasing uncertainty during annotation and potentially introducing label errors (Kondylatos et al., 2025). In this study, sensor artefacts are present but likely a secondary or minor source of uncertainty, whereas coherence variations are scientifically valid but likely negligible because the analysis is based on optical Landsat imagery rather than SAR data.

Spatial misalignment refers to inconsistencies in the geographic positioning of imagery and reference data. Such errors may arise from spatial mismatch between EO imagery (e.g., Landsat imagery shifted relative to reference polygons), reference labels (e.g., PRODES boundaries not perfectly aligned with image pixels), masks (e.g., cloud masks offset from actual cloud locations), field annotations (e.g., GPS positional errors in field observations), or in situ observations (e.g., ground observations recorded at incorrect coordinates). Even small positional discrepancies can cause pixels to be assigned incorrect labels, particularly along land-cover boundaries and narrow deforestation fronts (Kondylatos et al., 2025; Pelletier et al., 2017). In this study, masks, field annotations, and in situ observations are present but likely secondary or minor sources of uncertainty.

Temporal mismatch occurs when there is a difference between the acquisition date of the imagery and the date associated with the reference labels. For example, a Landsat image may be acquired before a forest-clearing event occurs (e.g., imagery showing intact forest immediately before clearing), whereas the corresponding reference label may be generated after the event has been detected (e.g., a PRODES label showing the subsequently detected deforestation). Because land cover can change rapidly, especially in active deforestation regions, temporal discrepancies can create apparent label errors even when both the imagery and the reference data are individually correct for their respective dates (Pelletier et al., 2017).

Reference-data and database inconsistencies arise when labels are derived from imperfect reference sources or when multiple datasets are combined. Examples include the use of proxy labels (e.g., using PRODES as a surrogate for ground truth), reference products used instead of direct observations (e.g., labels derived from MapBiomas instead of field surveys), differences in labelling strategies (e.g., different minimum mapping units used by two datasets), and inconsistencies introduced when combining multiple databases (e.g., merging PRODES and MapBiomas labels with different class definitions). Such inconsistencies may result in conflicting labels and introduce systematic noise into training datasets (Kondylatos et al., 2025; Pelletier et al., 2017). In this study, reference products used as substitutes for ground truth and differences in labelling strategies are present but likely secondary or minor sources of uncertainty, whereas inconsistencies introduced when combining multiple databases are scientifically valid but likely negligible as only PRODES labels are used.

Data encoding and communication errors originate from mistakes occurring during data storage, transfer, or management. These include data-entry mistakes (e.g., assigning class code 1 instead of class code 0), encoding errors (e.g., incorrect class labels introduced during raster conversion), communication problems (e.g., misunderstanding annotation guidelines among team members), and accidental feedback errors (e.g.,

an annotator unintentionally confirming an incorrect label). Although these errors are often unrelated to the imagery itself, they can nevertheless introduce incorrect labels and reduce dataset quality (Frénay & Verleysen, 2014). In this study, encoding errors are present but likely a secondary or minor source of uncertainty, whereas data-entry mistakes, communication problems, and accidental feedback errors are scientifically valid sources of label noise but likely negligible for explaining the observed label-noise and uncertainty patterns.

Some label uncertainty arises from the inherent stochasticity of the application domain itself. Similar observed conditions may not always lead to the same outcome (e.g., two forest areas with similar spectral signatures where one is subsequently cleared and the other remains intact), making certain labels inherently uncertain. In such cases, uncertainty is not necessarily caused by annotation mistakes or poor data quality but instead reflects genuine variability in the underlying processes being modelled. Consequently, some degree of label uncertainty may be unavoidable even when high-quality imagery and carefully curated reference data are available (Kondylatos et al., 2025). In this study, inherent stochasticity is present but likely a secondary or minor source of uncertainty.

2.10. Effects of Label Noise on Deep Learning

Because label noise alters the supervision provided during training, it can substantially affect the behaviour and performance of deep-learning models. The performance of supervised machine learning systems depends heavily on the availability of accurately labelled training data. However, large modern datasets frequently contain annotation errors due to human mistakes (e.g., an interpreter incorrectly labels an intact forest area as deforested), automated labelling procedures (e.g., labels inherited from an automated land-cover classification product), ambiguities in the data (e.g., cloud shadows, heterogeneous forest–pasture mosaics, or small clearings embedded within intact forest), or limitations in expert knowledge (e.g., difficulty distinguishing secondary forest from primary forest or identifying subtle deforestation events). As a result, noisy labels have become a common challenge in deep learning applications, particularly because deep neural networks often require substantially larger training datasets than traditional machine learning methods (Bernhardt et al., 2022; Karimi et al., 2020).

Numerous studies have shown that label noise can significantly reduce model generalisation performance. Deep neural networks possess sufficient representational capacity to fit highly complex patterns, including incorrect or randomly assigned labels. While neural networks typically learn meaningful patterns from correctly labelled examples during the early stages of training, they may eventually begin to memorise corrupted labels as training progresses. This memorisation process can lead to overfitting and reduced performance on unseen data, even when commonly used regularisation techniques such as data augmentation, dropout, weight decay, and batch normalisation are applied (Song et al., 2023; Karimi et al., 2020; Thulasidasan et al., 2019).

The severity of this effect depends on factors such as the proportion of noisy labels, the characteristics of the dataset, the model architecture, and the training procedure. Although some studies have demonstrated that deep neural networks can tolerate surprisingly high levels of label noise under certain conditions (Rolnick et al., 2017), others have reported substantial performance degradation when noisy labels are present (Zhang et al., 2016). In many cases, models trained on smaller but cleaner datasets outperform models trained on larger datasets containing substantial label noise. Consequently, label cleaning, noise detection, and robust training strategies have become important research directions for improving the reliability and generalisation capability of deep learning systems (Karimi et al., 2020; Song et al., 2023; Thulasidasan et al., 2019; Bernhardt et al., 2022).

2.11. Label Noise Mitigation

Given the detrimental effects of noisy labels on model performance, numerous methods have been developed to improve robustness during training. According to Song et al. (2023), methods for mitigating label noise can be grouped into five categories: robust architectures, robust regularisation techniques, robust loss functions, loss-adjustment methods, and sample-selection approaches. Although these methods differ in implementation, they share the common objective of improving the robustness of supervised deep learning models when trained on noisy datasets.

2.11.1. Robust Architecture

One approach to handling noisy labels is to modify the neural network architecture so that label corruption is explicitly modelled during training. Robust architecture methods aim to improve learning under label noise by modifying the neural network architecture itself. The central idea is that noisy labels are not generated randomly but arise through an underlying label corruption process that transforms clean labels into noisy labels. Rather than relying solely on loss functions or sample-selection strategies, robust architecture approaches explicitly model this corruption process within the network. Most methods achieve this by estimating a label transition matrix that describes the probability of one class label being corrupted into another, while more advanced approaches design specialised architectures to capture complex forms of label noise. By incorporating noise modelling directly into the network structure, these methods seek to improve generalisation despite the presence of incorrect annotations. These methods are divided into two categories: noise adaptation layer and dedicated architectures (Song et al., 2023).

2.11.1.1. Noise Adaptation Layer

The noise adaptation layer represents the most common architectural approach for handling label noise. Conceptually, it assumes that noisy labels are generated from clean labels through a label transition process. The objective is therefore to learn the transition probabilities between true and observed labels and use this information to correct predictions during training.

Mathematically, the noisy class posterior probability can be expressed as

$$p(\tilde{y} = j|x) = \sum_{i=1}^c p(\tilde{y} = j, y = i|x) = \sum_{i=1}^c T_{ij} p(y = i|x),$$

where $\tilde{y}$ denotes the observed noisy label, y denotes the latent clean label, x is the input sample, T_{ij} is the probability that class i is corrupted into class j , and c is the total number of classes.

To model this process within a neural network, an additional layer is placed above the softmax output. The base network predicts the clean-label probabilities, while the adaptation layer transforms these predictions according to the estimated transition matrix:

$$p(\tilde{y} = j|x; \theta, W) = \sum_{i=1}^c p(\tilde{y} = j|y = i; W) p(y = i|x; \theta),$$

where θ represents the parameters of the base neural network and W parameterises the transition matrix. During inference, the adaptation layer is removed and predictions are produced directly from the base model (Song et al., 2023).

The main contribution of noise adaptation methods is their ability to explicitly model label corruption within the learning process. Various methods have been proposed to estimate the transition matrix more effectively. Webly learning (Chen & Gupta, 2015) first trains on relatively reliable examples and

subsequently uses a confusion matrix to initialise the adaptation layer. Noise model (Sukhbaatar et al., 2014) approaches initialise the transition matrix as an identity matrix and gradually allow it to evolve during training through regularisation. Dropout Noise Model (Jindal et al., 2016) uses dropout regularisation to encourage diffusion of the transition matrix, while the s-model (Goldberger & Ben-Reuven, 2017) learns class-dependent label corruption and the c-model extends this approach to instance-dependent noise. Other methods, such as the Noisy Labels Neural-Network (NLNN) algorithm (Bekker & Goldberger, 2016), employ an expectation-maximisation framework that alternates between estimating the transition matrix and updating network parameters (Song et al., 2023).

The primary strength of noise adaptation layers is their direct modelling of label corruption, allowing the network to learn correction mechanisms during training. These methods are relatively straightforward to integrate into existing architectures and can improve robustness under both symmetric and asymmetric label noise. However, they have several limitations. Since all training examples are treated equally, they do not explicitly identify individual mislabelled samples. Consequently, transition matrix estimation can become unreliable when the noise rate is high or when only noisy labels are available. Expectation-maximisation-based approaches also suffer from high computational costs and may converge to local optima rather than globally optimal solutions (Song et al., 2023).

2.11.1.2. Dedicated Architectures

Dedicated architectures extend the idea of explicit noise modelling beyond simple transition matrices. These methods are designed to address more realistic and complex forms of label noise, including instance-dependent noise, where the probability of mislabelling varies across individual samples. Rather than merely adding an adaptation layer, dedicated architectures incorporate specialised modules or network structures to improve estimation of the noise-generation process (Song et al., 2023).

Conceptually, dedicated architectures seek to enhance the reliability of label-noise modelling by representing additional information about the data, annotation process, or label quality. Their goal is not only to estimate label transitions but also to capture more complex relationships between noisy and clean labels that cannot be adequately represented by a single transition matrix (Song et al., 2023).

The main contribution of dedicated architectures lies in their ability to model more sophisticated and realistic noise patterns. Probabilistic noise modelling (Xiao et al., 2015) employs separate networks to estimate noise types and transition probabilities, combining information from a small clean dataset with a larger noisy dataset. Masking methods (Han et al., 2018) incorporate human knowledge regarding invalid label transitions and use this information to constrain the noise modelling process, often through adversarial learning frameworks. Contrastive-additive noise networks (Yao et al., 2019) introduce quality embeddings that represent the trustworthiness of individual labels, enabling correction of inaccurately estimated transition probabilities. Other approaches, such as Robust Generative classifier (RoG) (Lee et al., 2019), build robust generative classifiers on top of discriminative neural networks that have already been trained on noisy data (Song et al., 2023).

The main strength of dedicated architectures is their improved robustness to diverse and realistic forms of label noise. By introducing specialised modelling components, these methods can better capture complex corruption patterns that simpler adaptation layers may fail to represent. This often results in more accurate estimation of label uncertainty and improved performance under challenging noise conditions. However, these advantages come at the cost of increased architectural complexity. Dedicated architectures are generally less flexible and more difficult to transfer across different neural network designs. As a result, their applicability may be restricted to specific architectures or problem settings, limiting their general use compared with simpler noise adaptation approaches (Song et al., 2023).

2.11.2. Robust Regularization

Another strategy is to improve robustness through regularization techniques that reduce the tendency of neural networks to memorise noisy labels. Robust regularization methods aim to improve learning under label noise by reducing overfitting and encouraging better generalization during training. The underlying assumption is that deep neural networks tend to memorize noisy labels as training progresses, leading to poor performance on unseen data. Regularization techniques counteract this behaviour by constraining the learning process or introducing controlled stochasticity, thereby making the model less sensitive to incorrect labels. Traditional regularization methods, such as data augmentation, weight decay, dropout, and batch normalization, have long been used to improve generalization. Although these techniques can provide robustness under moderate levels of label noise, their effectiveness decreases as noise levels increase. Consequently, more advanced regularization approaches have been developed specifically to address noisy-label learning. These methods can be divided into two categories: explicit regularization and implicit regularization (Song et al., 2023).

2.11.2.1. Explicit Regularization

Explicit regularization methods improve robustness by directly modifying the training objective, parameter updates, or model constraints. The goal is to guide the optimization process towards solutions that generalize better despite the presence of noisy labels. These approaches typically introduce additional loss terms, regularization constraints, or specialized parameter-update mechanisms that reduce the influence of corrupted annotations during learning (Song et al., 2023).

The principal contribution of explicit regularization methods is their ability to control overfitting to noisy labels through targeted modifications to the learning process. Several approaches have been proposed. Bilevel learning (Jenni & Favaro, 2018) introduces a nested optimization framework in which model updates are guided not only by the training data but also by performance on a clean validation set. Annotator confusion modelling (Tanno et al., 2019) assumes multiple annotators and employs a regularized expectation–maximisation framework to estimate the true confusion matrix of label providers. Pretraining-based approaches (Hendrycks et al., 2019) exploit representations learned from large datasets, thereby reducing the likelihood that noisy labels will drive parameter updates in undesirable directions. PHuber (Menon et al., 2020) introduces a composite loss function with gradient clipping to limit the influence of noisy examples, while robust early learning (Xia et al., 2021) distinguishes between parameters associated with clean and noisy labels and selectively penalizes the latter. Other methods, such as Open-set samples with Dynamic Noisy Labels (ODNL) (Wei et al., 2021), utilise auxiliary open-set data with randomly assigned labels to discourage overfitting to corrupted annotations (Song et al., 2023).

The main strength of explicit regularization methods is their ability to significantly improve robustness when appropriately configured. By directly constraining optimisation, these approaches can effectively reduce memorization of noisy labels and improve generalization. However, they often introduce additional hyperparameters that require careful tuning and may increase model complexity. In some cases, deeper or more sophisticated architectures are needed to compensate for the reduced flexibility imposed by the regularization mechanism. Consequently, performance gains may depend strongly on selecting suitable parameter settings (Song et al., 2023).

2.11.2.2. Implicit Regularization

Implicit regularization methods improve robustness without directly modifying the loss function or optimisation objective. Instead, they introduce stochasticity or alter the training data distribution in ways that naturally encourage more generalizable representations. These methods rely on the observation that certain training procedures inherently act as regularisers, reducing the tendency of neural networks to fit noisy labels (Song et al., 2023).

The primary contribution of implicit regularization methods is the promotion of smoother decision boundaries and more robust feature representations through modified training dynamics. One widely used approach is adversarial training (Goodfellow et al., 2014), which encourages the model to correctly classify both original inputs and deliberately perturbed examples. This improves resistance to label noise by making predictions less sensitive to small changes in the input space. Another common method is label smoothing (Zhang et al., 2017), which replaces hard one-hot labels with softened target distributions:

$$\bar{y}^{(i)} = (1 - \alpha) [\tilde{y} = i] + \frac{\alpha}{c},$$

where $\bar{y}^{(i)}$ is the smoothed target for class i , $\tilde{y}$ is the noisy label, α is the smoothing parameter, and c is the total number of classes. Rather than assigning full probability to a single class, label smoothing distributes a small amount of probability mass across all classes, reducing overconfidence and limiting the impact of incorrect labels (Song et al., 2023).

Another influential implicit regularization technique is mixup (Zhang et al., 2017), which generates synthetic training examples by linearly interpolating pairs of samples and their corresponding labels:

$$\begin{aligned} x_{\text{mix}} &= \lambda x_i + (1 - \lambda) x_j, \\ y_{\text{mix}} &= \lambda \tilde{y}_i + (1 - \lambda) \tilde{y}_j, \end{aligned}$$

where $(x_i, \tilde{y}_i)$ and $(x_j, \tilde{y}_j)$ are two training examples, and $\lambda \in [0, 1]$ controls the interpolation strength. By expanding the training distribution through these virtual examples, mixup encourages smoother decision boundaries and reduces the tendency to memorize noisy annotations (Song et al., 2023).

The main strength of implicit regularization methods is that they improve generalization without reducing the representational capacity of the model. Because they operate primarily on the training data or training process, they generally introduce fewer model-specific hyperparameters than explicit regularization methods. In addition, they can often be integrated easily into existing training pipelines. However, these methods also have limitations. By extending the feature space or label space, they increase the complexity of the optimisation problem and may slow convergence during training. Furthermore, although they reduce overfitting, they do not explicitly identify or correct mislabelled examples (Song et al., 2023).

2.11.3. Robust Loss Function

Robust loss-function methods address label noise by modifying the optimisation objective to reduce the influence of incorrect labels during training. The central idea is that a suitably designed loss function can prevent a deep neural network from overfitting noisy annotations and guide the model towards solutions that generalise well to unseen clean data. This category is grounded in statistical learning theory and has received considerable attention because the loss function directly determines how prediction errors influence parameter updates during optimisation (Song et al., 2023).

From a theoretical perspective, robust loss functions are motivated by the objective of learning a classifier that approaches the Bayes-optimal classifier despite being trained on noisy labels. Let

$$\hat{f} = \operatorname{argmin}_{f \in \mathcal{F}} \hat{R}_{\mathcal{L}, \tilde{\mathcal{D}}}(f),$$

Where,

$$\hat{R}_{\mathcal{L}, \tilde{\mathcal{D}}}(f) = \mathbb{E}_{\tilde{\mathcal{D}}}[\mathcal{L}(f(x; \theta), \tilde{y})].$$

Here, $\hat{f}$ is the classifier learned from noisy data, $\mathcal{F}$ is the hypothesis class, $\mathcal{L}$ is the loss function, and $\tilde{D}$ denotes the noisy training dataset. Under certain assumptions, including Lipschitz continuity and classification calibration, the difference between the risk of the learned classifier and the Bayes-optimal risk can be bounded by approximation and estimation errors:

$$R_D(\hat{f}) - R^* \leq \zeta_{\mathcal{L}} \left(\min_{f \in \mathcal{F}} R_{\mathcal{L}, D}(f) - \min_f R_{\mathcal{L}, D}(f) \right) + 4L_p RC(\mathcal{F}) + 2 \sqrt{\frac{\log(1/\delta)}{2|D|}}.$$

Here, $R_D(\hat{f})$ is the risk of the learned classifier on clean data, R^* is the Bayes-optimal risk, L_p is the Lipschitz constant of the loss function, $RC(\mathcal{F})$ is the Rademacher complexity of the hypothesis class, δ is a confidence parameter, and $|D|$ is the size of the clean dataset. This result provides the theoretical basis for designing loss functions that remain effective even when the training labels are corrupted (Song et al., 2023).

A key theoretical property studied in this literature is noise tolerance. For multiclass classification under symmetric label noise, a loss function is considered noise-tolerant when the noise rate satisfies

$$\tau < \frac{c-1}{c},$$

and the loss function obeys

$$\sum_{j=1}^c \mathcal{L}(f(x; \theta), y = j) = C,$$

where τ is the noise rate, c is the number of classes, and C is a constant. Under these conditions, the classifier trained on noisy data can achieve the same misclassification probability as one trained on clean data. Extensions of this theory have also been developed for multilabel classification and certain forms of asymmetric label noise (Song et al., 2023).

The principal contribution of this category is the design of loss functions that are inherently more resistant to corrupted labels than conventional objectives. Early theoretical work established sufficient conditions under which risk minimisation remains noise-tolerant. Building on this foundation, researchers proposed alternative losses that reduce the influence of noisy samples while preserving classification performance. Mean Absolute Error (MAE) (Ghosh et al., 2017) was one of the first robust alternatives shown to satisfy theoretical noise-tolerance conditions. However, although MAE exhibits robustness to label noise, its performance often deteriorates on complex datasets. To overcome this limitation, the Generalized Cross Entropy (GCE) loss (Zhang & Sabuncu, 2018) was introduced to combine the robustness of MAE with the optimisation advantages of Categorical Cross Entropy (CCE). Subsequent extensions, including temperature-based formulations derived from Tsallis divergence and the Bitempered Loss (Amid et al., 2019) based on Bregman divergence, further improved robustness while maintaining desirable optimisation properties (Song et al., 2023).

Additional robust-loss designs have focused on exploiting complementary theoretical properties. Symmetric Cross Entropy (SCE) (Wang et al., 2019) combines the standard cross-entropy objective with a reverse cross-entropy component to improve tolerance to incorrect labels. Curriculum Loss (CL) (Lyu & Tsang, 2019) introduces a surrogate objective that provides a tight approximation to the ideal but non-differentiable 0–1 loss while remaining suitable for multiclass classification. Active Passive Loss (APL) (Ma et al., 2020) combines two complementary objectives: an active component that increases confidence

in the assigned class and a passive component that simultaneously suppresses confidence in competing classes. Collectively, these methods demonstrate how modifications to the loss function can alter the learning dynamics of deep neural networks and improve their resilience to noisy supervision (Song et al., 2023).

The primary strength of robust loss-function methods is their strong theoretical foundation. Many approaches are supported by formal analyses that establish conditions under which learning remains tolerant to label noise. Because these methods operate directly through the optimisation objective, they can often be integrated into existing deep-learning frameworks without requiring architectural modifications or auxiliary datasets. Furthermore, they provide a principled mechanism for reducing the influence of corrupted labels while maintaining compatibility with standard training procedures (Song et al., 2023).

Despite these advantages, robust loss functions also have limitations. Empirical studies have shown that many of these methods perform best in relatively simple learning scenarios, particularly when the number of classes is small or the classification task is less complex. Their effectiveness may decrease as data complexity increases or when label noise exhibits highly realistic patterns. In addition, modified loss functions often complicate optimisation and can increase the number of training iterations required for convergence. Consequently, the theoretical robustness of these methods does not always translate into consistent performance improvements across all practical applications (Song et al., 2023).

2.11.4. Loss Adjustment

A related line of research seeks to improve robustness by adjusting how losses are computed or applied during training. Loss adjustment refers to a family of methods that reduce the negative effect of noisy labels by modifying the loss contribution of training examples before the deep neural network is updated. Unlike robust loss-function methods, which design new objective functions specifically for noisy-label learning, loss-adjustment methods usually retain the conventional training objective but alter how the loss is corrected, weighted, or relabelled during training. The central idea is to make the standard optimisation process more resistant to corrupted supervision by intervening in the update rule while training is in progress. These methods can be divided into four groups according to how the loss is modified: loss correction, loss reweighting, label refurbishment, and meta learning (Song et al., 2023).

2.11.4.1. Loss Correction

Loss correction methods adjust the loss by estimating how clean labels are transformed into noisy observed labels. This is usually represented through a noise transition matrix. The approach is related to noise adaptation layers, but the key difference is that the transition matrix is estimated separately from the main model rather than being learned as part of the network architecture. Once estimated, the transition matrix is used to correct the training loss or the model output so that optimisation better reflects the underlying clean-label distribution (Song et al., 2023).

In backward correction, the model is first trained without correction, and its outputs are then used to estimate the noise transition matrix. The model is subsequently retrained by correcting the original loss through the inverse of the estimated transition matrix. The corrected loss can be expressed as:

$$\mathcal{L}(f(x; \theta), \tilde{y}) = \hat{T}^{-1}[\mathcal{L}(f(x; \theta), 1), \dots, \mathcal{L}(f(x; \theta), c)],$$

where $\mathcal{L}$ is the loss function, $f(x; \theta)$ is the model prediction for input x , $\tilde{y}$ is the noisy label, $\hat{T}$ is the estimated noise transition matrix, and c is the number of classes. In this formulation, the loss for the observed noisy label is corrected using the estimated relationship between clean and corrupted labels (Song et al., 2023).

Forward correction instead modifies the model output before the loss is computed. Rather than applying the inverse transition matrix to the loss values, it multiplies the predicted class probabilities by the estimated transition matrix during the forward pass:

$$\mathcal{L}(f(x; \theta), \tilde{y}) = \mathcal{L}(\hat{T}f(x; \theta), \tilde{y}).$$

Here, $\hat{T}f(x; \theta)$ represents the model prediction transformed according to the estimated noise transition probabilities. The loss is then calculated using this corrected prediction and the noisy label (Song et al., 2023).

The main contribution of loss correction methods is that they provide a principled way to account for systematic label corruption. Backward correction (Patrini et al., 2017) applies the inverse of the estimated transition matrix to the loss values, whereas forward correction (Patrini et al., 2017) transforms the predicted class probabilities using the transition matrix before computing the loss. Gold loss correction (Hendrycks et al., 2018) improves this process by using clean validation data or anchor points to estimate the matrix more accurately. Later methods reduced the dependency on anchor points or improved matrix estimation. T-revision (Xia et al., 2019) estimates the transition matrix without anchor points, while Dual T (Yao et al., 2020) factorises the transition matrix into two simpler matrices to avoid estimating the noisy class posterior directly. Other extensions address instance-dependent label noise by modelling transition probabilities as dependent on the input example rather than assuming that noise is class-dependent only (Song et al., 2023).

The strength of loss correction is its theoretical clarity: if the noise transition matrix is estimated accurately, the training objective can be adjusted to better approximate learning from clean labels. However, this is also its main weakness. The robustness of these methods depends heavily on the quality of the estimated transition matrix. In practice, accurate estimation often requires prior knowledge, anchor points, or clean validation data, which may not be available in real-world noisy datasets (Song et al., 2023).

2.11.4.2. Loss Reweighting

Loss reweighting methods reduce the influence of noisy labels by assigning different weights to training examples. The guiding assumption is that examples likely to have correct labels should receive larger weights, whereas examples suspected to be mislabelled should receive smaller weights. This allows the model to continue using all examples while reducing the impact of those that are less reliable (Song et al., 2023).

The reweighted update rule for a mini-batch can be written as:

$$\theta_{t+1} = \theta_t - \eta \nabla \left(\frac{1}{|B_t|} \sum_{(x, \tilde{y}) \in B_t} w(x, \tilde{y}) \mathcal{L}(f(x; \theta_t), \tilde{y}) \right),$$

where θ_t represents the model parameters at iteration t , η is the learning rate, B_t is the mini-batch, $w(x, \tilde{y})$ is the weight assigned to example x with noisy label $\tilde{y}$, and $\mathcal{L}$ is the loss function. Examples with smaller weights have less influence on the parameter update, thereby reducing their ability to drive the model toward noisy-label memorisation (Song et al., 2023).

The contribution of loss reweighting methods lies in their flexible control over sample importance. Importance reweighting (Wang et al., 2018) estimates how much each noisy example should contribute by approximating the ratio between the clean and noisy joint data distributions. Active bias (Chang et al., 2017) takes a different view and increases the influence of uncertain examples whose predictions vary

during training, using prediction variance as a weighting signal. DualGraph (Zhang et al., 2021) uses graph neural networks to model structural relationships among labels and downweight abnormal examples that are likely to represent noise (Song et al., 2023).

The main strength of loss reweighting is that it offers a direct and interpretable way to reduce the influence of unreliable examples without removing them from training. However, it requires a weighting rule, and this rule often needs to be manually specified. Appropriate weighting schemes can vary substantially depending on the noise type, dataset structure, and training dynamics. As a result, these methods can be difficult to apply reliably in practice without careful tuning and additional hyperparameters (Song et al., 2023).

2.11.4.3. Label Refurbishment

Label refurbishment methods reduce overfitting to incorrect labels by replacing the noisy target with a revised label. The refurbished label is usually formed by combining the original noisy label with the model’s current prediction. Instead of backpropagating the loss using the noisy label alone, the model is trained using the refurbished target, which is expected to be closer to the clean label (Song et al., 2023).

The basic label-refurbishment formulation is:

$$\mathbf{y}_{\text{refurb}} = \alpha \tilde{\mathbf{y}} + (1 - \alpha) \hat{\mathbf{y}},$$

where $\mathbf{y}_{\text{refurb}}$ is the refurbished label, $\tilde{\mathbf{y}}$ is the noisy label, $\hat{\mathbf{y}}$ is the current prediction of the model, and $\alpha \in [0,1]$ represents the confidence assigned to the noisy label. A larger α places more trust in the original noisy label, while a smaller α gives more influence to the model prediction (Song et al., 2023).

The main contribution of label refurbishment is that it explicitly changes the target used during learning. Bootstrapping (Reed et al., 2014) introduced this idea by combining noisy labels with network predictions, with the confidence parameter estimated through cross-validation. Dynamic bootstrapping (Arazo et al., 2019) improves this by adjusting the confidence value for individual examples, using the loss distribution to estimate whether each sample is likely to be clean or noisy. Self-adaptive training (Huang et al., 2020) reduces instability by updating refurbished labels using an exponential moving average rather than relying only on the instantaneous prediction of the current model:

$$\mathbf{y}_{t+1}^{\text{refurb}} = \alpha \mathbf{y}_t^{\text{refurb}} + (1 - \alpha) \hat{\mathbf{y}}, \quad \mathbf{y}_0^{\text{refurb}} = \tilde{\mathbf{y}}.$$

Here, $\mathbf{y}_t^{\text{refurb}}$ is the refurbished label at iteration t , and $\mathbf{y}_0^{\text{refurb}}$ is initialised as the noisy label. This update makes the refurbished target more stable over training (Song et al., 2023).

Other methods refine the refurbishment process by deciding when and how labels should be updated. Dimensionality-Driven Learning (D2L) (Ma et al., 2018) uses local intrinsic dimensionality to estimate label confidence and prevent the representation space from expanding in ways that encourage noisy-label memorisation. Selectively Refurbish Unclean Samples (SELFIE) (Song et al., 2019) corrects only examples whose predictions remain consistent over time, based on the assumption that consistent predictions are more likely to reflect the true label. AdaCorr (Zheng et al., 2020) also selectively refurbishes noisy labels and provides a theoretical error bound. Self-Evolution Average Label (SEAL) (Chen et al., 2021) averages softmax predictions over the entire training process and then retrains the model using these averaged soft labels (Song et al., 2023).

The strength of label refurbishment is that it can directly repair noisy supervision rather than merely reducing its influence. When the dataset contains relatively few confusing classes, refurbished labels can be generated with high precision and can substantially improve robustness. However, the approach becomes

risky when many classes are visually or semantically similar. In such cases, wrongly refurbished labels may be introduced, and the model can overfit to these incorrect replacements (Song et al., 2023).

2.11.4.4. Meta-Learning

Meta-learning methods automate the process of loss adjustment. Rather than manually defining how losses should be corrected, weighted, or relabelled, these methods learn an adjustment rule from data. The central idea is “learning to learn”: a higher-level optimisation procedure guides the ordinary training process so that the model becomes more adaptable to different datasets and noise types (Song et al., 2023).

For loss reweighting, meta-learning methods aim to learn the weighting function $w(x, \tilde{y})$ in the reweighted update rule. Learning to Learn from Weak Supervision (L2LWS) (Dehghani et al., 2017b) and Controlled Weak Supervision (CWS) (Dehghani et al., 2017a) use a target network together with a meta-network. The meta-network is trained using a small clean validation set and then provides weight scores for examples used to train the target network. Automatic reweighting (Ren et al., 2018) uses gradient directions to determine which examples should receive larger or smaller weights, so that the resulting update reduces the loss on clean validation data. Meta-Weight-Net (Shu et al., 2019) parameterises the weighting function as a small multilayer perceptron and updates it through a meta-objective based on clean validation risk. Data coefficients (Zhang et al., 2020) methods estimate both exemplar weights and corrected labels through meta-optimisation, even when the clean validation set is very small (Song et al., 2023).

For label refurbishment, meta-learning methods can use external or auxiliary models to provide corrected targets. Knowledge distillation (Li et al., 2017) transfers information from an expert model trained on clean validation data to a target model, replacing the target model’s own prediction in the refurbishment process. MLC (Zheng et al., 2021) uses a meta-model to generate corrected labels for the target model, with both models trained together through bilevel optimisation (Song et al., 2023).

The key strength of meta-learning approaches is adaptability. Because the adjustment rule is learned rather than manually prescribed, these methods can generalise across different types of data and label noise more effectively than fixed reweighting or refurbishment strategies. Their main weakness is their reliance on clean and unbiased validation data. Such data are needed to optimise the auxiliary meta-objective, but in many real-world noisy-label settings, a representative clean validation set is unavailable or costly to obtain (Song et al., 2023).

2.11.5. Sample Selection

Rather than modifying losses or architectures, sample-selection methods focus on identifying and learning from examples that are likely to be correctly labelled. Sample-selection methods aim to improve learning under label noise by identifying and using training examples that are likely to be correctly labelled while excluding examples suspected of containing annotation errors. The central assumption is that noisy labels negatively influence model updates and that robustness can be improved by restricting learning to a subset of trusted samples. Rather than correcting labels or modifying the loss function directly, this category focuses on selecting reliable examples from a noisy training dataset and using only those examples to update the model parameters (Song et al., 2023).

Mathematically, let B_t denote the mini-batch at training iteration t , and let $C_t \subseteq B_t$ represent the subset of examples identified as clean. The model is updated using only the selected examples:

$$\theta_{t+1} = \theta_t - \eta \nabla \left(\frac{1}{|C_t|} \sum_{(x, \tilde{y}) \in C_t} \mathcal{L}(f(x; \theta_t), \tilde{y}) \right),$$

where η is the learning rate, $\mathcal{L}$ is the loss function, and $|\mathcal{C}_t|$ is the number of selected clean examples. Examples outside $\mathcal{C}_t$ are excluded from parameter updates because they are considered more likely to contain incorrect labels (Song et al., 2023).

The theoretical motivation for sample selection is closely related to the memorization behaviour of deep neural networks. Research has shown that neural networks typically learn simple and generalizable patterns before gradually memorizing noisy labels. Under assumptions regarding clusterable data, the distance between the initial model parameters W_0 and the parameters after t training iterations W_t can be bounded by

$$\|W_t - W_0\|_F \leq \sqrt{K + \left(\frac{K^2 \epsilon_0}{\|C\|^2}\right)t},$$

where $\|\cdot\|_F$ denotes the Frobenius norm, K is the number of clusters, C represents the cluster centres, and ϵ_0 is a neighbourhood parameter. This result suggests that model parameters remain relatively close to their initial values during early training, when the network primarily learns clean patterns, but move further away as the model begins fitting corrupted labels. Consequently, many sample-selection methods rely on the observation that correctly labelled examples often exhibit smaller training losses than mislabeled examples, particularly during the early stages of learning (Song et al., 2023).

A major contribution of this category is the development of mechanisms that exploit the memorization effect to identify trustworthy training samples. By selectively training on examples believed to be correctly labelled, these methods can substantially reduce the influence of label noise without requiring explicit noise modelling or label correction. Over time, sample selection has evolved into a broad family of approaches that includes multinet learning, multiround learning, and hybrid methods that combine sample selection with other robust-learning strategies (Song et al., 2023).

The label-noise mitigation strategy adopted in this study follows the general principle of sample selection by using uncertainty measures to identify candidate label errors, which are subsequently filtered from the training labels before model retraining. This approach was selected because it enables potentially erroneous labels to be identified explicitly before retraining, rather than addressing label noise indirectly through modifications to the model architecture, regularisation strategy, or loss function.

2.11.5.1. Multinet Learning

Multinet learning methods use multiple neural networks that cooperate during training to identify reliable examples. The underlying idea is that independent networks can provide complementary information, thereby reducing the risk that a single model reinforces its own mistakes. Sample selection is therefore guided either by a mentor network, a peer network, or interactions among multiple learners (Song et al., 2023).

The primary contribution of multinet methods is the reduction of confirmation bias during sample selection. Early work such as Decouple (Malach & Shalev-Shwartz, 2017) proposed maintaining two networks and updating them only on examples where their predictions disagreed. Subsequent methods exploited the small-loss principle, which assumes that correctly labelled examples tend to produce lower losses than mislabelled examples. MentorNet (Jiang et al., 2018) uses a pretrained mentor model to identify likely clean examples for a student network. Co-teaching (Han et al., 2018) and Co-teaching+ (Yu et al., 2019) employ two networks that exchange selected small-loss samples during training, thereby reducing the tendency of each model to reinforce its own errors. Other approaches, such as Joint Training with Co-Regularization (JoCoR) (Wei et al., 2020), introduce co-regularization mechanisms that encourage

the predictions of multiple networks to converge while still benefiting from collaborative learning (Song et al., 2023).

The main advantage of multinet learning is its ability to mitigate confirmation bias by leveraging multiple learners during the selection process. However, maintaining several networks simultaneously increases computational complexity and memory requirements. Furthermore, many methods depend heavily on the small-loss assumption, which becomes less reliable when the loss distributions of clean and noisy samples overlap substantially, such as under asymmetric label noise (Song et al., 2023).

2.11.5.2. Multi-round Learning

Multi-round learning methods iteratively improve the quality of the selected clean set through repeated rounds of training and refinement. Instead of relying on multiple networks, these approaches repeatedly identify clean examples, retrain the model using those examples, and then update the selection based on the improved model. The quality of the selected subset is therefore expected to improve progressively across training rounds (Song et al., 2023).

The key contribution of multi-round learning is iterative refinement. Methods such as Iterative Trimmed Loss Minimization (ITLM) (Shen & Sanghavi, 2019) repeatedly select low-loss examples and retrain the model using only those examples. Iterative Noisy Cross-Validation (INCV) (Chen et al., 2019) combines repeated data partitioning and cross-validation to identify reliable samples, while O2U-Net (Huang et al., 2019) repeatedly retrains the model using cyclical learning rates to accumulate loss statistics that can distinguish clean and noisy examples. Other methods extend the selection process beyond simple loss-based criteria. For example, iterative detection (Wang et al., 2018) uses outlier-detection techniques in feature space, MORPH (Song et al., 2021) progressively expands a trusted set of examples through self-transitional learning, TopoFilter (Wu et al., 2020) exploits topological properties of learned representations, and Noisy Graph Cleaning (NGC) (Wu et al., 2021) performs geometry-based selection using nearest-neighbour graphs and latent representations (Song et al., 2023).

The major strength of multi-round learning is that the selected clean set becomes progressively larger and more accurate through iterative refinement. This often results in improved robustness compared with single-pass selection approaches. However, each additional training round increases computational cost, causing training time to grow approximately in proportion to the number of refinement iterations (Song et al., 2023).

2.11.5.3. Hybrid Approaches

Hybrid approaches combine sample selection with complementary noisy-label learning strategies to address a fundamental limitation of pure sample-selection methods: the exclusion of unselected examples. Rather than discarding potentially useful data, hybrid approaches seek to exploit both selected and unselected samples by integrating sample selection with techniques such as semi-supervised learning, loss correction, label correction, or robust loss functions (Song et al., 2023).

The principal contribution of hybrid methods is the ability to utilise the entire training dataset while still benefiting from clean-sample identification. A common strategy is to treat selected examples as labelled data and the remaining examples as unlabelled data within a semi-supervised learning framework. Self-Ensemble Label Filtering (SELF) (Nguyen et al., 2019) progressively filters unreliable examples while incorporating consistency-based learning through a mean-teacher architecture. DivideMix (Li et al., 2020) separates data into clean and noisy subsets using Gaussian mixture models and subsequently applies semi-supervised learning through MixMatch (Berthelot et al., 2019). Robust Curriculum Learning (RoCL) (Zhou et al., 2021) combines supervised learning on selected clean samples with semi-supervised learning

on relabelled noisy samples. Other hybrid methods integrate sample selection with loss correction or robust loss functions. For example, SELFIE (Song et al., 2019) combines sample selection and label correction, while Curriculum Loss (Lyu & Tsang, 2019) incorporates robust-loss principles to improve clean-sample extraction (Song et al., 2023).

The primary strength of hybrid approaches is their ability to achieve substantially higher robustness by combining complementary strategies. Unlike conventional sample-selection methods, they make use of information contained in examples that would otherwise be discarded. However, integrating multiple techniques introduces additional hyperparameters and design choices, increasing sensitivity to dataset characteristics and noise patterns. Furthermore, the combination of multiple learning components inevitably increases computational complexity and training cost (Song et al., 2023).

3. STUDY AREA AND DATASETS

3.1. Study Area

According to Rodrigues et al. (2025), the state of Pará in northern Brazil covers approximately 1.25 million km², representing roughly one-third of the Brazilian Amazon biome. The region has a tropical climate and contains a heterogeneous landscape that includes dense rainforest, extensive river systems, and large areas of agricultural and pastoral land (IBGE, 2019; Martorano et al., 1993). Figure 1 illustrates the location of the state of Pará within Brazil, while Figure 2 illustrates the spatial extent of the study area and the distribution of the three classification classes (deforestation, non-deforestation, and cloud) used in this study.

Pará has undergone significant land-use change over recent decades. Rodrigues et al. (2025) report that more than 292,000 km² of forest have been cleared in the state since monitoring began, making it one of the most deforested states in the Brazilian Legal Amazon. The authors further report that more than 18,000 km² of primary forest have been identified as degraded due to logging and fire since 2016, consistent with broader trends reported by INPE (2023). The state also contains extensive protected areas, covering approximately 457,000 km² (Fundação Amazônia de Amparo à Estudos e Pesquisas (FAPESPA), 2020).

This combination of deforestation, forest degradation, rapid land-use transitions, and the persistent cloud cover typical of the Amazon Basin (Nunes et al., 2016) creates spatial and temporal conditions in which mismatches between reference labels and satellite imagery are likely to occur. Consequently, the study area provides an appropriate setting for investigating label noise and uncertainty in machine-learning-based deforestation mapping.

Despite the large spatial extent of Pará, this study focuses on a smaller region comprising approximately four Landsat scenes. The study area extends from 144,585 to 576,015 m E and from 9,403,885 to 9,795,715 m N in SIRGAS 2000 / UTM Zone 22S (EPSG:31982). This subset was selected to provide a sufficiently large and heterogeneous study area while remaining computationally manageable for iterative model development, evaluation, and refinement. The selected extent captures diverse land-cover types and varying degrees of deforestation, while enabling repeated model training, hyperparameter optimisation, Monte Carlo-based uncertainty estimation, and retraining experiments to be conducted within the available computational resources and research timeframe. Consequently, the selected area provides a practical and heterogeneous environment for evaluating uncertainty-based approaches to label-noise identification.



Figure 1: Location of the state of Pará, Brazil

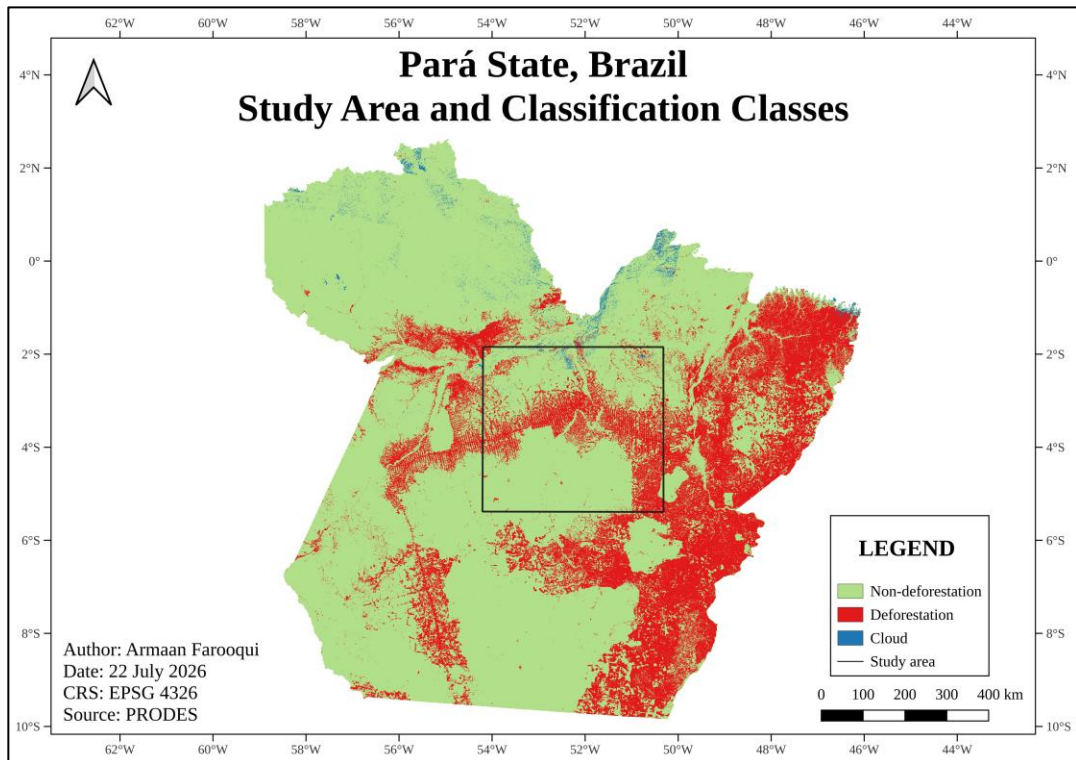


Figure 2: Study area and classification classes

3.2. Datasets

3.2.1. PRODES Label

The Brazilian Amazon Rainforest Monitoring Program by Satellite (**PRODES**) (Instituto Nacional de Pesquisas Espaciais [INPE], 2024), operated by the National Institute for Space Research (INPE), was used as the source of reference labels in this study. PRODES has monitored primary-forest loss in the Brazilian Legal Amazon since 1988 and is designed to estimate annual deforestation caused by the clear-cut removal of primary forest. Within the Legal Amazon, the programme maps forest loss in areas classified as part of the forest vegetation domain rather than providing a complete land-use and land-cover classification of all surface types (Almeida et al., 2021).

Within the PRODES framework, deforestation refers to the anthropogenic removal of primary forest and the conversion of the original forest cover to another land-cover type. The programme maps clear-cut deforestation, where the original forest cover has been completely removed. Areas affected by partial degradation or selective logging are not classified as deforestation until a clear-cut pattern becomes detectable in the satellite imagery (Almeida et al., 2021).

PRODES is produced through expert visual interpretation of multitemporal satellite imagery. Trained specialists compare images from different years and identify changes in forest cover using visual characteristics such as tone, colour, shape, texture, and spatial context. The resulting deforestation polygons are subjected to specialist auditing across the full set of monitored Landsat scenes, with additional attention given to areas where deforestation is particularly difficult to detect or where deforestation activity is concentrated (Almeida et al., 2021).

The programme uses medium-resolution optical satellite imagery from sensors including Landsat 8, Sentinel-2, and CBERS-4/4A. These sensors provide imagery at spatial resolutions of approximately 10–30 m, depending on the sensor and spectral band, with revisit intervals of approximately 5–16 days, and include spectral information in the green, red, and infrared regions. PRODES only maps deforestation polygons larger than 6.25 ha, meaning that small forest clearings visible in individual satellite pixels may not be represented in the reference dataset (Almeida et al., 2021).

PRODES follows an annual and incremental mapping approach. For each PRODES year, new deforestation increments occurring between the previous and current observation periods are identified, while areas mapped as deforested in earlier years are excluded from being mapped again. The PRODES calendar year extends from 1 August to 31 July, with 1 August used as the annual reference date. Therefore, the temporal meaning of a PRODES label differs from that of a single satellite acquisition date and should not be interpreted as indicating the exact date on which deforestation occurred (Almeida et al., 2021).

Image selection is concentrated in the dry season because vegetation conditions are more favourable for detecting forest loss and cloud cover is generally reduced. Nevertheless, persistent clouds can leave some areas unobserved. In such cases, imagery from different dates or sensors may be combined to complete the interpretation of a scene. As a result, an annual PRODES product may incorporate observations collected on multiple dates rather than correspond to one uniform image acquisition date across the entire study area (Almeida et al., 2021).

The official PRODES raster dataset for Brazil was downloaded in GeoTIFF format and has a spatial resolution of 30 m. The dataset records deforestation mapped up to 2024. The original PRODES classes

and their reclassification for this study are summarised in Table 1. The annual deforestation and residual deforestation classes were assigned to the deforestation class (1), while hydrography (91) and native vegetation (100) were assigned to the non-deforestation class (0).

Table 1: Original PRODES classes and their reclassification for this study

Original PRODES class	Meaning	Study class
dYYYY	Deforestation mapped in year YYYY (e.g., d2024 = deforestation mapped in 2024)	Deforestation (1)
rYYYY	Residual deforestation associated with year YYYY (e.g., r2024 = residual deforestation associated with 2024)	Deforestation (1)
Hydrography (91)	Water	Non-deforestation (0)
Native vegetation (100)	Remaining native vegetation	Non-deforestation (0)

3.2.2. Landsat Imagery

Landsat (Earth Resources Observation and Science (EROS) Center, 2020) is a joint Earth observation programme operated by the National Aeronautics and Space Administration (NASA) and the U.S. Geological Survey (USGS) to provide continuous multispectral observations of the Earth's land surface. Since the launch of the first Landsat satellite in 1972, the programme has produced the longest continuous satellite record of Earth's terrestrial surface, enabling the detection, monitoring, and analysis of both natural and human-induced land-cover changes over time. This long-term archive has made Landsat one of the most widely used sources of satellite imagery for environmental monitoring and land-cover mapping (U.S. Geological Survey, 2022, 2015).

This study used Landsat 8 and Landsat 9 Collection 2 Level-2 Science Products. The Level-2 products are generated from terrain-corrected Level-1 imagery through atmospheric correction to produce physically meaningful surface reflectance measurements that approximate the reflectance observed directly above the Earth's surface. In addition to Surface Reflectance (SR), the Level-2 products include Surface Temperature (ST), quality-assessment (QA) layers, and auxiliary information required for scientific analyses. The products are distributed as Cloud Optimized GeoTIFF (COG) files, facilitating efficient storage and access for large-scale geospatial analyses (U.S. Geological Survey, 2022).

The Operational Land Imagers onboard Landsat 8 and Landsat 9 acquire multispectral imagery at a spatial resolution of 30 m in the visible, near-infrared (NIR), and shortwave infrared (SWIR) regions, together with a 15 m panchromatic band. In this study, the multispectral imagery comprised the red, green, blue, near-infrared (NIR), shortwave infrared 1 (SWIR-1), and shortwave infrared 2 (SWIR-2) bands. The QA_PIXEL quality-assessment band was also used to derive cloud information during preprocessing. Each satellite follows a 16-day revisit cycle, while the combined Landsat 8 and Landsat 9 constellation provides an effective 8-day repeat coverage of the Earth's land surface (U.S. Geological Survey, 2015, 2022).

4. METHODOLOGY

4.1. Methodological Overview

The overall methodology adopted in this study is illustrated in Figure 3. The workflow comprises four main stages: (1) baseline semantic segmentation and uncertainty estimation, (2) manual sample validation, (3) uncertainty-based label-error detection and feature analysis, and (4) retraining and performance comparison.

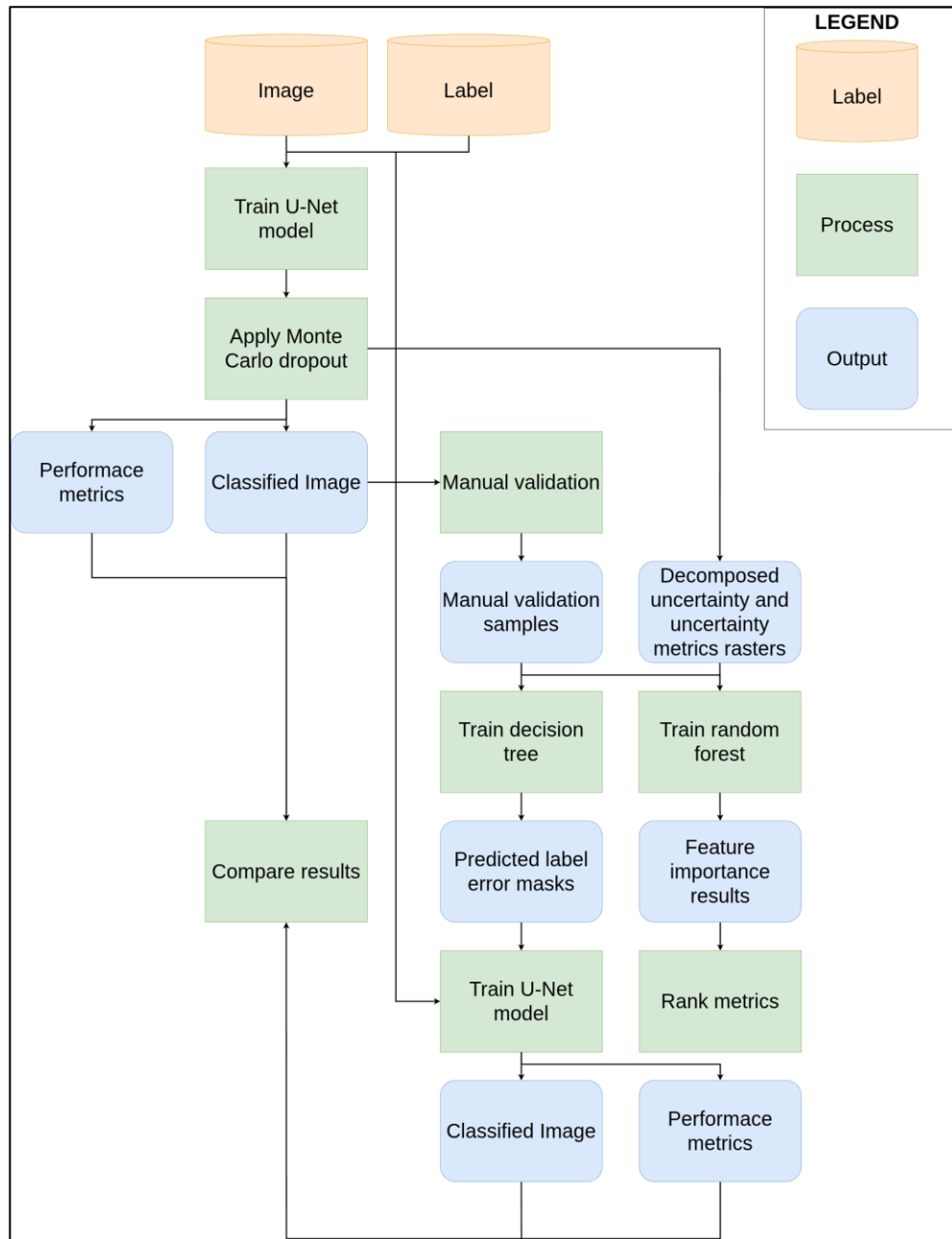


Figure 3: Overall methodology

The workflow begins with the preprocessed Landsat multispectral imagery and the corresponding reference-label dataset. These datasets are used to train a baseline U-Net semantic segmentation model, which produces pixel-wise land-cover predictions for the study area. Monte Carlo Dropout inference is subsequently applied to the trained model to obtain the final classified image together with predictive, aleatoric, and epistemic uncertainty estimates and additional uncertainty metrics (e.g., predictive entropy, Monte Carlo disagreement, and Brier score) derived from the stochastic predictions. The classified image is also evaluated against the reference labels to establish the baseline segmentation performance.

The classified image is then manually interpreted to construct a reference dataset containing correctly labelled samples, label errors, and model errors. These manually interpreted samples are used as the training data for a decision tree classifier, while the uncertainty decomposition and uncertainty metric rasters serve as the predictor variables for both the decision tree and random forest analyses. The decision tree classifies pixels as label errors, model errors, or correct samples and generates binary masks of pixels predicted as label errors, whereas the random forest quantifies the relative importance of the uncertainty metrics for classifying pixels into these three sample classes.

Finally, the predicted label-error masks are used to filter the original training labels before retraining the U-Net model. The retrained model is evaluated using the same validation and test datasets as the baseline model, enabling a controlled comparison of segmentation performance.

4.2. Data Preprocessing

Prior to model training, the Landsat imagery and reference labels were preprocessed to produce a multispectral image and a corresponding three-class reference-label raster sharing a common coordinate reference system, spatial extent, and pixel grid.

Landsat scenes were selected using the Earth Search SpatioTemporal Asset Catalog (STAC). The study area boundary was transformed to WGS 84 (EPSG:4326) and used as the spatial search extent for querying Landsat Collection 2 Level-2 imagery acquired during 2024. The returned scenes were restricted to Landsat 8 and Landsat 9 imagery based on their Landsat scene identifiers. Because multiple Landsat acquisitions were available for the same spatial locations during 2024, the scenes were grouped by their Landsat grid codes, with each grid code representing a particular scene location. Within each group, the acquisition with the lowest reported cloud-cover percentage was selected. This procedure produced a set of least-cloudy scenes covering the study area while avoiding the selection of multiple acquisitions for the same scene location. The identifiers of the selected scenes were subsequently exported for acquisition and further preprocessing.

The official PRODES raster was first reprojected to the SIRGAS 2000 / UTM Zone 22S coordinate reference system (EPSG:31982) and cropped to the study area boundary using the GDAL *gdalwarp* utility. The resulting raster was subsequently reclassified into a binary land-cover map in which the annual deforestation and residual deforestation classes were assigned to the deforestation class (1), while hydrography (91) and native vegetation (100) were assigned to the non-deforestation class (0).

A binary cloud-mask layer was generated from the Landsat Collection 2 Level-2 QA_PIXEL quality-assessment band by extracting pixels flagged as high-confidence cloud (bit 3). The cloud mask was subsequently reprojected, cropped, and resampled to match the binary PRODES label raster. Cloud pixels were then incorporated into the reference labels by assigning all cloud-mask pixels to class 2 while preserving the original deforestation (1) and non-deforestation (0) labels elsewhere.

The Landsat multispectral imagery was prepared in a similar manner. The red, green, blue, near-infrared (NIR), shortwave infrared 1 (SWIR-1), and shortwave infrared 2 (SWIR-2) bands were combined into a virtual raster and subsequently reprojected, cropped, and resampled to match the three-class reference-label raster.

4.3. Baseline U-Net Model

A baseline semantic segmentation model was developed using the U-Net architecture implemented with the *segmentation_models_pytorch* library.

The U-Net architecture was selected because it is a well-established and widely adopted semantic segmentation model for remote sensing. Its encoder–decoder architecture with skip connections effectively integrates contextual and fine spatial information, making it well suited to pixel-wise land-cover classification. Consequently, U-Net provides a robust and reproducible baseline for evaluating the proposed uncertainty-based label-noise mitigation methodology (Ronneberger et al., 2015; Lv et al., 2023). The original U-Net architecture proposed by Ronneberger et al. (2015) is reproduced in Figure 4 to illustrate the network design adopted as the basis for this study.

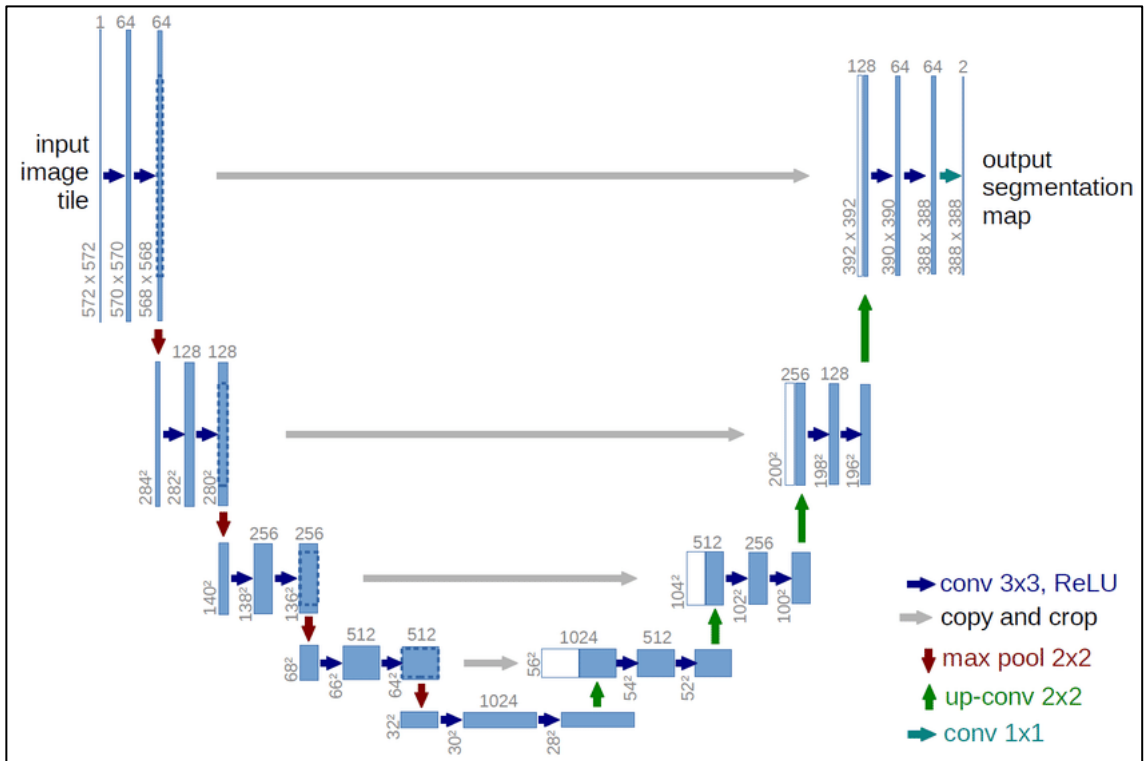


Figure 4: Original U-Net architecture

Original U-Net architecture reproduced from Ronneberger et al. (2015), “U-Net: Convolutional Networks for Biomedical Image Segmentation”. The diagram illustrates the encoder–decoder structure with skip connections as proposed by the original authors.

The network employed a ResNet34 encoder without pretrained weights. A ResNet encoder was adopted because its residual learning framework employs shortcut connections that alleviate gradient degradation, enabling efficient training of deep convolutional networks while maintaining strong feature extraction capabilities for semantic segmentation (He et al., 2016; Shehab et al., 2021; Shafiq & Gu, 2022). Among

the available ResNet variants, ResNet34 was selected because it has been reported to provide a good balance between computational complexity and segmentation performance (Karvonen, 2024). Pretrained ImageNet weights were not used because the six-channel Landsat imagery employed in this study differs from the three-channel RGB imagery on which ImageNet models are trained. Previous research has shown that ImageNet pretrained weights cannot be directly transferred to non-RGB remote-sensing datasets and that transfer learning performs poorly for multispectral imagery because of substantial domain differences between natural photographs and remote-sensing data (Ghanbarzadeh & Soleimani, 2024). Moreover, even for RGB aerial imagery, ImageNet pretraining has been shown to introduce a large domain gap and systematic bias arising from differences in viewpoint, colour, texture, object characteristics, and scene layout, which can limit downstream performance on remote-sensing tasks (Wang et al., 2023). Consequently, the network was trained from randomly initialised weights rather than using ImageNet pretraining. The encoder accepted the six-channel Landsat imagery as input.

The model produced pixel-wise predictions for the three target classes: non-deforestation, deforestation, and cloud. The distribution of pixels among the three classes in the reference-label dataset is presented in Table 2. A dropout layer was incorporated into the network architecture to enable Monte Carlo Dropout inference during the subsequent uncertainty estimation stage.

Table 2: Distribution of reference-label classes

Class	Pixel count	Percentage
Non-deforestation	145,807,965	75.63%
Deforestation	45,875,339	23.80%
Cloud	1,106,936	0.57%
Total	192,790,240	100.00%

The preprocessed imagery and reference labels were divided into non-overlapping image patches of 256×256 pixels. Patch-based training was adopted because large remote-sensing images cannot be processed directly by convolutional neural networks due to GPU memory limitations, requiring the imagery to be divided into smaller patches for efficient model training. However, patch size represents a trade-off between computational efficiency and the preservation of spatial context, as excessively small patches may lose important semantic information while larger patches impose greater memory requirements (Liu et al., 2018). Accordingly, a patch size of 256×256 pixels was selected. An ablation study by Hu et al. (2024) demonstrated that this patch size provided a favourable balance between preserving semantic information and computational efficiency, achieving better segmentation performance than both smaller (128×128) and larger (512×512) patches under the same training time. Furthermore, the 256×256 patch size enabled a larger training batch size than 512×512 patches, thereby improving training stability (Hu et al., 2024).

To minimise spatial autocorrelation, the patches were first grouped into larger spatial blocks before being partitioned into the training, validation, and test datasets using a target split of 70%, 15%, and 15%, ensuring that neighbouring patches were not distributed across different subsets. The resulting dataset contained 2,294 patches, comprising 1,522 training patches, 349 validation patches, and 423 test patches. The distribution of patches among the three subsets is summarised in Table 3, while their spatial distribution across the study area is shown in Figure 5. The realised proportions differed slightly from the

target proportions because the partitioning was performed using spatial blocks rather than individual patches.

Table 3: Distribution of image patches among the training, validation, and test datasets

Dataset	Number of patches	Percentage
Training	1,522	66.35%
Validation	349	15.21%
Test	423	18.44%
Total	2,294	100.00%

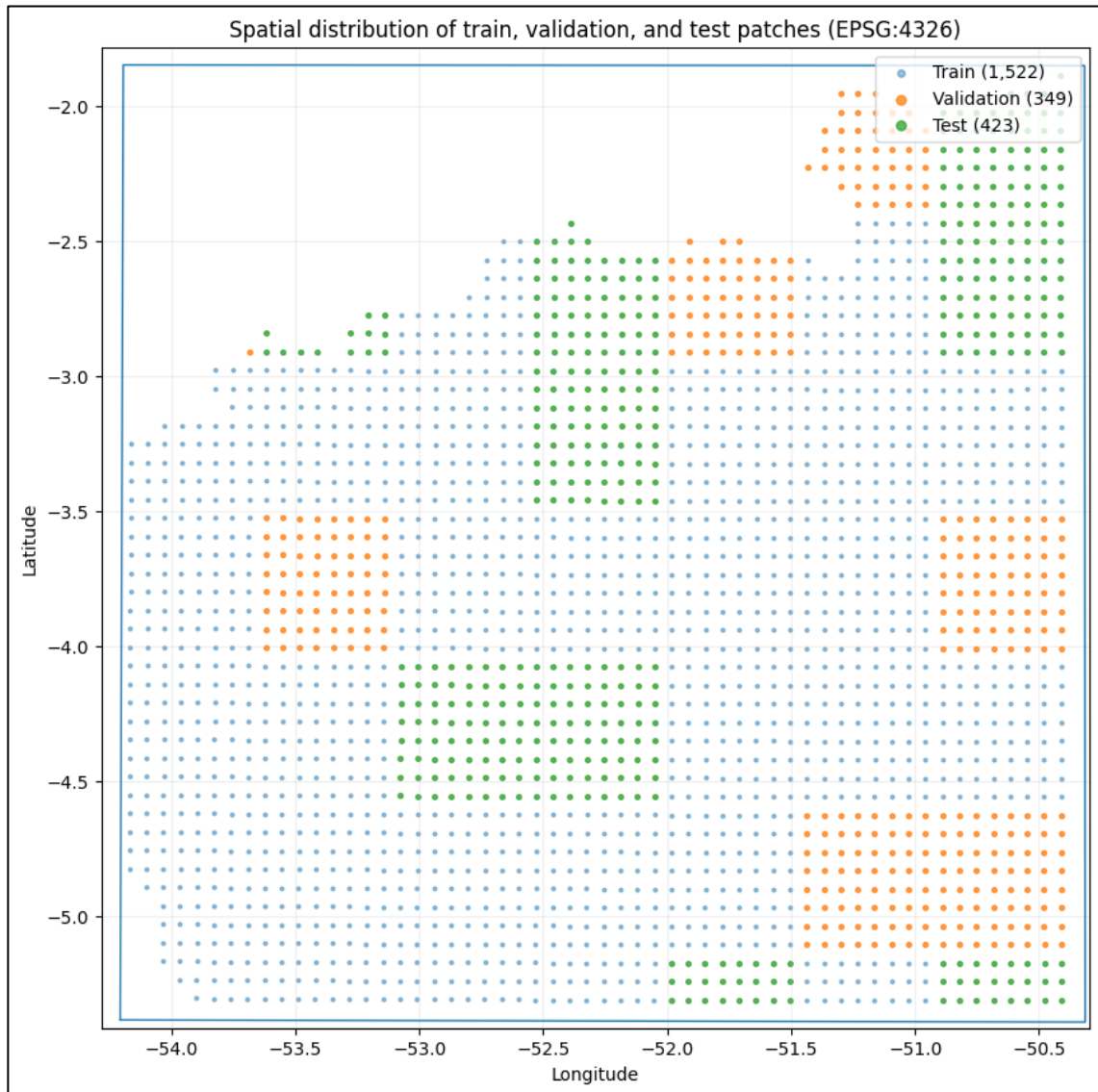


Figure 5: Spatial distribution of the training, validation, and test datasets

Input imagery was normalised using the per-band mean and standard deviation calculated exclusively from the training dataset. Data augmentation was applied only to the training patches and consisted of random

horizontal and vertical flipping, random rotations in multiples of 90° , and additive Gaussian noise. The validation and test datasets were not augmented.

The network was trained using the AdamW optimiser and a weighted cross-entropy loss function to account for class imbalance. Class weights were derived from the class frequencies of the training dataset using an inverse-frequency weighting scheme. Hyperparameters, including the learning rate, weight decay, batch size, augmentation probability, and Gaussian noise level, were selected through Optuna-based optimisation. Model training employed early stopping based on the validation mean Intersection over Union (mIoU), and the checkpoint achieving the highest validation mIoU was retained for the subsequent uncertainty estimation.

4.4. Uncertainty Estimation

Uncertainty estimation was performed using the trained baseline U-Net model with dropout layers enabled during inference. To estimate predictive uncertainty, Monte Carlo Dropout inference was performed by maintaining dropout and executing multiple stochastic forward passes through the network. The predictive distribution was approximated by averaging the outputs from repeated stochastic forward passes, while predictive uncertainty was estimated from the variability among these predictions (Gal & Ghahramani, 2016). Fifty stochastic forward passes were performed. Although Gal and Ghahramani (2016) demonstrated that reliable uncertainty estimates can be obtained using a relatively small number of stochastic forward passes, Milanés-Hermosilla et al. (2021) noted that $T = 50$ is considered a safe choice for uncertainty estimation and further demonstrated experimentally that predictive performance stabilised before 50 forward passes. Consequently, $T = 50$ was adopted in this study as a literature-supported choice for obtaining stable Monte Carlo estimates.

For each stochastic forward pass, the U-Net produced a probability distribution over the three output classes through the softmax activation layer. The resulting class probabilities were averaged across the 50 stochastic forward passes to obtain the mean predictive probability distribution, which was used to generate the final class prediction. The Monte Carlo predictions were additionally used to estimate predictive uncertainty as the predictive variance, aleatoric uncertainty as the mean predicted data variance across the stochastic forward passes, and epistemic uncertainty as the variance of the predicted class probabilities across the stochastic forward passes.

The mean predictive probability distribution and the complete set of Monte Carlo probability distributions were subsequently used to derive the remaining uncertainty and calibration metrics. Predictive entropy was calculated from the mean predictive probability distribution, while softmax confidence was obtained as the maximum predicted class probability. Mutual information was computed as the difference between predictive entropy and the expected entropy of the Monte Carlo predictions, and Monte Carlo disagreement fraction (variation ratio) was calculated as the proportion of stochastic predictions that differed from the modal class prediction. Expected calibration error was obtained by comparing predicted confidences with observed accuracies across confidence bins. Cross-entropy loss was calculated from the mean predicted class probabilities and the corresponding reference labels, whereas the Brier score was computed as the mean squared difference between the predicted class probabilities and the one-hot encoded reference labels. Finally, a derived feature computed as the difference between cross-entropy loss and predictive entropy ($CE - \text{entropy}$) was subsequently calculated to investigate whether combining information about prediction error and predictive uncertainty provided additional discriminatory information for distinguishing manually interpreted label errors, model errors, and correct samples.

All uncertainty metrics were exported as GeoTIFF raster layers using the spatial reference, extent, and pixel grid of the original reference-label dataset.

4.5. Manual Validation Dataset

To train the decision tree and random forest models, a training dataset was constructed from 294 polygons manually delineated using the Landsat imagery, corresponding reference labels, and baseline model predictions within QGIS. The dataset comprised 98 polygons for each of the three sample categories: correct samples, label errors, and model errors. Correct-sample polygons represented regions where both the reference labels and model predictions were consistent with the visual interpretation of the satellite imagery. Label-error polygons represented regions where the reference annotation was considered incorrect based on visual interpretation of the Landsat imagery, while the model prediction was consistent with the interpreted land-cover class. Model-error polygons represented regions where the reference label was consistent with the visual interpretation but the model prediction was incorrect. The spatial distribution of the manually delineated polygons is presented in Figure 6.

The selected polygons were subsequently overlaid with the uncertainty metric rasters. For every pixel contained within the selected polygons, the values of all uncertainty metrics were extracted. Each pixel inherited the sample category of its parent polygon, thereby producing a pixel-level dataset in which every observation consisted of a vector of predictor variables and an associated manually assigned sample class. To prevent class imbalance from influencing the subsequent analyses, the extracted pixels were balanced by randomly subsampling each category to match the number of pixels in the smallest class. The polygon and pixel sample sizes before and after balancing are presented in Table 4.

Table 4: Distribution of manually interpreted polygons and extracted pixels

Sample category	Polygons	Pixels before balancing	Pixels after balancing
Correct samples	98	4,111,448	351,862
Label errors	98	753,194	351,862
Model errors	98	351,862	351,862
Total	294	5,216,504	1,055,586

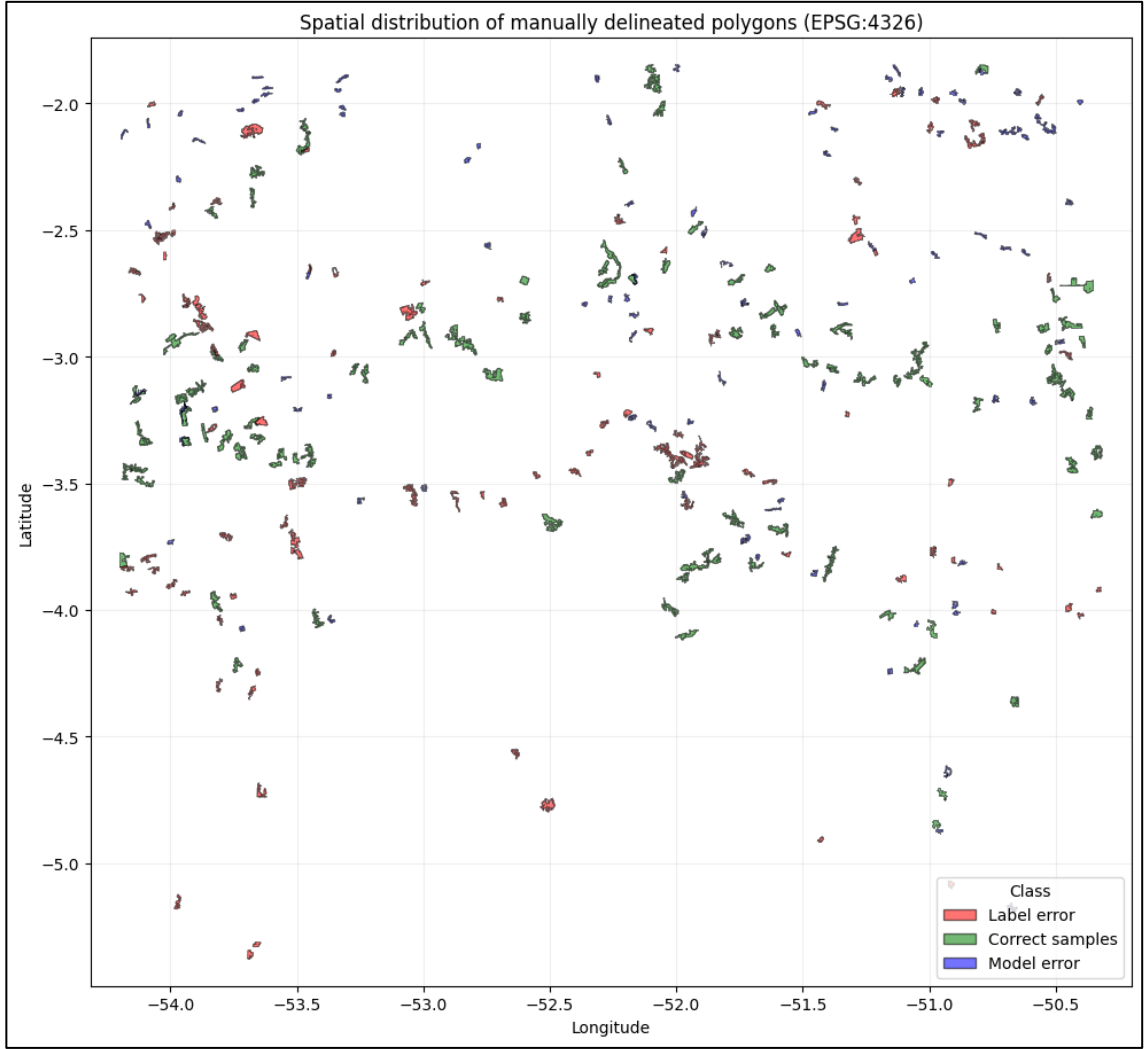


Figure 6: Spatial distribution of the manually interpreted sample polygons

4.6. Decision Tree-Based Label Error Detection

A decision tree classifier was trained on the balanced pixel-level dataset described in the preceding section to distinguish label errors, model errors, and correctly labelled samples based on their uncertainty metrics.

A decision tree was selected because its hierarchical decision rules provide an interpretable framework for identifying uncertainty thresholds associated with label errors, model errors, and correct samples. Unlike black-box models, decision trees represent the classification process as a hierarchy of explicit decision rules based on threshold splits, enabling direct interpretation of how individual uncertainty metrics contribute to the separation of the three sample categories. The learned decision rules provide explicit uncertainty thresholds that can be translated into a practical sample-selection strategy for identifying potentially mislabelled samples prior to model retraining (Blockeel et al., 2023; Höhl et al., 2024). The workflow used to apply the trained decision tree for label-error identification and subsequent sample filtering is illustrated in Figure 7.

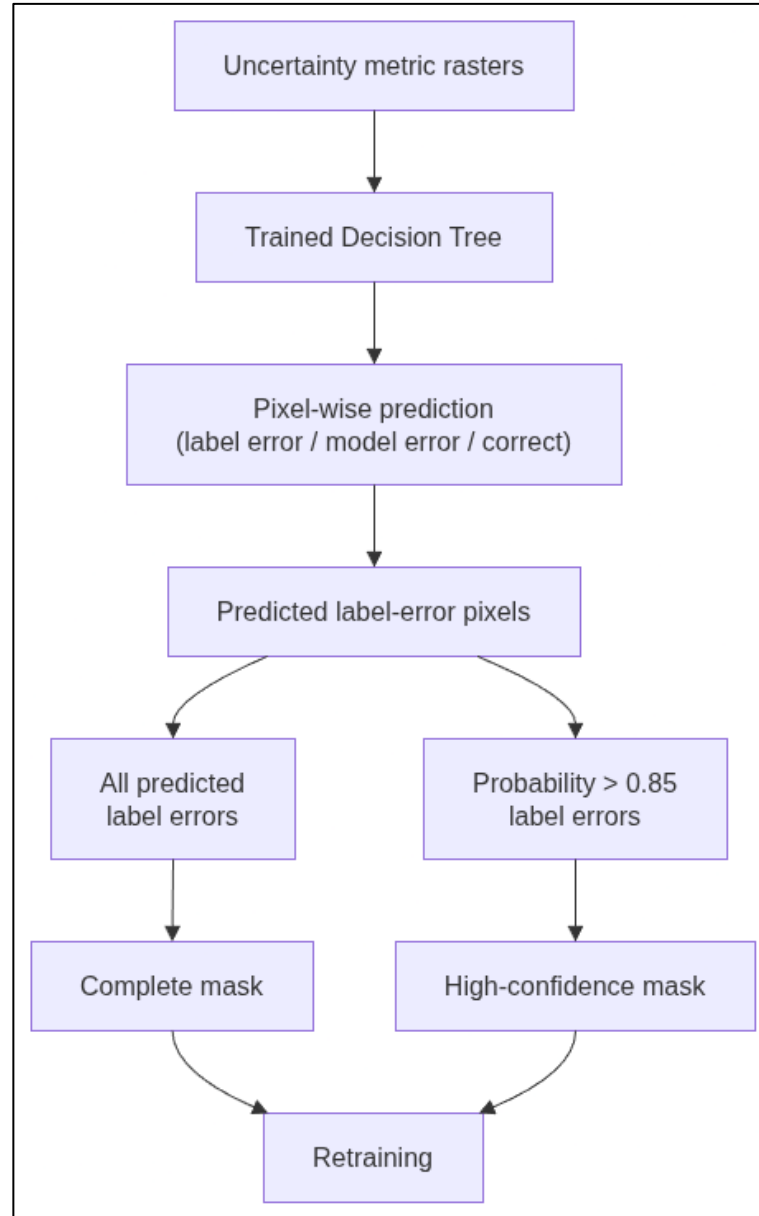


Figure 7: Decision tree-based label-error identification workflow

Following model training, the uncertainty metric values for each pixel across the study area were used as inputs to the trained decision tree. For each pixel, the decision tree predicted one of the three sample categories (label error, model error, or correct sample) and estimated the probability of membership in each category. These predictions were reconstructed spatially to produce a three-class classification raster and a probability raster representing the predicted probability of each pixel belonging to the label-error class.

The three-class classification raster was first used to identify all pixels classified as label errors, producing a complete predicted label-error mask. A second, more conservative mask was generated by retaining only pixels classified as label errors with a predicted label-error probability greater than 0.85. The threshold was selected from the precision–recall analysis presented in Table 8, in which thresholds ranging from 0 to 1 were evaluated. At a threshold of 0.85, precision for the label-error class reached 1.000. Precision was prioritised because false-positive identification of correctly labelled pixels could result in valid training

information being unnecessarily removed. The complete label-error mask and the high-confidence (> 0.85) label-error mask were therefore retained as two alternative filtering strategies for the retraining experiments.

An independent manually audited point dataset was subsequently used to evaluate the decision tree classifier. A total of 150 validation points were selected, comprising 50 points from each of the three predicted decision-tree classes (label error, model error, and correct sample). Within each predicted class, the points were spatially distributed across the study area to provide representative spatial coverage. Each sampled point was manually interpreted using the Landsat imagery and assigned a reference land-cover class. Based on the agreement between the manual interpretation, original reference label, and baseline U-Net prediction, each validation point was subsequently categorised as a label error, model error, correct sample, or both a label error and model error where both conditions were satisfied. The spatial distribution of the validation points is presented in Figure 8.

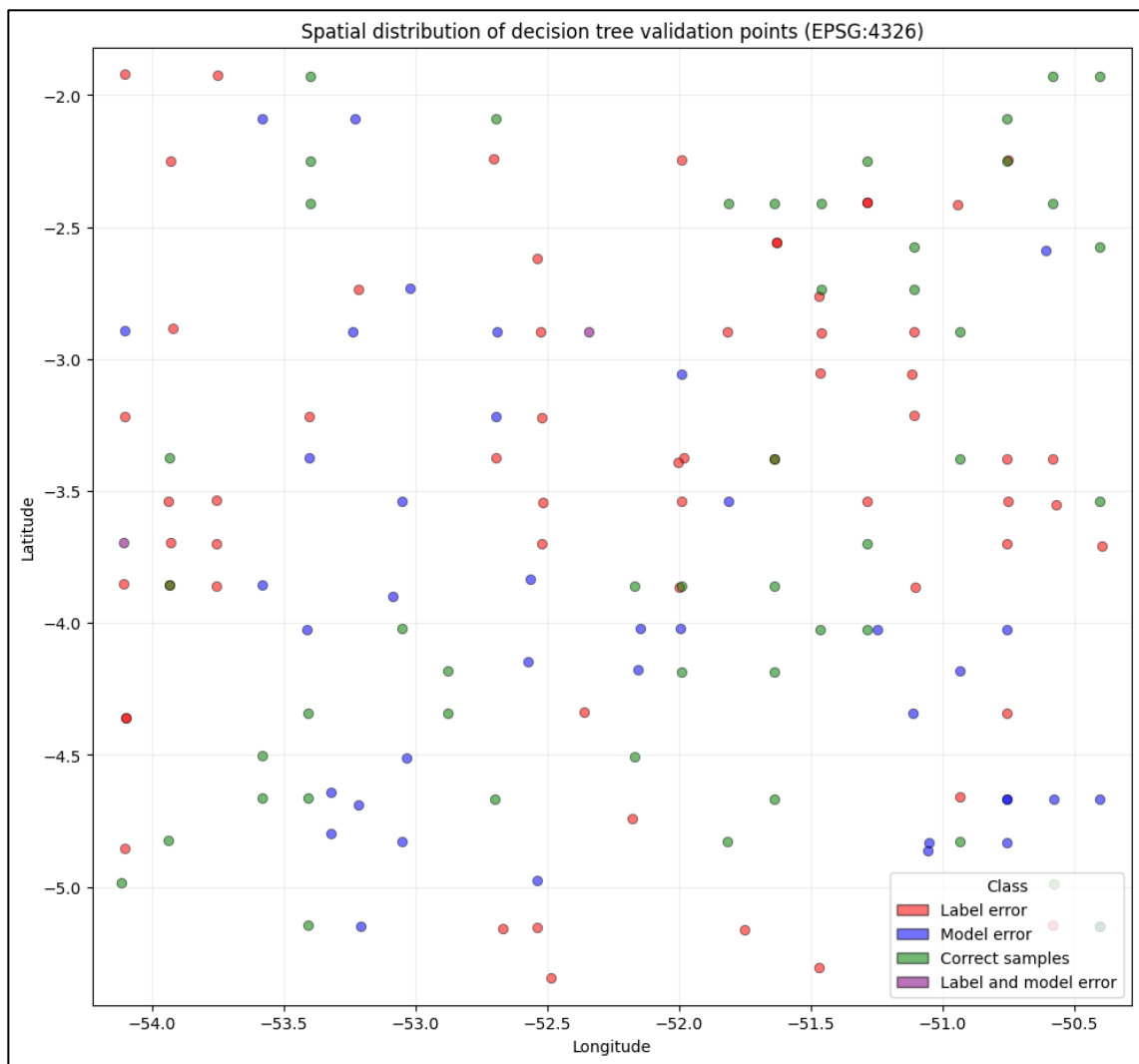


Figure 8: Spatial distribution of decision tree validation points

4.7. Random Forest Feature Importance Analysis

A random forest classifier was trained using the same training dataset as the decision tree. The random forest analysis was included to complement the decision tree by providing an additional ranking of the uncertainty metrics according to their importance for distinguishing label errors, model errors, and correctly labelled samples. To account for class imbalance, the random forest was trained using balanced subsampling, with an ensemble of 500 decision trees constructed from bootstrap samples of the training data.

Unlike a single decision tree, a random forest aggregates the predictions of multiple decision trees, providing more robust estimates of variable importance across the ensemble (Breiman, 2001). Furthermore, random forest variable-importance measures have been widely used in remote-sensing applications to identify and rank the variables that contribute most strongly to classification performance (Belgiu & Drăguț, 2016).

Following model training, feature-importance scores were calculated to quantify the relative contribution of each uncertainty metric to the classification of label errors, model errors, and correctly labelled samples. The resulting rankings complemented the threshold-based interpretation provided by the decision tree analysis.

To complement the model-based feature-importance rankings obtained from the decision tree and random forest, the Hilbert–Schmidt Independence Criterion (HSIC) was computed for every predictor variable. HSIC provided an independent measure of the nonlinear statistical dependence between each uncertainty metric and the manually assigned sample classes. This analysis was used to assess whether the metrics identified as important by the tree-based models also exhibited strong statistical dependence with the sample classes, thereby providing additional evidence for addressing RQ2 concerning which uncertainty metrics were most useful for distinguishing label errors, model errors, and correct samples.

As a kernel-based criterion for evaluating statistical dependence, HSIC provides a measure of dependence between predictor variables and class labels and has been successfully applied to supervised feature selection in remote sensing (Gretton et al., 2005; Camps-Va et al., 2010). A class-balanced subset of the training dataset was used for the HSIC analysis to reduce computational cost while maintaining equal representation of the sample classes.

4.8. Retraining with the Filtered Dataset

To assess whether the removal of suspected label errors improved segmentation performance, the baseline U-Net model was retrained over the same study area used for the baseline experiment, using the filtered training datasets generated from the decision-tree label-error masks. All retraining experiments used the same network architecture, training procedure, data augmentation strategy, optimisation algorithm, hyperparameter configuration, and training, validation, and test patch partitions as the baseline model. The only modification was the application of the decision-tree label-error masks to the training labels.

During retraining, pixels identified as suspected label errors by the decision-tree masks were excluded from the loss computation, while all remaining pixels retained their original class labels. The validation and test datasets were left unchanged to provide an independent basis for comparison with the baseline model. Separate retraining experiments were conducted using the complete decision-tree label-error mask and the high-confidence decision-tree mask generated using a label-error probability threshold of 0.85.

To account for the stochasticity of neural network training, the baseline model and each retraining experiment were trained independently using the same ten random seeds. For every training run, early stopping based on the validation (mIoU) was employed, and the checkpoint achieving the highest validation mIoU was retained. Performance metrics were subsequently aggregated across the ten runs prior to comparing the baseline and retrained models.

5. RESULTS AND DISCUSSION

5.1. Distinguishing Label Errors, Model Errors, and Correct Samples

Research Question 1. Can uncertainty measures distinguish label errors, model errors, and correct samples?

Table 5 summarises the descriptive statistics of the uncertainty metrics for the three manually interpreted sample classes, while Figure 9 illustrates their distributions using boxplots. Overall, the uncertainty metrics exhibited distinct patterns across the three sample classes, indicating that different uncertainty measures capture different characteristics of label errors, model errors, and correct samples.

Table 5: Descriptive statistics by sample class

Metric	Class name	n	Mean	Std	Median	IQR
Predictive uncertainty	Label error	351862	0.49	0.12	0.56	0.22
Predictive uncertainty	Model error	351862	0.53	0.09	0.58	0.08
Predictive uncertainty	Correct samples	351862	0.44	0.13	0.42	0.25
Aleatoric uncertainty	Label error	351862	0.47	0.12	0.53	0.23
Aleatoric uncertainty	Model error	351862	0.52	0.10	0.57	0.10
Aleatoric uncertainty	Correct samples	351862	0.42	0.13	0.40	0.26
Epistemic uncertainty	Label error	351862	0.01	0.01	0.01	0.02
Epistemic uncertainty	Model error	351862	0.01	0.01	0.01	0.01
Epistemic uncertainty	Correct samples	351862	0.02	0.01	0.02	0.01
Predictive entropy	Label error	351862	0.83	0.16	0.93	0.28
Predictive entropy	Model error	351862	0.88	0.12	0.94	0.11
Predictive entropy	Correct samples	351862	0.76	0.17	0.74	0.32
Disagreement fraction	Label error	351862	0.04	0.09	0.00	0.02
Disagreement fraction	Model error	351862	0.11	0.16	0.02	0.18
Disagreement fraction	Correct samples	351862	0.03	0.08	0.00	0.00
Mutual information	Label error	351862	0.03	0.01	0.03	0.03
Mutual information	Model error	351862	0.02	0.01	0.02	0.02
Mutual information	Correct samples	351862	0.03	0.02	0.03	0.02
Calibration error	Label error	351862	0.63	0.14	0.58	0.29
Calibration error	Model error	351862	0.57	0.12	0.51	0.16
Calibration error	Correct samples	351862	0.32	0.14	0.27	0.30
Cross entropy loss	Label error	351862	1.55	0.67	1.27	1.36
Cross entropy loss	Model error	351862	1.23	0.53	0.96	0.50
Cross entropy loss	Correct samples	351862	0.41	0.22	0.31	0.46
Softmax confidence	Label error	351862	0.63	0.14	0.58	0.29
Softmax confidence	Model error	351862	0.57	0.12	0.51	0.16

Softmax confidence	Correct samples	351862	0.68	0.14	0.73	0.30
CE - entropy	Label error	351862	0.72	0.82	0.33	1.64
CE - entropy	Model error	351862	0.35	0.64	0.01	0.58
CE - entropy	Correct samples	351862	-0.35	0.08	-0.39	0.14
Brier score	Label error	351862	1.00	0.39	0.88	0.82
Brier score	Model error	351862	0.82	0.32	0.65	0.39
Brier score	Correct samples	351862	0.20	0.16	0.11	0.34

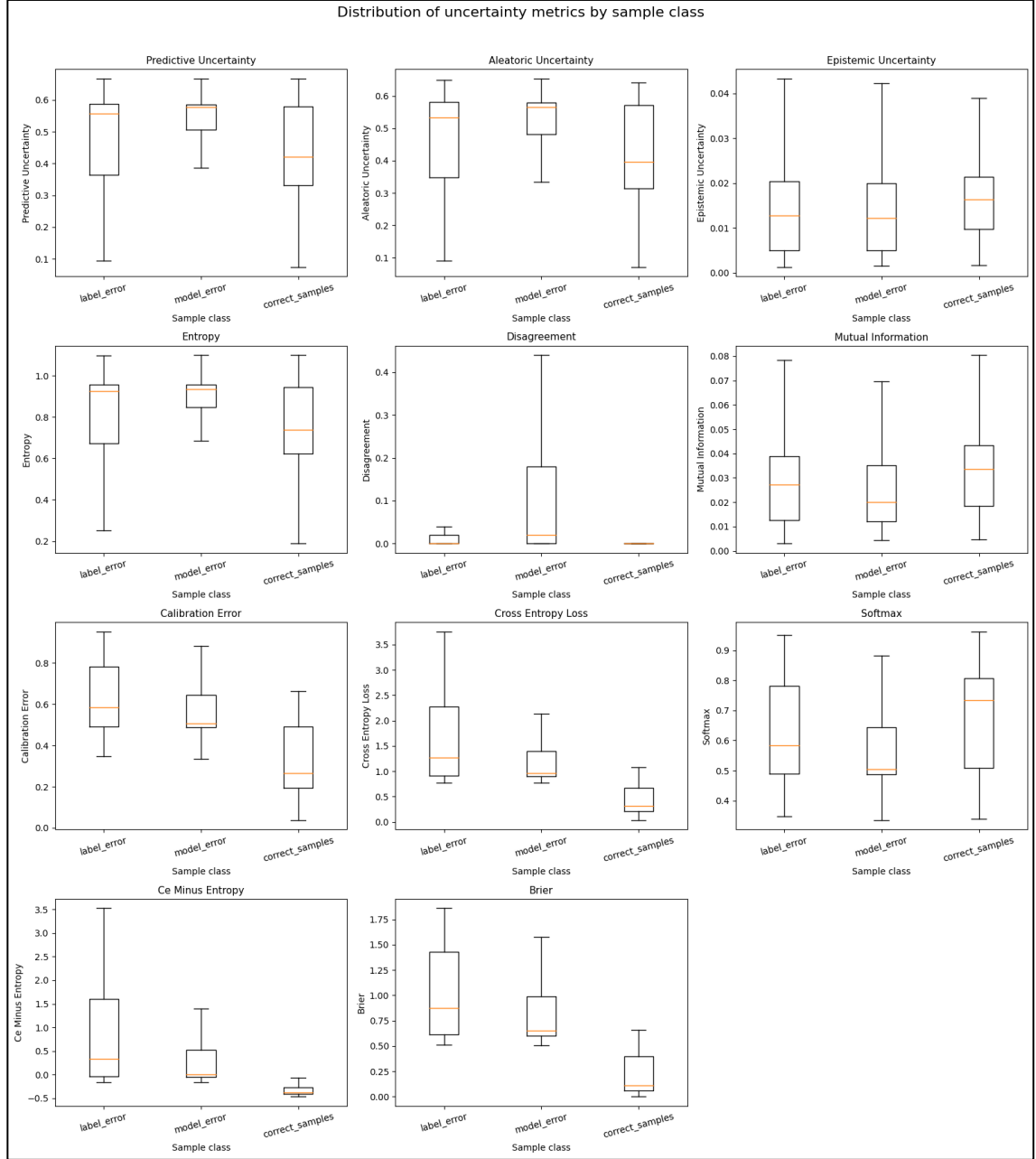


Figure 9: Boxplots of the uncertainty metrics by sample class

Among the decomposed uncertainty quantities, predictive uncertainty and aleatoric uncertainty exhibited the highest mean and median values for the model-error class, followed by label errors and correct samples. Predictive entropy showed a similar class pattern. The corresponding boxplots show that model

errors generally exhibited higher median values and narrower interquartile ranges than the other sample classes, indicating that model errors were consistently associated with higher predictive uncertainty.

Epistemic uncertainty exhibited a different pattern, with the highest values for the correct-sample class, while label errors and model errors showed similar lower distributions. Mutual information showed a broadly similar pattern. These metrics therefore provided less separation between the two error classes than predictive uncertainty, aleatoric uncertainty, and predictive entropy.

Monte Carlo disagreement exhibited the clearest separation of model errors from the other two sample classes. Model errors showed substantially higher disagreement than both label errors and correct samples, whereas the distributions of label errors and correct samples largely overlapped. Calibration error, cross-entropy loss, CE – entropy, and the Brier score exhibited a different pattern, with the highest values occurring for label errors, intermediate values for model errors, and the lowest values for correct samples. Conversely, softmax confidence was highest for correct samples and lowest for model errors.

These differences can be explained by the different information captured by the uncertainty metrics. The higher predictive uncertainty, aleatoric uncertainty, predictive entropy, and Monte Carlo disagreement observed for model errors indicate that the model assigned less decisive class probabilities and showed greater variability across stochastic predictions for these pixels. This suggests that model errors were associated with pixels for which the model had greater difficulty consistently assigning the correct land-cover class. In contrast, epistemic uncertainty was slightly higher for correct samples than for either error class, indicating that higher epistemic uncertainty did not correspond to classification error in the same way as the other uncertainty measures in this dataset. One possible explanation is that some correctly classified pixels represented spectrally heterogeneous or transitional areas, such as forest–deforestation boundaries, where predictions varied across stochastic forward passes while the final predicted class nevertheless remained correct. However, the present analysis does not establish the specific cause of this pattern. Label errors, meanwhile, produced particularly high values for reference-dependent metrics such as cross-entropy loss and the Brier score because these metrics directly compare the predicted probabilities with the reference label. Consequently, when the model prediction was consistent with the imagery but disagreed with an incorrect reference label, these metrics could produce large error values even though the prediction itself was relatively stable. This helps explain why reference-dependent metrics provided stronger separation of label errors from correct samples, whereas predictive uncertainty and Monte Carlo disagreement more clearly characterised model errors. These results indicate that different uncertainty metrics characterise different aspects of prediction behaviour. Predictive uncertainty, aleatoric uncertainty, predictive entropy, and Monte Carlo disagreement primarily distinguished model errors from the other sample classes, whereas calibration error, cross-entropy loss, CE – entropy, and the Brier score more effectively distinguished label errors from correct samples. However, the overlap observed in several boxplots indicates that no individual uncertainty metric completely separated the three sample classes. The key implication is that label errors and model errors produce different but overlapping model-derived signatures: model errors are more strongly characterised by predictive uncertainty and instability, whereas label errors are more strongly characterised by disagreement between the model prediction and the reference label.

To evaluate whether the uncertainty metrics could collectively distinguish the sample classes, a decision tree classifier was trained using all uncertainty metrics as predictor variables. The resulting confusion matrix is presented in Figure 10, while the corresponding classification performance metrics are summarised in Table 6. The combined uncertainty measures distinguished the three sample classes with a macro-average precision of 0.78, macro-average recall of 0.78, macro-average F1-score of 0.77, macro-average IoU of 0.65, and an overall accuracy of 0.84.

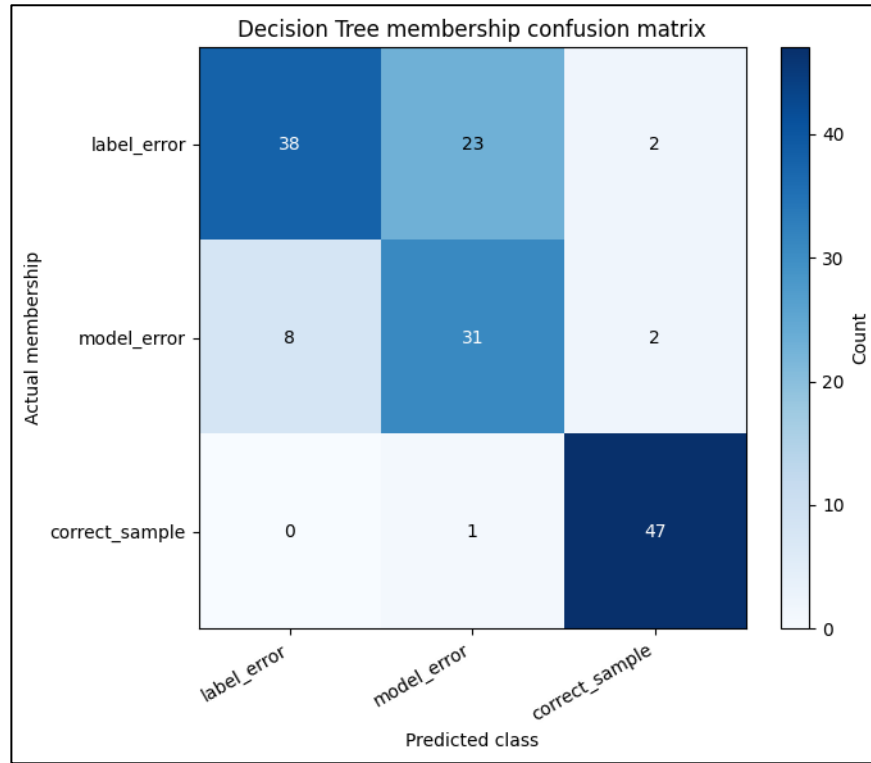


Figure 10: Decision tree confusion matrix

Table 6: Decision tree classification performance

Class	Precision	Recall	F1	IoU	Accuracy	Support
Label error	0.83	0.60	0.70	0.54	0.78	63
Model error	0.56	0.76	0.65	0.48	0.77	41
Correct sample	0.96	0.98	0.97	0.94	0.98	48
Macro average	0.78	0.78	0.77	0.65	0.84	152

The classifier achieved the highest performance for the correct sample class ($F1 = 0.97$), followed by the label error class ($F1 = 0.70$) and the model error class ($F1 = 0.65$). Most misclassifications occurred between the label error and model error classes, whereas only three correct samples were incorrectly classified as error samples. Similarly, only four error samples were incorrectly classified as correct samples. These results indicate that the uncertainty metrics more reliably distinguished correct samples from erroneous samples than they distinguished between label errors and model errors.

The greater confusion between label errors and model errors is likely to reflect the fact that both categories involve disagreement between the model prediction and the reference label. Although their underlying causes differ, both can therefore generate elevated error-related and uncertainty values. Correct samples, by contrast, are characterised by agreement between the prediction, reference label, and visual interpretation, producing a more distinctive uncertainty profile. The remaining overlap between the two

error classes suggests that uncertainty information alone does not always provide sufficient information to determine whether a disagreement originates from the model or from the reference data.

The observed association between uncertainty and model errors is consistent with previous studies showing that predictive uncertainty can provide information about prediction reliability and classification errors. Using Monte Carlo Dropout for medical image segmentation, Nair et al. (2020) found that uncertainty measures were associated with incorrect predictions and that predictions with greater uncertainty were more likely to be erroneous. Similarly, Klanecek et al. (2023) reported a strong inverse relationship between an MC-Dropout-derived uncertainty measure and segmentation performance ($r = -0.76$), with greater uncertainty corresponding to lower Dice similarity scores. Their uncertainty measure also successfully identified unacceptable segmentations, demonstrating that uncertainty can provide useful information for detecting model failures. These findings are consistent with the present results, in which model errors generally exhibited higher predictive uncertainty, aleatoric uncertainty, predictive entropy, and Monte Carlo disagreement than correct samples (Klanecek et al., 2023; Nair et al., 2020).

However, the relationship between uncertainty and prediction error depends on the uncertainty measure employed. Nair et al. (2020) showed that different uncertainty measures capture different aspects of prediction behaviour, including consistency across stochastic predictions and prediction confidence. In a remote-sensing semantic-segmentation application, Westphal et al. (2025) similarly found differences among uncertainty measures in their ability to identify misclassified pixels, with some measures assigning high uncertainty to correctly classified pixels. This is consistent with the present study, where the uncertainty metrics exhibited different distributions across the three sample classes and epistemic uncertainty was slightly higher for correct samples than for either error class. The results therefore support a more nuanced relationship between uncertainty and classification error: uncertainty provides useful information about prediction reliability, but high uncertainty does not necessarily indicate an erroneous prediction, and its discriminatory value depends on the metric considered (Nair et al., 2020; Westphal et al., 2025).

Research on learning with noisy labels further demonstrates that model-derived information can provide evidence about label reliability, rather than only model reliability. Low confidence in an assigned label can indicate a potential annotation error, while prediction–label consistency can similarly provide information about whether a sample is likely to be clean (Jakubik et al., 2025; Zhang et al., 2024). Uncertainty estimates have also been incorporated alongside conventional confidence or loss information for identifying reliable samples. Ji et al. (2021), for example, combined MC-dropout-based epistemic uncertainty with loss information for clean-sample selection, while Jakubik et al. (2025) showed that incorporating predictive uncertainty into confidence-based label-error detection improved its performance. Zhang et al. (2024) likewise incorporated epistemic uncertainty into noisy-label learning to account for differences in the model's learning state across samples. These studies support the present finding that label errors, like model errors, can exhibit distinctive model-derived characteristics, providing a basis for investigating whether the two sources of disagreement can be distinguished (Jakubik et al., 2025; Ji et al., 2021; Zhang et al., 2024).

Nevertheless, abnormal model behaviour does not uniquely identify whether an error originates from the reference label or from difficulty experienced by the model. Difficult but correctly labelled samples and noisy-label samples can exhibit similar characteristics. Guan et al. (2023) demonstrated this specifically for loss, showing that hard positive and noisy samples can both produce large loss values. Deng et al. (2024) similarly found that hard clean and noisy samples can both exhibit high learning difficulty, characterised by high loss and low prediction probability, despite different underlying causes. The difficulty of hard clean samples was associated with irregular feature patterns, whereas that of noisy samples was primarily

associated with incorrect labels. Zhou et al. (2026) further showed that uncertainty can characterise hard samples and that sample-quality factors can produce ambiguous model behaviour. Thus, high loss, low prediction probability, low confidence, or high uncertainty can indicate that a sample is problematic without necessarily revealing the source of that difficulty (Deng et al., 2024; Guan et al., 2023; Zhou et al., 2026).

This ambiguity provides a useful explanation for the present study's greater confusion between label errors and model errors than between error samples and correct samples. Both error classes involve abnormal prediction behaviour, but the literature indicates that similar loss, probability, confidence, or uncertainty patterns can arise from different underlying causes. The observed overlap between label errors and model errors is therefore consistent with the limitations of model-derived indicators for determining the source of abnormal prediction behaviour (Deng et al., 2024; Guan et al., 2023; Zhou et al., 2026).

Taken together, these findings provide context for the three-class distinction evaluated in the present study. The decision tree distinguished label errors, model errors, and correct samples with a macro-average F1-score of 0.77. The scientific contribution of this analysis is the explicit treatment of label errors and model errors as separate competing classes within the same classification framework. This distinction is important because prediction–reference disagreement can result either from an incorrect reference label or from an incorrect model prediction. The present results demonstrate that these two sources of disagreement can be distinguished to a meaningful but imperfect degree using model-derived information. At the same time, the remaining confusion between label errors and model errors provides empirical evidence that the ambiguity identified in previous noisy-label research also persists when these two sources of prediction–reference disagreement are explicitly separated (Deng et al., 2024; Guan et al., 2023; Zhou et al., 2026).

Answer to Research Question 1. The results demonstrate that model-derived uncertainty and error-related measures can distinguish label errors, model errors, and correct samples, although the distinction is imperfect. The combined metrics enabled the decision tree to classify the three sample classes with a macro-average F1-score of 0.77 and an overall accuracy of 0.84. Correct samples were distinguished most reliably, while greater confusion occurred between label errors and model errors, indicating that model-derived information can identify different sources of prediction–reference disagreement to a meaningful degree but cannot always determine whether the disagreement originates from the reference label or the model prediction. This distinction provides a basis for identifying candidate label errors for subsequent inspection or filtering, the effectiveness and downstream implications of which are evaluated in Research Questions 3 and 4.

5.2. Importance of Uncertainty Metrics

Research Question 2. Which uncertainty metrics are most useful for distinguishing these sample classes?

Table 7 compares the importance of the uncertainty metrics using three complementary approaches: decision tree feature importance, random forest feature importance, and the Hilbert–Schmidt Independence Criterion (HSIC) score. Although each method quantifies feature importance differently, all three were used to identify the uncertainty metrics that contributed most to distinguishing the manually interpreted sample classes.

Table 7: Comparison of uncertainty metric importance

Metric	DT	RF	HSIC
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Brier score	0.8617	0.2189	0.4142
Cross-entropy loss	0.0586	0.1730	0.3657
CE – entropy	0.0003	0.1878	0.3508
Calibration error	0.0001	0.0761	0.3137
Predictive entropy	0.0285	0.0476	0.0586
MC disagreement	0.0225	0.0318	0.0714
Mutual information	0.0105	0.0765	0.0211
Epistemic uncertainty	0.0177	0.0716	0.0102
Aleatoric uncertainty	0.0000	0.0433	0.0608
Softmax confidence	0.0002	0.0339	0.0604
Predictive uncertainty	0.0000	0.0395	0.0607

The decision tree assigned the greatest importance to the Brier score (0.8617), while all remaining metrics received substantially lower importance values. Cross-entropy loss (0.0586), predictive entropy (0.0285), Monte Carlo disagreement (0.0225), epistemic uncertainty (0.0177), and mutual information (0.0105) contributed modestly to the classification, whereas the remaining metrics had little or no influence on the fitted decision tree.

The random forest distributed feature importance more evenly across the uncertainty metrics than the decision tree. The Brier score again received the highest importance (0.2189), followed by CE – entropy (0.1878) and cross-entropy loss (0.1730). Mutual information (0.0765), calibration error (0.0761), and epistemic uncertainty (0.0716) also contributed to the classification, while predictive entropy, predictive uncertainty, aleatoric uncertainty, Monte Carlo disagreement, and softmax confidence exhibited comparatively lower importance values.

The HSIC analysis similarly identified the Brier score (0.4142), cross-entropy loss (0.3657), CE – entropy (0.3508), and calibration error (0.3137) as having the strongest statistical dependence with the manually interpreted sample classes. The remaining uncertainty metrics exhibited substantially lower HSIC scores, indicating weaker associations with the sample classes.

Across all three approaches, the Brier score consistently emerged as the most informative uncertainty metric for distinguishing the sample classes. Cross-entropy loss and CE – entropy were also consistently assigned relatively high importance by both the random forest and HSIC analyses. In contrast, predictive uncertainty, aleatoric uncertainty, predictive entropy, softmax confidence, Monte Carlo disagreement, mutual information, and epistemic uncertainty generally received lower importance scores, although their relative rankings differed among the three methods. Overall, the three analytical approaches consistently indicated that reference-dependent metrics incorporating disagreement between model predictions and reference labels contributed more strongly to distinguishing label errors, model errors, and correct samples than metrics based solely on predictive uncertainty.

This pattern can be explained by the information incorporated into the highest-ranking metrics. The Brier score and cross-entropy loss directly compare the model predictions with the reference labels and therefore respond strongly when the predicted class probabilities disagree with the assigned reference class. This property is particularly relevant for label errors, where the model prediction may be consistent with the satellite imagery but inconsistent with an incorrect reference label. CE – entropy additionally combines information about prediction error with predictive uncertainty, which may help distinguish disagreement with the reference label from uncertainty in the model prediction itself. In contrast, metrics based primarily on variability among Monte Carlo predictions characterise uncertainty in the model predictions without directly incorporating agreement with the reference label. This may explain their generally lower importance for distinguishing among the three sample classes.

The high importance of reference-dependent metrics in the present study is consistent with previous noisy-label research showing that prediction loss and prediction–label agreement contain useful information about label reliability. Tatjer et al. (2024) found that noisy samples tend to exhibit larger loss values than clean samples and demonstrated this behaviour specifically for cross-entropy loss, while also noting that the corresponding distributions can overlap. Cordeiro et al. (2023) similarly used classification loss, including cross-entropy loss, to distinguish samples considered clean from those considered noisy, with loss behaviour across multiple epochs used to improve the reliability of this separation. These findings are consistent with the relatively high importance of cross-entropy loss observed in the present study (Cordeiro et al., 2023; Tatjer et al., 2024).

Prediction–label consistency provides a related source of information. Zhang et al. (2024) used the consistency between predicted probabilities and observed labels to distinguish clean and noisy samples, with greater consistency indicating a greater likelihood that a sample was correctly labelled. Their approach used Jensen–Shannon divergence to quantify this relationship. Although Jensen–Shannon divergence differs mathematically from the Brier score, both are reference-dependent measures that quantify disagreement between model outputs and an assigned label. The consistently high importance of the Brier score in the present study is therefore consistent with the broader finding that prediction–label disagreement can provide useful information about label reliability. However, Zhang et al. (2024) also noted that erroneous model predictions can themselves lead to incorrect clean/noisy assignments. This is particularly relevant here because prediction–reference disagreement may arise from either a label error or a model error, meaning that high Brier or cross-entropy values should not be interpreted as uniquely identifying erroneous labels (Zhang et al., 2024).

The comparatively lower importance of several conventional uncertainty measures is also consistent with literature showing that uncertainty metrics are not interchangeable. Van Aalst et al. (2025) found that predictive entropy, mutual information, and variance differed significantly in their reliability, with predictive entropy performing better for calibration and error localisation in their application. Sahlsten et al. (2024) likewise observed substantial differences among eight uncertainty measures, including predictive entropy, expected entropy, mutual information, coefficient of variation, and structure-based variants, demonstrating that different formulations can have markedly different relationships with segmentation performance. These studies indicate that differences in metric importance can reflect the particular uncertainty component or prediction characteristic captured by each measure rather than indicating that any one metric is universally superior (Sahlsten et al., 2024; van Aalst et al., 2025).

The usefulness of an uncertainty measure can also depend on the downstream objective. Westphal et al. (2025), in a remote-sensing semantic-segmentation application, found that feature conformal prediction produced uncertainty estimates that were most closely aligned with overall network performance, whereas network probability was more useful for identifying misclassified pixels. Van Aalst et al. (2025) similarly

argued that particular uncertainty metrics may be more appropriate for specific downstream applications. These findings provide context for the different rankings observed in the present study: the objective here was not simply to estimate predictive uncertainty or identify misclassified pixels, but to distinguish among label errors, model errors, and correct samples. Metrics that are useful for general prediction reliability therefore need not be the most informative for this more specific three-class distinction (van Aalst et al., 2025; Westphal et al., 2025).

Previous noisy-label research further suggests that combining complementary signals can be useful when loss or learning difficulty alone is ambiguous. Deng et al. (2024) showed that hard clean samples and noisy samples can both exhibit high learning difficulty, but that combining learning difficulty with feature similarity improved their separation. Guan et al. (2023) similarly found that hard and noisy samples could both produce large losses and therefore supplemented loss information with differences among predicted class probabilities. Park et al. (2023) incorporated uncertainty information into an uncertainty-aware robust-learning framework and showed that combining this framework with the loss-based DivideMix approach improved clean/noisy partitioning under several noise settings. Together, these studies support the broader principle that combining complementary model-derived information can improve sample discrimination when individual indicators are ambiguous (Deng et al., 2024; Guan et al., 2023; Park et al., 2023).

This literature provides a useful context for the relatively high importance of CE – entropy in the present study. Cross-entropy directly incorporates disagreement between predicted probabilities and the reference label, whereas predictive entropy characterises uncertainty in the predictive distribution without requiring the reference label. CE – entropy therefore combines reference-dependent prediction-error information with predictive uncertainty. The cited studies do not evaluate CE – entropy itself, so its usefulness should not be presented as previously established. Rather, the present study provides empirical evidence that CE – entropy, together with the Brier score and cross-entropy loss, was particularly informative for distinguishing label errors, model errors, and correct samples. The contribution of this analysis is therefore the empirical comparison of the usefulness of reference-dependent and uncertainty-based metrics for this specific three-class distinction. While previous studies have established the value of loss, uncertainty, and complementary model-derived signals for noisy-label learning, the present results demonstrate which types of metrics were most informative when label errors and model errors were treated as separate competing classes (Deng et al., 2024; Guan et al., 2023; Park et al., 2023).

Answer to Research Question 2. The feature-importance analyses consistently identified the Brier score as the most informative metric for distinguishing the three manually interpreted sample classes. Cross-entropy loss and CE – entropy were also among the most informative metrics across the random forest and HSIC analyses. Overall, reference-dependent metrics incorporating prediction–reference disagreement were more informative for this specific three-class distinction than metrics based solely on predictive uncertainty, indicating that direct information about disagreement with the reference label was particularly useful when distinguishing label errors, model errors, and correct samples.

5.3. Uncertainty-Guided Label-Error Identification

Research Question 3. How effectively can uncertainty metrics be used to identify label errors in semantic segmentation?

The decision tree classification results presented previously demonstrate that the uncertainty metrics distinguished the manually interpreted sample classes, achieving a precision of 0.83 and a recall of 0.60 for the label-error class (Table 6). The corresponding confusion matrix (Figure 10) shows that most label

errors were correctly identified, although some were misclassified as model errors. In contrast, only two label-error samples were incorrectly classified as correct samples, indicating that the uncertainty metrics more reliably distinguished error samples from correct samples than they distinguished between the two error types.

Representative examples of the manually interpreted sample classes are presented in the following figures. Figure 11 shows examples of label errors in which areas labelled as deforestation in the reference dataset appear as non-deforested areas in the Landsat imagery. A likely explanation for several of these apparent label errors is temporal mismatch between the historical deforestation information represented by PRODES and the more recent Landsat imagery used in this study. Many of the affected areas were associated with deforestation mapped in earlier years, while the corresponding Landsat imagery showed vegetation cover at the time of image acquisition. These cases may therefore represent areas that were correctly mapped as deforested historically but subsequently experienced vegetation regrowth, causing the historical deforestation label to disagree with the current land-cover appearance. Figure 12 shows examples of model errors in which areas predicted as deforestation by the model appear as non-deforested areas in the Landsat imagery. These model errors indicate locations where the spectral and spatial information available to the U-Net produced a deforestation prediction despite the area being visually interpreted as non-deforested. Such errors may arise where non-deforested land-cover characteristics resemble those learned by the model for deforestation. However, the visual analysis conducted here does not establish the specific cause of each model error. Figure 13 shows correct samples, including both correctly labelled deforestation and non-deforestation areas.

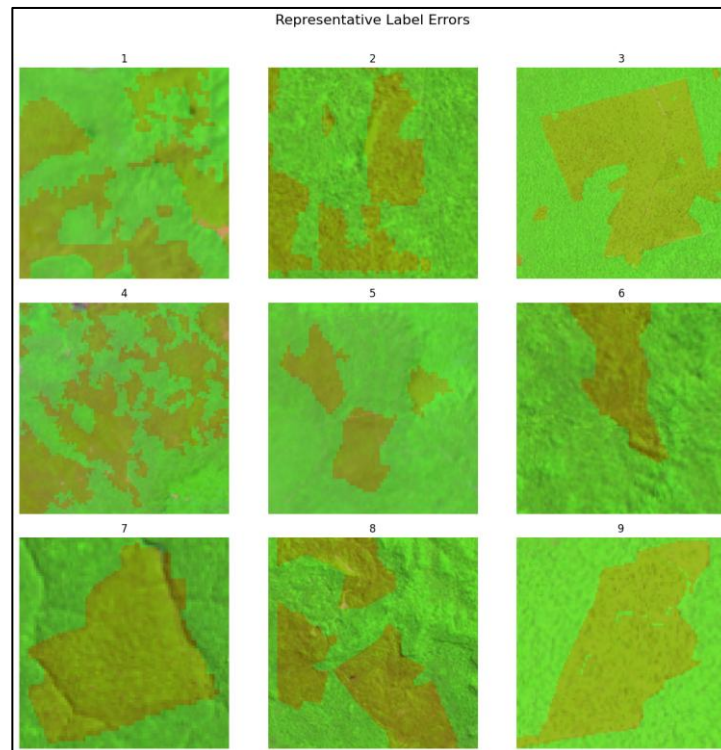


Figure 11: Representative label errors

Red areas indicate pixels labelled as deforestation in the reference dataset. In all examples, these labelled areas appear as non-deforested land in the corresponding Landsat imagery and were therefore manually interpreted as label errors.

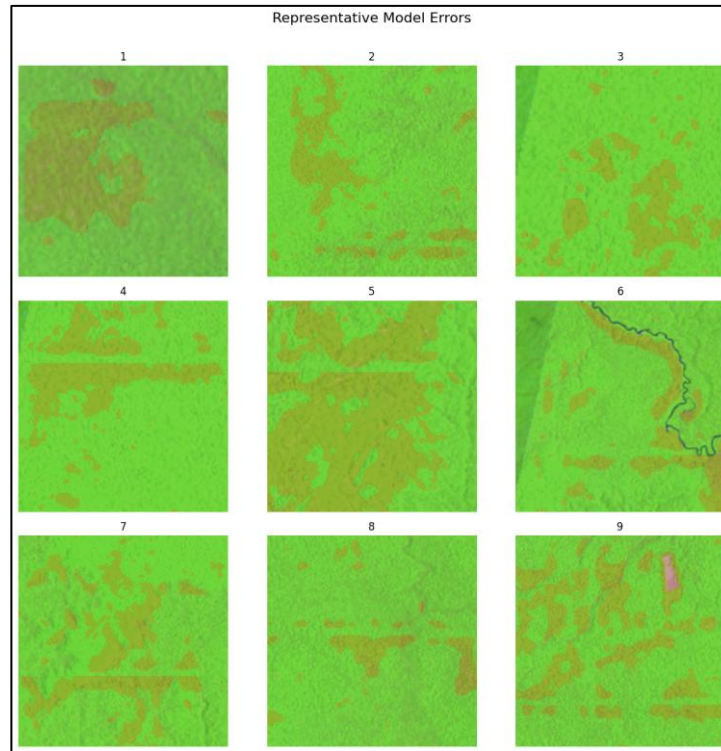


Figure 12: Representative model errors

Red areas indicate pixels predicted as deforestation by the model. In all examples, these predicted deforestation areas appear as non-deforested land in the corresponding Landsat imagery and were therefore manually interpreted as model errors.



Figure 13: Representative correct samples

Red areas indicate pixels labelled as deforestation in the reference dataset. Examples 1–5 show correctly labelled deforestation areas, where the reference labels correspond to deforested areas visible in the Landsat imagery. Examples 6–9 show correctly labelled non-deforestation areas, where the absence of a deforestation label is consistent with the non-deforested land cover visible in the imagery.

For uncertainty-guided label-error identification, precision is particularly important because incorrectly identifying correctly labelled pixels as label errors may introduce unnecessary modifications during subsequent retraining. The decision tree therefore provides a practical mechanism for prioritising candidate label errors while reducing false-positive identifications. The observed trade-off between precision and recall suggests that increasing the confidence threshold allows fewer but more reliable candidate label errors to be selected for further processing.

The ability of the classifier to identify label errors can partly be explained by the characteristic disagreement between the model predictions and the reference labels in these samples. As demonstrated by the feature-importance analysis, reference-dependent metrics such as the Brier score and cross-entropy loss were particularly informative for distinguishing the sample classes because they directly quantify disagreement between predicted probabilities and the assigned reference labels. The relatively high precision of 0.83 indicates that this combination of uncertainty metrics produced a sufficiently distinctive pattern for many of the samples classified as label errors, such that most samples predicted as label errors were confirmed as genuine label errors. However, the lower recall of 0.60 indicates that this pattern was not present or sufficiently distinctive for all label errors. Some label errors may exhibit uncertainty characteristics similar to model errors or correct samples, particularly where the model itself is uncertain or where the model prediction does not provide clear evidence that the reference label is incorrect. Moreover, disagreement between the prediction and reference label does not provide an unambiguous indicator of its source: it may result either from an incorrect reference label or from an incorrect model prediction. This overlap in the uncertainty characteristics of the two error types helps explain the observed confusion between the label-error and model-error classes. The results therefore suggest that uncertainty-guided classification can prioritise samples that are likely to contain label errors but cannot independently determine the source of every prediction–reference disagreement.

Overall, the quantitative classification results and representative examples indicate that uncertainty metrics provide a practical basis for prioritising candidate label errors for further inspection or filtering. The relatively high precision indicates that most samples identified as label errors were genuine label errors, while the moderate recall and confusion with model errors show that not all label errors could be reliably distinguished. Uncertainty-guided classification should therefore be interpreted as a method for narrowing the set of potentially erroneous labels rather than as a definitive means of determining whether every prediction–label disagreement originates from the reference data or the model.

Previous studies demonstrate that model-derived uncertainty and confidence can be operationalised for identifying potentially erroneous labels. Jakubik et al. (2025) developed uncertainty-quantification-based label-error detection methods combining confidence-based detection with predictive uncertainty information derived from approaches including Monte Carlo dropout, entropy, and ensemble learning. Their experiments showed that incorporating predictive uncertainty improved label-error detection compared with conventional confident-learning approaches. This provides evidence that uncertainty estimates can contribute useful information for identifying erroneous labels rather than serving only as measures of prediction reliability (Jakubik et al., 2025).

However, automated label-error detection remains imperfect, and its effectiveness depends on the relative importance assigned to identifying errors and avoiding false-positive detections. Nazaretyan et al. (2025) found substantial variation in both precision and recall across noise-filtering methods and observed a

characteristic trade-off between these measures. Methods with comparatively high precision could exhibit lower recall, whereas methods that identified a larger proportion of noisy samples could produce more false positives. They further found that naturally occurring label noise was more difficult to identify than artificially introduced noise and that detection performance depended on the dataset and the type and amount of noise. Label-error detection should therefore be evaluated in relation to its intended application rather than under the expectation that a single method will simultaneously maximise error detection and minimise false positives (Nazaretyan et al., 2025).

Evidence from remote-sensing semantic segmentation provides an important qualification to the usefulness of uncertainty for this purpose. Ulman et al. (2023) investigated pixel-wise predictive uncertainty for identifying omission errors in building-footprint labels and found that, although label noise affected uncertainty, the uncertainty distributions of clean and noisy pixels overlapped substantially. Consequently, uncertainty alone did not reliably separate the two groups, with a maximum IoU of 0.27 for noisy-pixel identification. Nevertheless, differences between the uncertainty distributions remained apparent under some noise conditions, leading the authors to suggest that uncertainty could still contribute to label cleaning by providing prior information about potentially noisy samples or identifying particularly trustworthy subsets of the data (Ulman et al., 2023).

The present results are consistent with this literature while demonstrating the potential of uncertainty-guided classification for candidate label-error identification. The decision tree achieved a precision of 0.83 and recall of 0.60 for the label-error class, indicating that most samples identified as label errors were genuine label errors, although a substantial proportion of manually verified label errors remained unidentified. Compared with the limited separation between clean and noisy pixels reported by Ulman et al. (2023), the present results demonstrate that model-derived information can provide a useful basis for identifying candidate label errors, although the moderate recall confirms that such identification remains incomplete (Jakubik et al., 2025; Nazaretyan et al., 2025; Ulman et al., 2023).

The practical value of such a classifier does not necessarily require exhaustive identification of every erroneous label. Model-derived probabilities can instead be used to rank or prioritise observations according to their likelihood of containing label errors. Northcutt et al. (2021), for example, incorporated ranking into Confident Learning to select observations considered more likely to contain label errors. Similarly, Bernhardt et al. (2022) demonstrated that prioritising samples likely to be clearly mislabelled can make label correction substantially more resource-efficient than random inspection. Their active label-cleaning approach required fewer re-annotations to obtain comparable improvements in label quality and, under some conditions, corrected labels up to four times more effectively than random selection. These findings support using label-error detection to reduce the search space and direct inspection or relabelling towards the most promising candidates rather than requiring every erroneous observation to be identified (Bernhardt et al., 2022; Northcutt et al., 2021).

A selective approach is particularly important because false-positive label-error detections can themselves reduce dataset quality. Lin et al. (2025) found that automated label cleaning could incorrectly identify and modify correctly labelled observations and that excessive repeated cleaning could decrease overall label accuracy. They also associated ambiguous observations with increased false labelling and proposed manual confirmation based on ranked label-quality scores as one possible strategy. In the present study, the relatively high precision and moderate recall can therefore be interpreted as a useful operating characteristic for candidate selection: the classifier identifies a smaller but comparatively reliable subset of label errors, while stricter probability thresholds can further trade coverage for confidence when selecting pixels for inspection, filtering, or subsequent retraining (Lin et al., 2025; Northcutt et al., 2021).

The scientific contribution of the present analysis is therefore the empirical quantification of how effectively uncertainty-guided classification can identify label errors at the pixel level when model errors are retained as a competing source of prediction–reference disagreement. Within the three-class framework established in RQ1, the decision tree achieved a precision of 0.83 and a recall of 0.60 for label errors, indicating that candidate label errors could be identified with comparatively high reliability, but that a substantial proportion of manually verified label errors remained undetected. These results extend previous work on uncertainty-based label-error detection by quantifying the effectiveness of candidate label-error identification when disagreement caused by incorrect model predictions is not subsumed within the clean or non-label-error class. The findings therefore show that uncertainty-guided classification can provide useful but incomplete pixel-level label-error identification under this more explicit attribution setting (Jakubik et al., 2025; Ulman et al., 2023).

Answer to Research Question 3. Uncertainty-guided classification was moderately effective for identifying candidate label errors at the pixel level. The decision tree achieved a precision of 0.83 and a recall of 0.60 for the label-error class, indicating that most samples identified as label errors were genuine label errors, although a substantial proportion of manually verified label errors remained undetected. The representative examples further demonstrate that the identified samples correspond to meaningful annotation inconsistencies visible in the imagery. When model errors were explicitly retained as a competing source of prediction–reference disagreement, the results therefore show that uncertainty-guided classification can provide useful but incomplete identification of candidate label errors and is better suited to prioritising pixels for subsequent inspection or filtering than to definitively determining the source of every prediction–reference disagreement.

5.4. Uncertainty-Guided Model Retraining

Research Question 4. Does filtering identified label errors improve model retraining?

Table 8 presents the effect of varying the decision tree probability threshold on the identification of candidate label errors. As the probability threshold increased, the number of selected pixels decreased substantially, while precision generally increased and recall decreased. At lower thresholds, a larger proportion of candidate label-error pixels was retained, resulting in higher recall but lower precision. Conversely, higher thresholds produced fewer candidate pixels with greater confidence but substantially reduced recall.

Table 8: Decision tree probability threshold analysis

Probability threshold (%)	Selected pixels	Precision	Recall
0%	150	0.42	1.00
5%	100	0.61	0.97
10%	100	0.61	0.97
15%	100	0.61	0.97
20%	73	0.75	0.87
25%	73	0.75	0.87
30%	73	0.75	0.87
35%	70	0.76	0.84
40%	63	0.78	0.78
45%	62	0.79	0.78

50%	46	0.83	0.60
55%	43	0.86	0.59
60%	26	0.85	0.35
65%	26	0.85	0.35
70%	6	0.83	0.08
75%	6	0.83	0.08
80%	6	0.83	0.08
85%	3	1.00	0.05
90%	2	1.00	0.03
95%	2	1.00	0.03
100%	0	NaN	0.00

A probability threshold of 85% was selected for generating the high-confidence label-error mask used for retraining. At this threshold, the decision tree achieved a precision of 1.00, indicating that all candidate label errors selected at this threshold were confirmed as label errors in the validation dataset, although the corresponding recall decreased to 0.05. This threshold was selected to minimise false-positive label-error identifications during retraining, thereby prioritising precision over recall.

The effect of uncertainty-guided sample filtering on model performance is summarised in Table 9. Three retraining strategies were evaluated: the baseline U-Net trained using the original reference labels, retraining after filtering the complete set of decision tree-identified label errors, and retraining using only the candidate label errors identified with an 85% decision tree probability threshold.

Table 9: Performance comparison of the retraining strategies

Method	Test F1 (mean $\pm$ SD)	Test IoU (mean $\pm$ SD)	Test Accuracy (mean $\pm$ SD)	Validation F1 (mean $\pm$ SD)	Validation IoU (mean $\pm$ SD)	Validation Accuracy (mean $\pm$ SD)
Baseline	0.82 $\pm$ 0.19	0.73 $\pm$ 0.17	0.92 $\pm$ 0.06	0.83 $\pm$ 0.21	0.75 $\pm$ 0.20	0.90 $\pm$ 0.11
Decision tree	0.82 $\pm$ 0.19	0.74 $\pm$ 0.17	0.92 $\pm$ 0.06	0.83 $\pm$ 0.21	0.75 $\pm$ 0.20	0.90 $\pm$ 0.11
Decision tree (85% probability)	0.82 $\pm$ 0.19	0.74 $\pm$ 0.17	0.92 $\pm$ 0.06	0.83 $\pm$ 0.21	0.76 $\pm$ 0.20	0.90 $\pm$ 0.11

Overall, retraining did not produce a meaningful improvement in segmentation performance. Retraining using all decision tree-identified label errors produced a mean test F1-score of 0.82, mean test IoU of 0.74, and mean test accuracy of 0.92, compared with 0.82, 0.73, and 0.92, respectively, for the baseline model. Similarly, retraining using the 85% probability threshold produced a mean test F1-score of 0.82, mean test IoU of 0.74, and mean test accuracy of 0.92. Validation performance was also comparable across the three models, with a mean F1-score of 0.83 and accuracy of 0.90 for all strategies, while validation IoU was 0.75 for the baseline and complete decision-tree filtering strategies and 0.76 for the 85% probability-threshold strategy. These differences were small relative to the variability across the ten training runs, as reflected by

the corresponding standard deviations. Overall, neither filtering strategy produced a meaningful improvement or deterioration in segmentation performance compared with the baseline model.

These results indicate that uncertainty-guided filtering of candidate label errors had little influence on the overall segmentation performance achieved by the U-Net under either retraining strategy. The similar performance of the two filtering strategies suggests that removing a larger set of decision tree-identified label errors did not provide a substantial advantage over removing only the highest-confidence candidates. In particular, although the 85% probability threshold achieved very high precision, its recall of 0.05 meant that only a small proportion of the label errors were identified and consequently available for filtering. The high reliability of the selected samples therefore came at the cost of substantially reducing the number of potentially erroneous training pixels removed.

Several factors may explain why neither filtering strategy produced a meaningful improvement over the baseline. One possible explanation is the high quality of the PRODES reference dataset. As described previously, PRODES is produced through expert visual interpretation of multitemporal satellite imagery and is subjected to specialist auditing and quality-control procedures (Almeida et al., 2021). Furthermore, PRODES has been reported to achieve an overall mapping accuracy of approximately 95% (Vieira et al., 2022). Consequently, the prevalence of annotation errors within the training data may already have been relatively low, limiting the potential benefit of filtering additional label errors. A further possible explanation is that the baseline U-Net was sufficiently robust to the remaining label noise. Deep neural networks have been shown to tolerate substantial levels of label noise under some conditions (Rolnick et al., 2017), and the similar performance of the baseline and retrained models in the present study is consistent with the possibility that the remaining label errors did not substantially affect the patterns learned by the model. Together, these findings suggest that successfully identifying candidate label errors does not necessarily translate into improved downstream segmentation performance, particularly when the original reference dataset is already of relatively high quality and the remaining label noise has limited influence on model learning.

Previous studies have demonstrated that identifying and correcting noisy labels can improve downstream semantic-segmentation performance, although these improvements depend on the severity of the label noise and the correction strategy employed. In remote-sensing semantic segmentation, Liu et al. (2024) showed that correcting incomplete building annotations could substantially improve performance under severe simulated label corruption, while the benefit was smaller under less severe corruption because models trained directly on the noisy annotations experienced less degradation. Similar findings have been reported in medical image segmentation. Gao et al. (2024) demonstrated that identifying and subsequently correcting noisy annotations generally improved segmentation performance compared with directly training on the uncorrected labels, including an increase in mean Dice score from 86.11% to 90.00% for real-world optic-cup annotations. At the pixel level, Pethmunee et al. (2025) likewise found that models retrained using refined segmentation masks generally outperformed models trained on the original annotations. However, these studies also showed that improvement was not automatic, with downstream performance depending on the severity of the noise and the reliability, extent, and implementation of the correction procedure (Gao et al., 2024; Liu et al., 2024; Pethmunee et al., 2025).

The absence of a meaningful improvement in the present study therefore differs from studies in which label correction produced substantial gains, but is consistent with evidence that the benefit of noisy-label handling depends strongly on how severely the original annotations affect model learning. Liu et al. (2024) found that models trained directly on less severely corrupted annotations experienced comparatively little damage and that the advantage of explicit noise handling became more pronounced as corruption increased. Similarly, Zhang et al. (2023) observed progressively poorer direct-training performance as

segmentation-label noise increased, with particularly strong benefits from noise-tolerant learning under severe corruption. These findings suggest that removing genuine label errors may have little effect when those errors constitute only a small or weakly influential component of the training data. Given the previously reported high quality of PRODES and its specialist quality-control procedures, the remaining annotation errors in the present study may therefore have been insufficiently prevalent or influential to substantially affect the baseline U-Net. Consequently, successfully identifying erroneous pixels did not necessarily imply that those pixels represented an important source of baseline model error (Liu et al., 2024; Zhang et al., 2023).

The amount of filtering also cannot be considered independently of the information retained for learning. Wu et al. (2024) found that downstream performance varied non-monotonically with filtering intensity and emphasised the need to balance the removal of unreliable observations against the retention of sufficient useful training information. This issue is particularly relevant to semantic segmentation because filtering may remove informative correctly labelled pixels, including high-loss pixels around semantic boundaries, thereby reducing useful information available to the model (Zhang et al., 2023). Thus, increasing the number of filtered pixels does not necessarily improve model performance, even when the filtering procedure successfully removes genuine annotation errors (Wu et al., 2024; Zhang et al., 2023).

The probability-threshold experiment further demonstrates a related trade-off between the reliability and coverage of label-error filtering. Tatjer et al. (2024) showed that the threshold used for clean/noisy sample partitioning changes the composition of the data used for subsequent learning: stricter thresholds increase selectivity but retain fewer clean observations, whereas more permissive thresholds admit a greater number of noisy observations. Both outcomes can affect downstream learning because noisy samples retained as clean can introduce incorrect information, while clean samples treated as noisy can result in the loss of relevant information. Similarly, Zhang et al. (2025) used a progressive strategy in which a small, reliable clean subset was first identified and subsequently expanded to improve model performance and robustness, illustrating that highly reliable sample selection must also provide sufficient useful information for subsequent learning (Tatjer et al., 2024; Zhang et al., 2025).

This trade-off provides an important interpretation of the two filtering strategies evaluated in the present study. The complete decision-tree label-error mask provided greater coverage of potential label errors but included more false-positive identifications, whereas the 85% probability threshold produced a highly reliable set of label errors at the expense of substantially reduced coverage. Although the latter achieved a precision of 1.00, its recall of only 0.05 meant that only a small proportion of the manually verified label errors were identified at this threshold. The absence of a meaningful performance improvement under either strategy suggests that neither increasing the coverage of filtering at the expense of precision nor maximising precision at the expense of coverage was sufficient to materially alter the patterns learned by the U-Net. This is consistent with previous evidence that filtering intensity and downstream performance are not monotonically related and that effective noisy-label learning requires a balance between noise removal and the preservation of useful training information (Tatjer et al., 2024; Wu et al., 2024; Zhang et al., 2025).

The scientific contribution of these findings is the empirical evaluation of whether uncertainty-guided pixel-level label-error detection translates into improved downstream model performance. Previous studies have shown that label correction can improve semantic segmentation while also demonstrating that its benefit depends on the severity of annotation noise and the correction procedure employed (Gao et al., 2024; Liu et al., 2024; Pethmunee et al., 2025). The present study provides evidence that, under the conditions evaluated here, identifying and filtering genuine label errors did not translate into improved model retraining. Neither the broader decision tree filtering strategy nor the more selective high-

confidence strategy produced a meaningful improvement over the baseline U-Net. These findings extend the existing literature by demonstrating this outcome for uncertainty-guided pixel-level filtering of a high-quality remote-sensing reference dataset and support the interpretation that the downstream benefit of label-error filtering is conditional on whether the removed errors materially affect model learning.

Answer to Research Question 4. Filtering identified label errors did not meaningfully improve model retraining performance. Both retraining using all decision tree-identified label errors and retraining using only high-confidence candidate label errors produced performance comparable to the baseline model in terms of F1-score, IoU, and overall accuracy. These results demonstrate that successfully identifying candidate label errors did not translate into improved downstream segmentation performance under the conditions evaluated in this study. This may reflect the relatively limited influence of the remaining label noise in the high-quality reference dataset, the possibility that the baseline model was sufficiently tolerant of this noise, and the trade-off between the reliability and coverage of the pixels selected for filtering.

6. CONCLUSION

6.1. Conclusion

This study investigated the use of uncertainty estimation for label-error identification in semantic segmentation of deforestation using Landsat imagery from the Brazilian Amazon. A four-stage methodology was developed comprising (1) baseline semantic segmentation and uncertainty estimation, (2) manual sample validation, (3) uncertainty-based label-error detection and feature analysis, and (4) uncertainty-guided model retraining. The objective was to evaluate whether uncertainty measures could distinguish label errors, model errors, and correct samples, identify the most informative uncertainty metrics, assess the effectiveness of uncertainty-guided label-error identification, and determine whether filtering identified label errors improved semantic segmentation performance.

The first stage demonstrated that uncertainty measures contain useful information for distinguishing the three manually interpreted sample classes. Although no individual uncertainty metric completely separated label errors, model errors, and correct samples, the combined uncertainty metrics enabled the decision tree to classify the sample classes with a macro-average F1-score of 0.77 and an overall accuracy of 0.84. These results indicate that different uncertainty measures capture complementary characteristics of prediction behaviour and provide useful discriminatory information when used together.

The feature analysis identified the Brier score as the most informative uncertainty metric for distinguishing the manually interpreted sample classes. Cross-entropy loss and CE – entropy were also identified as highly informative by the random forest and HSIC analyses, indicating that reference-dependent metrics incorporating prediction–reference disagreement contributed more strongly to distinguishing the sample classes than metrics based solely on predictive uncertainty.

The uncertainty-guided label-error detection stage demonstrated that uncertainty measures can be used to identify candidate label errors. The decision tree classifier achieved a precision of 0.83 and a recall of 0.60 for the label-error class, indicating that uncertainty-guided classification provides a practical mechanism for prioritising candidate label errors while maintaining relatively few false-positive identifications.

The final retraining stage demonstrated that uncertainty-guided filtering of candidate label errors did not meaningfully improve semantic segmentation performance. Retraining using either the complete set of decision tree-identified candidate label errors or only the high-confidence candidates produced F1-score, IoU, and overall accuracy comparable to those of the baseline model. The absence of a measurable improvement may reflect the relatively high quality of the PRODES reference dataset, the possibility that the baseline U-Net was sufficiently tolerant of the remaining label noise, or the limited number of erroneous pixels removed through filtering, particularly under the high-confidence strategy. These findings indicate that successfully identifying candidate label errors does not necessarily translate into improved downstream segmentation performance.

Overall, this study demonstrates that uncertainty estimation provides a practical framework for distinguishing label errors, model errors, and correct samples and for identifying candidate label errors within semantic segmentation datasets. Although uncertainty-guided retraining did not meaningfully improve segmentation performance under the conditions evaluated in this study, the proposed methodology provides a reproducible workflow for analysing uncertainty, identifying potential label errors, evaluating the contribution of individual uncertainty metrics, and assessing the impact of uncertainty-

guided sample filtering on semantic segmentation performance. Consequently, this work contributes to the growing field of data-centric artificial intelligence by demonstrating how uncertainty estimation can be used not only to quantify prediction reliability but also to support the systematic analysis of label quality in remote-sensing datasets.

6.2. Limitations

This study has several limitations that should be considered when interpreting the results. First, the manually interpreted reference dataset was constructed from a limited number of representative samples rather than the complete study area. Although these samples were carefully selected to represent label errors, model errors, and correct samples, they may not capture the full range of error characteristics present throughout the Landsat imagery. Consequently, the decision tree classifier was trained using a relatively small manually interpreted dataset, which may have limited its ability to generalise to all types of label and model errors.

Second, the study was conducted using a single semantic segmentation architecture (U-Net) and a single uncertainty estimation method based on Monte Carlo Dropout. The reported findings therefore reflect the behaviour of this particular modelling framework and may differ for alternative segmentation architectures or uncertainty estimation approaches, such as deep ensembles, evidential neural networks, or conformal prediction.

Third, the uncertainty-guided label-error detection framework relied on manually interpreted sample classes as reference data. Although manual interpretation provides a practical approach for constructing validation data, the interpretation process inevitably involves expert judgement, particularly in regions containing mixed land cover, gradual land-cover transitions, cloud contamination, or temporal inconsistencies between the satellite imagery and the reference labels.

Fourth, the retraining experiments were performed by filtering candidate label errors identified by the decision tree classifier rather than manually verifying and correcting the affected reference labels. Consequently, the retraining results depend on the accuracy of the automatically identified candidate label errors. Although a high probability threshold was selected to maximise precision, some true label errors remained undetected, while the limited number of selected samples restricted the amount of training data affected by the filtering procedure.

Finally, the absence of a meaningful improvement following retraining should not necessarily be interpreted as evidence that uncertainty-guided label-error detection is ineffective. The PRODES reference dataset is produced through expert visual interpretation and specialist quality-control procedures (Almeida et al., 2021) and has been reported to achieve an overall mapping accuracy of approximately 95% (Vieira et al., 2022). Consequently, the prevalence of annotation errors within the training data may already have been relatively low, limiting the potential benefit of correcting additional label errors. The absence of a meaningful improvement in this study may therefore reflect the high quality of the reference dataset rather than a limitation of uncertainty-guided label-error identification itself.

6.3. Future Work

The uncertainty-guided framework presented in this study provides a foundation for future research on data-centric semantic segmentation and label-error identification in remote sensing.

First, future studies could evaluate the proposed framework using larger manually interpreted reference datasets covering a broader range of land-cover conditions, seasons, and geographical regions. Increasing the diversity and number of manually interpreted samples would enable more comprehensive evaluation of uncertainty-guided label-error identification and improve the generalisability of the feature-importance analyses and classification models.

Second, the methodology could be extended to alternative semantic segmentation architectures and uncertainty estimation methods. While this study employed a U-Net with Monte Carlo Dropout, future work could compare the proposed framework with uncertainty estimates derived from deep ensembles, evidential neural networks, Bayesian neural networks, or conformal prediction. Such comparisons would help determine whether the observed relationships between uncertainty measures and label errors are consistent across different modelling approaches.

Third, future research could investigate alternative machine-learning models for uncertainty-guided label-error detection. Although this study employed an interpretable decision tree classifier, other supervised learning approaches, such as gradient boosting, support vector machines, or neural-network classifiers, may improve the discrimination between label errors and model errors while maintaining acceptable levels of interpretability.

Fourth, the retraining strategy could be extended by incorporating human verification of automatically identified candidate label errors before retraining. Rather than filtering candidate samples directly, an interactive human-in-the-loop workflow could combine uncertainty-guided prioritisation with expert validation to construct higher-quality reference datasets and assess the resulting impact on semantic segmentation performance.

Finally, the proposed methodology could be evaluated using additional remote sensing datasets, land-cover products, and application domains beyond Amazonian deforestation mapping. Applying the framework to different sensors, spatial resolutions, and land-cover classification problems would help determine the extent to which uncertainty-guided label-error identification generalises across remote sensing applications.

ETHICAL CONSIDERATIONS

This research did not involve human participants or the collection, processing, or analysis of personal data. The study relied exclusively on existing Earth-observation and geospatial datasets and therefore did not require participant recruitment, informed consent, or procedures for protecting the confidentiality of personal information.

The principal ethical considerations concern the interpretation and potential application of the research results. The uncertainty-guided approach developed in this study identifies candidate label errors rather than definitively establishing that the corresponding reference labels are incorrect. Incorrect deforestation predictions or incorrectly identified label errors could therefore lead to misleading interpretations if the outputs were treated as authoritative environmental information. In particular, the identified candidate label errors should not be interpreted as definitive corrections to the operational PRODES deforestation dataset without additional expert verification.

Ethical considerations also arise from uncertainties inherent in the research methodology. Manual interpretation of the satellite imagery involves some degree of interpretative uncertainty, while temporal differences between the Landsat imagery and PRODES reference data may produce apparent disagreements that do not necessarily represent annotation errors. Furthermore, the decision tree classifier does not perfectly distinguish label errors from model errors, and the identification and filtering of candidate label errors does not necessarily result in improved segmentation performance.

These risks were mitigated through manual interpretation of the selected samples, independent validation of the decision tree classifications, transparent reporting of model performance and uncertainty estimates, and explicit acknowledgement of the limitations of the methodology. Automatically identified label errors are therefore reported as candidate label errors, and the conclusions are restricted to the datasets, study area, and experimental conditions evaluated in this research. The proposed approach should consequently be regarded as a method for supporting label-quality assessment and prioritising potentially erroneous samples for further inspection rather than as an automated replacement for expert validation or established operational deforestation-monitoring products.

DATA MANAGEMENT PLAN

Research Data

1. Primary Data

Table: Overview of Primary Data

Data type or name	Data form	File format	Contains personal data?	Rights claimed by a third party?
Training, validation, and test data derived from secondary datasets	Geospatial/tabular data	GeoTIFF (.tif), CSV (.csv)	No	No
Manual interpretation and validation datasets	Geospatial vector/tabular data	GeoPackage (.gpkg), CSV (.csv), Excel (.xlsx)	No	No
Uncertainty, prediction, and error-analysis datasets	Geospatial raster/vector/tabular data	GeoTIFF (.tif), GeoPackage (.gpkg), CSV (.csv), Parquet (.parquet)	No	No
Trained models and model-training, filtering, and retraining results	Machine-learning models and tabular/array outputs	PyTorch model files (.pt), Joblib files (.joblib), NumPy files (.npy), CSV (.csv), SQLite database files (.db)	No	No
Processing, modelling, and analysis code	Software/code	Jupyter Notebook (.ipynb)	No	No

Note: Generated and derived research data may incorporate information from the secondary datasets listed in Table 2 and therefore remain subject, where applicable, to the terms and licences of the underlying source data.

2. Secondary Data

Table: Overview of Secondary Data

Data type or name	Data form	File format	Data owner	Contains personal data?	Openly accessible? If yes, under what licence?	Access restrictions
Landsat 8–9 Collection 2	Satellite Earth	GeoTIFF (.TIF),	U.S. Geological	No	Yes. Public Domain	None; acknowledgement/citati

Level-2 imagery	observation data	with associated metadata (.json, .xml, .txt)	Survey (USGS), Earth Resources Observation and Science (EROS) Center			on of the data source is requested
PRODES deforestation data (2024)	Geospatial deforestation data	GeoTIFF (.tif)	Brazilian National Institute for Space Research (INPE)	No	Yes. Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)	Attribution and share-alike requirements apply

Data Organising and Documentation

1. Folder structure

The research data are organised primarily according to data-processing stages and individual experiments. Source and processed datasets are stored separately from generated outputs. Model and analysis outputs are further organised within experiment-specific directories, with subfolders used where appropriate to distinguish different types of outputs, including raster data, tabular results, figures, GeoPackages, model outputs, and statistical analyses.

Because the research involved iterative model development and exploratory analysis, the working directory also contains intermediate datasets, alternative analyses, and earlier versions of notebooks and outputs. These files are retained to preserve the development history of the research rather than being comprehensively reorganised after completion.

For final archiving, documentation will identify the principal datasets, notebooks, models, and outputs corresponding to the methodology and results reported in the thesis. This will allow the existing research directory structure to be retained while providing users with guidance on which files are required to understand and reproduce the main analyses.

2. File naming convention

Files are generally named descriptively according to their purpose, processing stage, model or experiment, dataset subset, uncertainty metric, or type of analysis. File and directory names frequently use underscores to separate descriptive terms. Generated outputs are also grouped into format-specific or analysis-specific directories where appropriate, such as folders containing figures, CSV tables, GeoPackages, and raster outputs.

Because the research was conducted iteratively, some files and notebooks contain suffixes or identifiers indicating different working versions. These version identifiers are retained where necessary to distinguish iterations. The README accompanying the archived data will identify the principal files used for the final thesis analyses so that users do not need to infer the final workflow from file names alone.

3. Version control

Formal version control was not used consistently during the research. Instead, development versions of Jupyter notebooks, datasets, models, and generated outputs were retained as separate files and directories. This resulted in multiple intermediate and experimental versions being preserved throughout the project.

For archiving, the files and notebooks corresponding to the analyses reported in the final thesis will be identified in the accompanying README. Intermediate versions may also be retained where they provide a record of the analytical development, but they will not be presented as part of the principal reproducibility workflow.

4. Data documentation

A README file will accompany the archived research data and serve as a guide to the existing project directory. Rather than documenting every intermediate file individually, it will identify the principal files and directories required to understand and reproduce the research. The README will document the main data sources, processing and analysis steps, relevant class definitions, and geospatial information, and the principal notebooks and outputs corresponding to the analyses reported in the thesis.

Where applicable, existing geospatial metadata embedded within GeoTIFF and GeoPackage files, including coordinate reference systems, spatial extents, raster dimensions, resolution, data types, and attribute information, will be retained. No additional formal metadata standard will be applied.

Data Storage, Security and Sharing

1. Storage of research data

All research data and files were primarily stored on ITC's Geospatial Computing Platform, which was also used for the computational processing and analysis conducted during the research. The research did not involve personal or confidential data.

2. Additional copies

Copies of important research files were stored on the encrypted hard drive of the researcher's personal computer as an additional backup against accidental data loss.

3. Security measures

Access to the research data stored on the Geospatial Computing Platform was controlled through the researcher's institutional account. The additional copies stored on the personal computer were protected by hard-drive encryption and the computer's access controls. No anonymisation or pseudonymisation was required because the research did not involve personal data.

4. Data sharing during the research

Research data, intermediate results, and final outputs were shared with the thesis supervisors where required for supervision and evaluation. The secondary Landsat and PRODES datasets used in the research are publicly accessible.

Data Archiving

1. Data selected for archiving

The research data, generated products, trained models, analysis code, and accompanying documentation required to understand, verify, and reproduce the research will be deposited in the ITC MSc research data archive for long-term preservation. The research does not contain personal, confidential, or otherwise sensitive data.

Table: Research data and materials selected for long-term archiving

Data type or file	Preservation file format	Can be made publicly available?	Access restrictions
Training, validation, and test data derived from secondary datasets	GeoTIFF (.tif), CSV (.csv)	Yes*	No
Manual interpretation and validation datasets	GeoPackage (.gpkg), CSV (.csv), Excel (.xlsx)	Yes	No
Uncertainty, prediction, and error-analysis datasets	GeoTIFF (.tif), GeoPackage (.gpkg), CSV (.csv), Parquet (.parquet)	Yes*	No
Trained models and model-training, filtering, and retraining results	PyTorch (.pt), Joblib (.joblib), NumPy (.npy), CSV (.csv), SQLite (.db)	Yes*	No
Processing, modelling, and analysis code	Jupyter Notebook (.ipynb)	Yes	No
Research documentation	README (.txt) and metadata sheet	Yes	No

*Public reuse or redistribution of generated or derived data remains subject, where applicable, to the terms and licences of the underlying secondary datasets.

2. Data not selected for archiving

The original Landsat 8–9 Collection 2 Level-2 imagery and PRODES deforestation data will not be duplicated in the archive because these secondary datasets are publicly available from their respective data providers. Their sources and relevant dataset information will be documented to allow the data to be reacquired for research verification and reproduction.

USE OF GENERATIVE AI

ChatGPT (OpenAI) was used to assist with debugging and commenting code and with language editing, including improvements to clarity, spelling, and grammar. The tool was used in a supporting capacity, while the research design, analyses, and interpretation remain the work of the author. The author takes full responsibility for the content and integrity of this thesis.

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